ENGINEERING REFERENCE MANUAL · PART 14 OF 14

Space Exploration & Aerospace Engineering

Part XIV

Appendices and Reference Data

Constants, equations, planetary data, frequency bands, mission databases, a timeline, acronyms, a glossary, standards and sources, and image credits — the quick-reference tools an engineer keeps within reach.



Appendices A–J · pages 1071–1183 of the complete manual

First Edition · Current to October 2026 · Written in plain language with deep technical detail

ENGINEERING REFERENCE MANUAL

Space Exploration & Aerospace Engineering

A complete, illustrated guide to the space ecosystem — from stratospheric balloons to interstellar probes — explaining how every major class of space system works, how it is engineered, and where artificial intelligence is changing it.

First Edition · Information current to October 2026
A4 format · Fourteen Parts · Fifty-three chapters · Ten appendices
Version 1.0 · October 2026

About this edition

Space Exploration & Aerospace Engineering — An Engineering Reference Manual. First Edition, Version 1.0, October 2026.

This manual was written to be read by students, engineers, analysts, managers and curious readers. Each topic starts with plain-language explanations and builds up to the equations, numbers and design trade-offs that practising engineers use. Sentences are kept short and every technical term is explained in plain words when it first appears; the glossary (Appendix H) and the list of acronyms (Appendix G) collect them. Facts, mission status and programme milestones are current to October 2026; space programmes change quickly, so dates of future events should be read as plans, not certainties.

Figures, photographs and data

Diagrams, charts and graphs were drawn specifically for this manual from published physical relationships and public data. Photographs are from the NASA Image and Video Library and are used under NASA's media usage guidelines (NASA material is generally not subject to copyright in the United States); individual credits appear with each image and in Appendix J. NASA does not endorse this publication. Numerical values (masses, power levels, performance) are representative published figures and may differ slightly between sources and mission phases.

Safety and responsibility

This is an educational reference. It is not a design standard, a flight rule, or a substitute for the requirements, analyses and verification processes of a qualified engineering organisation. Chapters dealing with military and intelligence systems use only openly published information and describe principles, not operational or classified details.

How numbers are written

SI units are used throughout, with common aerospace units (nautical miles, feet, pounds-force) shown where the industry uses them. Large numbers use thin spacing (e.g., 35 786 km). "≈" means approximately, "~" means order-of-magnitude. Dates follow the ISO format (YYYY-MM-DD) in tables.

Typography and production

Set in Source Serif 4, Inter and JetBrains Mono. Equations typeset with MathJax. Produced from Markdown sources with an automated pipeline (pandoc, Chromium, pypdf); the sources and build scripts accompany the PDF.

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This volume is one Part of *Space Exploration & Aerospace Engineering — An Engineering Reference Manual*. Page numbers, chapter numbers and figure numbers are the same as in the complete manual, so cross-references to other chapters (for example “Chapter 24”) and to the appendices refer to the complete book.



PART XIV

Appendices and Reference Data

Constants, equations, planetary data, frequency bands, mission databases, a timeline, acronyms, a glossary, standards and sources, and image credits — the quick-reference tools an engineer keeps within reach.

A Constants, Units and Time Systems
B Key Equations
C Solar System Data
D Radio-Frequency Bands and Spectrum
E Mission and Vehicle Databases
F Timeline of Space Exploration and Engineering
G Acronyms and Abbreviations
H Glossary
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J Image Credits

Image: NASA/GSFC

APPENDIX

A Constants, Units and Time Systems

The physical and astronomical constants used throughout this manual, the SI units and prefixes, conversion factors for the non-SI units still common in aerospace, decibel conventions used in link budgets, and the time scales and epochs used to timestamp orbits, telemetry and observations. Values follow the 2019 SI, CODATA 2022 recommended values, the IAU 2012 and 2015 resolutions, and the WGS 84 Earth model.

IN THIS CHAPTER

A.1 The SI and Its Defining Constants
A.2 Physical Constants
A.3 Astronomical and Earth Constants
A.4 Conversion Factors
A.5 Decibels
A.6 Time Scales and Epochs

This appendix collects the numbers an engineer reaches for most often. Exact values (by definition) are marked **exact**; other values are given to the precision useful in engineering work, with the last digits uncertain as in the source. Where two sources differ slightly — for example the Earth's gravitational parameter in WGS 84 and in the latest geodetic models — the value used throughout this manual is given, and the difference is far smaller than any engineering margin.

A.1 The SI and Its Defining Constants

Since 20 May 2019 all seven SI base units are defined by fixing the numerical values of seven constants of nature (Table A.1). The kilogram is no longer a metal cylinder in Paris; it is defined through the Planck constant. For practical engineering nothing changed, but every unit can now in principle be realised anywhere — including in orbit.

TABLE A.1 The seven defining constants of the SI (2019).

Defining constant	Symbol	Value (exact)	Unit it defines
Caesium hyperfine transition frequency	$\Delta\nu_{Cs}$	9 192 631 770 Hz	second
Speed of light in vacuum	c	299 792 458 m/s	metre
Planck constant	h	$6.626\,070\,15 \times 10^{-34}$ J s	kilogram
Elementary charge	e	$1.602\,176\,634 \times 10^{-19}$ C	ampere
Boltzmann constant	k	$1.380\,649 \times 10^{-23}$ J/K	kelvin
Avogadro constant	N_A	$6.022\,140\,76 \times 10^{23}$ mol^{-1}	mole
Luminous efficacy of 540 THz light	K_{cd}	683 lm/W	candela

TABLE A.2 SI prefixes. Ronna, quetta, ronto and quecto were added in 2022 — partly so that data volumes and planetary masses could be written compactly (the Earth's mass is about 6 Rg).

Factor	Prefix	Symbol	Factor	Prefix	Symbol
10^{30}	quetta	Q	10^{-1}	deci	d
10^{27}	ronna	R	10^{-2}	centi	c
10^{24}	yotta	Y	10^{-3}	milli	m
10^{21}	zetta	Z	10^{-6}	micro	μ
10^{18}	exa	E	10^{-9}	nano	n
10^{15}	peta	P	10^{-12}	pico	p
10^{12}	tera	T	10^{-15}	femto	f
10^{9}	giga	G	10^{-18}	atto	a
10^{6}	mega	M	10^{-21}	zepto	z
10^{3}	kilo	k	10^{-24}	yocto	y
10^{2}	hecto	h	10^{-27}	ronto	r
10^{1}	deca	da	10^{-30}	quecto	q

For data storage the binary prefixes are also standardised (IEC 80000-13): 1 KiB (kibibyte) = 1 024 bytes, 1 MiB = 1 048 576 bytes, 1 GiB = 2^{30} bytes, 1 TiB = 2^{40} bytes. Space-mission documents usually quote downlink rates in decimal units (1 Mbit/s = 10^{6} bit/s) and on-board memory in binary units, so a "256 GB" recorder may hold 256 GiB = 275 × 10^{9} bytes. Always check which convention a specification uses.

A.2 Physical Constants

Table A.3 lists the physical constants used in this manual. The gravitational constant G is known only to about 22 parts per million, which is why orbital mechanics always uses the product GM (the *gravitational parameter* μ) — it can be measured from spacecraft tracking about a million times more precisely than G or M separately.

TABLE A.3 Physical constants (CODATA 2022 recommended values).

Constant	Symbol	Value	Notes
Newtonian constant of gravitation	G	$6.674\,30 \times 10^{-11}$ m^3 kg^{-1} s^{-2}	± 0.000 15
Stefan–Boltzmann constant	σ	$5.670\,374\,419 \times 10^{-8}$ W m^{-2} K^{-4}	exact (derived)
Wien displacement constant	b	$2.897\,771\,955 \times 10^{-3}$ m K	$\lambda_{peak} = b/T$
Molar gas constant	R	8.314 462 618 J mol^{-1} K^{-1}	exact (derived)
Vacuum permittivity	ε_0	$8.854\,187\,8188 \times 10^{-12}$ F/m	
Vacuum permeability	μ_0	$1.256\,637\,061\,27 \times 10^{-6}$ N/A^2	$\approx 4\pi \times 10^{-7}$
Impedance of free space	Z_0	376.730 Ω	$\mu_0 c$
Electron mass	m_e	$9.109\,383\,7139 \times 10^{-31}$ kg	0.511 MeV/c^2
Proton mass	m_p	$1.672\,621\,925\,95 \times 10^{-27}$ kg	938.272 MeV/c^2
Atomic mass constant	m_u	$1.660\,539\,068\,92 \times 10^{-27}$ kg	1 Da
Electronvolt	eV	$1.602\,176\,634 \times 10^{-19}$ J	exact

Constant	Symbol	Value	Notes
Standard acceleration of gravity	g_o	9.806 65 m/s²	exact; used for I_{sp}
Standard atmosphere	atm	101 325 Pa	exact
Boltzmann constant in decibels	10 log k	−228.6 dBW K^{-1} Hz^{-1}	link budgets

A.3 Astronomical and Earth Constants

TABLE A.4 Astronomical and Earth constants.

Quantity	Value	Source / note
Astronomical unit (au)	149 597 870 700 m	exact (IAU 2012)
Light-time for 1 au	499.005 s (8 min 19 s)	
Light-year (Julian year × c)	9.460 730 × 10^{15} m = 63 241 au	exact
Parsec	3.085 678 × 10^{16} m = 3.262 ly = 206 265 au	exact via au
Sun, $GM_\odot$	1.327 124 400 41 × 10^{20} m^3/s^2	planetary ephemerides
Sun, mass	1.988 4 × 10^{30} kg	GM/G
Sun, nominal radius	695 700 km	IAU 2015
Sun, nominal luminosity	3.828 × 10^{26} W	IAU 2015
Sun, effective temperature	5 772 K	IAU 2015
Total solar irradiance at 1 au	≈ 1 361 W/m²	varies ± 0.1 % with the cycle
Earth, $GM_\oplus$	3.986 004 418 × 10^{14} m^3/s^2	WGS 84
Earth, mass	5.972 2 × 10^{24} kg	
Earth, equatorial radius a	6 378 137 m	WGS 84
Earth, flattening f	1/298.257 223 563	WGS 84
Earth, polar radius b	6 356 752.3 m	WGS 84
Earth, mean radius	6 371.0 km	
Earth, J_2 zonal harmonic	1.082 63 × 10^{-3}	drives nodal regression
Earth, rotation rate $\omega_\oplus$	7.292 115 × 10^{-5} rad/s	
Sidereal day	86 164.091 s (23 h 56 min 4.09 s)	
Mean solar day	86 400 s	
Tropical year	365.242 19 days	seasons
Sidereal year	365.256 36 days	
Obliquity of the ecliptic (J2000)	23.439°	
Moon, GM	4 902.800 km^3/s^2	
Moon, mean distance	384 400 km (light-time 1.28 s)	varies 356 400–406 700 km
Moon, mean radius	1 737.4 km	

USEFUL ORBITAL NUMBERS FOR EARTH

TABLE A.5 Circular-orbit period, speed and horizon distance (spherical Earth, *GM* from WGS 84). One degree of longitude along the geostationary arc is 736 km.

Altitude	Period	Circular speed	Distance to horizon	Typical use
200 km	88.5 min	7.78 km/s	1 610 km	Lowest practical orbit, parking orbits
400 km	92.6 min	7.67 km/s	2 290 km	ISS, Tiangong
550 km	95.6 min	7.59 km/s	2 710 km	Starlink shells, EO satellites
800 km	100.9 min	7.45 km/s	3 290 km	Sun-synchronous weather and EO
1 200 km	109.4 min	7.25 km/s	4 090 km	OneWeb, Jason/Sentinel-6
20 200 km	11 h 58 min	3.87 km/s	25 800 km	GPS (half a sidereal day)
35 786 km	23 h 56 min	3.07 km/s	41 680 km	Geostationary orbit (radius 42 164 km)

Escape speed from the Earth's surface is 11.19 km/s; from a 400 km orbit it is 10.85 km/s, only 3.18 km/s more than circular speed.

A.4 Conversion Factors

Aerospace still mixes SI with US customary units — rocket thrust in pounds-force, altitude in feet, pressures in psi, antenna gain in decibels. Table A.6 lists the conversions used most often; all are exact except where three or fewer significant figures are shown as a rounded value.

TABLE A.6 Conversion factors.

Quantity	Unit	SI value
Length	inch / foot	0.025 4 m / 0.304 8 m (exact)
	statute mile / nautical mile	1 609.344 m / 1 852 m (exact)
Mass	pound (lb)	0.453 592 37 kg (exact)
	slug	14.593 9 kg
	metric tonne (t)	1 000 kg
Force	pound-force (lbf)	4.448 222 N
	kilogram-force (kgf)	9.806 65 N (exact)
Pressure	psi (lbf/in²)	6 894.757 Pa
	bar / atm / torr	10^5 Pa / 101 325 Pa / 133.322 Pa
Energy	kilowatt-hour	3.6 MJ (exact)
	British thermal unit (BTU)	1 055.06 J
	thermochemical calorie	4.184 J (exact)
	kilotonne of TNT	4.184×10^{12} J
Power	horsepower (mechanical)	745.700 W
Speed	knot / mile per hour	0.514 444 m/s / 0.447 04 m/s
	Mach 1 at sea level (15 °C)	340.3 m/s

Quantity	Unit	SI value
Angle	degree / arcminute / arcsecond	17.453 3 mrad / 290.888 µrad / 4.848 14 µrad
	milliarcsecond (mas)	4.848 14 nrad
	steradian (sphere = 4π sr)	1 sr = 3 282.8 deg²
Temperature	°C / °F / °R	$T_K = T_{°C} + 273.15$; $T_{°F} = 1.8\ T_{°C} + 32$; $T_{°R} = 1.8\ T_K$
Radiation dose	rad / rem	0.01 Gy / 0.01 Sv
Magnetic flux density	gauss (G) / gamma (γ)	10^{-4} T / 1 nT
Specific impulse	second (s)	effective exhaust speed $v_e = I_{sp} \times g_0$ (300 s ≈ 2.94 km/s)
Thrust-specific fuel consumption	lb/(lbf h)	28.33 mg/(N s)
Data rate	bit/s (decimal)	1 Gbit/s = 10^9 bit/s

A.5 Decibels

Radio engineers express ratios that span many orders of magnitude in decibels: a power ratio P_2/P_1 is $10 \log_{10}(P_2/P_1)$ dB. A factor of two is 3.01 dB, ten is 10 dB, a thousand is 30 dB, and a million is 60 dB. Multiplications in a link budget become additions. A suffix tells you the reference (Table A.7).

TABLE A.7 Decibel conventions used in link budgets (Chapter 7).

Unit	Reference	Example
dBW	1 W	20 W transmitter = 13.0 dBW
dBm	1 mW	1 W = 30 dBm; GPS signal at the ground ≈ −128.5 dBm
dBi	isotropic antenna	1 m dish at 8.4 GHz with 60 % efficiency ≈ 36.7 dBi
dBic	isotropic, circular polarisation	common for GNSS and S-band patch antennas
dBK	1 kelvin	290 K system noise temperature = 24.6 dBK
dB/K	gain-to-noise temperature ratio G/T	ground station 35 dB/K
dBHz	1 Hz	1 MHz bandwidth = 60 dBHz; C/N_0 in dBHz
dBW/m²	flux density	saturated flux density of a GEO transponder
dBsm	1 m² (radar cross-section)	a 1U CubeSat ≈ −20 dBsm at high frequency
dBc	carrier power	spurious emission 60 dBc below the carrier

A.6 Time Scales and Epochs

A spacecraft clock, a ground station log and a planetary ephemeris may each use a different time scale. Mixing them up produces errors of a minute or more — enough to put a planetary encounter or a collision-avoidance manoeuvre in the wrong place. Table A.8 summarises the

scales in use.

TABLE A.8 Time scales used in space operations.

Time scale	What it is	Relation (2026)
TAI (International Atomic Time)	Weighted average of about 450 atomic clocks worldwide	Reference
UTC (Coordinated Universal Time)	Civil time; stays within 0.9 s of Earth rotation by inserting leap seconds	UTC = TAI − 37 s (since 1 January 2017)
UT1	Time defined by the Earth's actual rotation angle	UT1 − UTC within ± 0.9 s, published by the IERS
GPS time	Continuous atomic time of the GPS system, started 6 January 1980	GPS = TAI − 19 s = UTC + 18 s
Galileo System Time / BeiDou Time	Continuous GNSS time scales	GST = GPS time; BDT = UTC + 4 s
TT (Terrestrial Time)	Ideal time on the geoid, used for ephemerides	TT = TAI + 32.184 s = UTC + 69.184 s
TDB (Barycentric Dynamical Time)	Time at the Solar System barycentre, used by planetary ephemerides	TDB − TT within ± 1.7 ms (periodic)
Spacecraft clock (SCLK)	On-board counter; drifts	Correlated to UTC by the ground (SPICE kernels)

Leap seconds. Because the Earth's rotation is irregular, 27 leap seconds were added to UTC between 1972 and 2016. None has been needed since, because the Earth has recently been spinning slightly faster. In 2022 the General Conference on Weights and Measures decided that the tolerance between UTC and UT1 will be relaxed by 2035, effectively ending leap seconds. Software that assumes every minute has 60 seconds, or that GPS time equals UTC, has caused real outages.

Julian Date. Astronomers count days continuously from noon on 1 January 4713 BC (Julian calendar). The Julian Date (JD) at 00:00 UTC on 1 January 2026 is 2 461 041.5. The Modified Julian Date is MJD = JD − 2 400 000.5, so a day starts at midnight and the numbers are shorter (MJD 61 041 for the same instant). The standard epoch **J2000.0** is JD 2 451 545.0 TT — 12:00 TT on 1 January 2000 (11:58:55.816 UTC). Orbital elements in two-line element sets use a two-digit year and day-of-year with fraction.

GNSS weeks. GPS broadcasts a week number that originally rolled over every 1 024 weeks (19.6 years); rollovers on 21 August 1999 and 6 April 2019 broke unprepared receivers. Modernised GPS messages use a 13-bit week number that will not roll over until 2137.

Relativity matters. A GPS satellite clock runs fast by about 38 µs per day compared with a clock on the ground — about +45.9 µs from the weaker gravity at 20 200 km and −7.2 µs from its orbital speed. Uncorrected, this would cause ranging errors growing by about 11 km per day; the clocks are deliberately set slightly slow before launch. A clock on the Moon runs fast by about 56 µs per day relative to Earth. In 2024 the US government asked NASA to establish a Coordinated Lunar Time, and the International Astronomical Union adopted resolutions calling for a lunar reference system and time scale, so that lunar navigation systems such as LunaNet and ESA's Moonlight can share a common clock.

Mars time. A Martian solar day (*sol*) lasts 24 h 39 min 35.244 s (88 775.244 s). Mars rover teams work on "Mars time" for the first weeks of a mission, shifting their working day by about 40 minutes every Earth day. A Mars year lasts 668.6 sols (687.0 Earth days).

KEY TAKEAWAYS

- Seven constants now define the SI exactly; c, h, e and k have no uncertainty.
- Use the gravitational parameter $\mu = GM$, not G and M separately — it is known far more precisely.
- $g_0 = 9.806\,65\ \text{m/s}^2$ exactly converts specific impulse in seconds to exhaust speed.
- In link budgets, add decibels; −228.6 dBW/(K Hz) is Boltzmann's constant.
- UTC = TAI − 37 s, GPS = UTC + 18 s, TT = TAI + 32.184 s in 2026 — always state which time scale a timestamp uses.

APPENDIX

B Key Equations

The essential equations of space engineering gathered in one place — orbital mechanics and manoeuvres, rocket propulsion, atmospheric flight and entry, power, thermal and attitude control, communication links, remote sensing and radar, the space environment and astronomy — each with its symbols defined and a pointer to the chapter where it is derived and used.

IN THIS CHAPTER

B.1 Orbital Mechanics
B.2 Rocket Propulsion
B.3 Atmospheric Flight, Drag and Entry
B.4 Power, Thermal and Attitude Control
B.5 Communication Links
B.6 Remote Sensing, Optics and Radar
B.7 Space Environment and Astronomy

This appendix is a formula sheet. Each table gives the equation, defines its symbols briefly and names the chapter where it is explained. SI units are assumed throughout unless stated: metres, kilograms, seconds, kelvin, watts and radians. μ = GM is the gravitational parameter of the central body (Appendix C), r a distance from its centre, g_0 = 9.806 65 m/s^2, σ the Stefan–Boltzmann constant and k the Boltzmann constant (Appendix A).

B.1 Orbital Mechanics

TABLE B.1 Orbital mechanics.

Quantity	Equation	Symbols and notes
Vis-viva (speed anywhere on an orbit)	$v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right)$	a semi-major axis; Chapter 3
Circular and escape speed	$v_c = \sqrt{\mu/r},\ v_{esc} = \sqrt{2\mu/r}$	
Orbital period	$T = 2\pi\sqrt{a^3/\mu}$	Kepler's third law
Specific orbital energy	$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a}$	negative for bound orbits
Specific angular momentum	$h = rv\cos\gamma = \sqrt{\mu a(1-e^2)}$	γ flight-path angle
Orbit equation (conic)	$r = \frac{a(1-e^2)}{1+e\cos\nu}$	e eccentricity, ν true anomaly
Periapsis and apoapsis	$r_p = a(1-e),\ r_a = a(1+e)$	
Kepler's equation	$M = E - e\sin E,\ M = n(t-t_p),\ n = \sqrt{\mu/a^3}$	E eccentric anomaly, n mean motion
True from eccentric anomaly	$\tan\frac{\nu}{2} = \sqrt{\frac{1+e}{1-e}}\tan\frac{E}{2}$	

Quantity	Equation	Symbols and notes
Hohmann transfer Δv	$\Delta v_1 = \sqrt{\frac{\mu}{r_1}}\left(\sqrt{\frac{2r_2}{r_1+r_2}}-1\right)$, $\Delta v_2 = \sqrt{\frac{\mu}{r_2}}\left(1-\sqrt{\frac{2r_1}{r_1+r_2}}\right)$	circular, coplanar orbits
Hohmann transfer time	$t_H = \pi\sqrt{(r_1+r_2)^3/(8\mu)}$	half the transfer-orbit period
Pure plane change	$\Delta v = 2v\sin(\Delta i/2)$	cheapest at low speed (apoapsis)
Combined manoeuvre	$\Delta v^2 = v_1^2 + v_2^2 - 2v_1 v_2\cos\Delta i$	GTO-to-GEO insertion
Nodal regression (J_2)	$\dot\Omega = -\frac{3}{2}nJ_2\left(\frac{R}{p}\right)^2\cos i$, $p = a(1-e^2)$	Earth J_2 = 1.0826 × 10^{-3}
Sun-synchronous condition	$\dot\Omega = 360^\circ/365.2422\ \text{d} = 0.9856^\circ/\text{day}$	gives i ≈ 97–99° in LEO
Apsidal precession (J_2)	$\dot\omega = \frac{3}{4}nJ_2\left(\frac{R}{p}\right)^2(5\cos^2 i - 1)$	zero at i = 63.4° (Molniya)
Ground-track shift per orbit	$\Delta\lambda \approx T\,(\omega_\oplus - \dot\Omega)$	westward, in radians
Sphere of influence	$r_{SOI} = a_p(m_p/M_\odot)^{2/5}$	patched conics; Chapter 24
Hyperbolic excess and C3	$v_\infty^2 = C_3 = v^2 - \frac{2\mu}{r}$	departure energy
Δv to depart from a circular orbit	$\Delta v = \sqrt{v_\infty^2 + 2\mu/r} - \sqrt{\mu/r}$	Oberth effect: burn deep in the well
Gravity-assist turn angle	$\sin\frac{\delta}{2} = \frac{1}{1 + r_p v_\infty^2/\mu}$	velocity change $2v_\infty\sin(\delta/2)$
Clohessy–Wiltshire (relative motion)	$\ddot x - 3n^2x - 2n\dot y = 0$, $\ddot y + 2n\dot x = 0$, $\ddot z + n^2 z = 0$	x radial, y along-track; Chapters 21 and 40
Synodic period	$\frac{1}{T_{syn}} = \left\lvert\frac{1}{T_1} - \frac{1}{T_2}\right\rvert$	interval between launch windows

B.2 Rocket Propulsion

TABLE B.2 Rocket propulsion.

Quantity	Equation	Symbols and notes
Rocket equation (Tsiolkovsky)	$\Delta v = v_e\ln\frac{m_0}{m_f} = I_{sp}\,g_0\ln\frac{m_0}{m_f}$	m_0 initial, m_f final mass; Chapter 35
Propellant for a given Δv	$m_p = m_f\left(e^{\Delta v/v_e} - 1\right)$	
Thrust	$F = \dot m v_e + (p_e - p_a)A_e$	p_e exit, p_a ambient pressure, A_e exit area
Specific impulse	$I_{sp} = \frac{F}{\dot m g_0}$	seconds
Characteristic velocity	$c^* = \frac{p_c A_t}{\dot m}$	combustion efficiency; A_t throat area
Thrust coefficient	$C_F = \frac{F}{p_c A_t}$, $F = C_F\,p_c A_t$	nozzle efficiency; 1.4–2.0
Ideal exhaust velocity	$v_e = \sqrt{\frac{2\gamma}{\gamma-1}\frac{R_u T_c}{\mathcal{M}}\left[1 - \left(\frac{p_e}{p_c}\right)^{(\gamma-1)/\gamma}\right]}$	γ ratio of specific heats, $\mathcal{M}$ molar mass, T_c chamber temperature

Quantity	Equation	Symbols and notes
Nozzle area ratio	$\frac{A_e}{A_t} = \frac{1}{M_e}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M_e^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}$	M_e exit Mach number
Staged rocket Δv	$\Delta v = \sum_i v_{e,i} \ln \frac{m_{0,i}}{m_{f,i}}$	Chapter 34
Losses in ascent	$\Delta v_{total} = \Delta v_{orbit} + \Delta v_{gravity} + \Delta v_{drag} + \Delta v_{steering}$	≈ 9.3–9.6 km/s to LEO
Electric thruster power	$P_{jet} = \frac{F v_e}{2} = \frac{\dot{m} v_e^2}{2},\ F = \frac{2\eta P}{v_e}$	η thruster efficiency; Chapters 6 and 42
Burn time at constant thrust	$t_b = \frac{m_p v_e}{F}$	

B.3 Atmospheric Flight, Drag and Entry

TABLE B.3 Atmospheric flight and entry.

Quantity	Equation	Symbols and notes
Exponential atmosphere	$\rho = \rho_0 e^{-h/H},\ H = \frac{R_u T}{\mathcal{M} g}$	H scale height (8.5 km on Earth); Chapter 2
Dynamic pressure	$q = \frac{1}{2}\rho v^2$	"max-Q" during ascent
Drag force	$D = \frac{1}{2}\rho v^2 C_D A$	C_D ≈ 2.2 for satellites in free-molecular flow
Ballistic coefficient	$\beta = \frac{m}{C_D A}$	kg/m²; higher β → longer orbital life
Decay of a circular orbit	$\frac{da}{dt} = -\frac{\sqrt{\mu a}\,\rho}{\beta}$	per orbit $\Delta a = -2\pi \rho a^2/\beta$; Chapter 21
Peak deceleration (ballistic entry, Allen–Eggers)	$a_{max} = \frac{v_E^2 \sin\gamma_E}{2eH}$	v_E, γ_E entry speed and angle; e = 2.718; Chapter 10
Stagnation-point heat flux (Sutton–Graves)	$\dot{q} = k\sqrt{\rho/r_n}\,v^3$	$k \approx 1.74 \times 10^{-4}$ (Earth), 1.90×10^{-4} (Mars) in SI; r_n nose radius
Terminal velocity	$v_t = \sqrt{\frac{2mg}{\rho C_D A}}$	parachute sizing; Chapter 27
Lift-to-drag ratio and cross-range	L/D; cross-range grows roughly with $(L/D)^2$	capsule ≈ 0.3, Shuttle ≈ 1–4.5
Mach number	$M = v/a,\ a = \sqrt{\gamma R_u T/\mathcal{M}}$	340 m/s at sea level

B.4 Power, Thermal and Attitude Control

TABLE B.4 Power, thermal and attitude control.

Quantity	Equation	Symbols and notes
Solar array power	$P = S\,\eta\,A\cos\theta \cdot L_d$	$S \approx 1\,361$ W/m² at 1 au ($\propto 1/d^2$); η cell efficiency; L_d degradation; Chapter 5
Array sizing with eclipses	$P_{sa} = \frac{\frac{P_e T_e}{X_e} + \frac{P_d T_d}{X_d}}{T_d}$	e eclipse, d daylight; X path efficiencies
Battery capacity	$C = \frac{P_e T_e}{\mathrm{DoD} \cdot n \cdot \eta_b}$	DoD depth of discharge

Quantity	Equation	Symbols and notes
Radiated power	$P = \varepsilon \sigma A T^4$	ε emissivity
Equilibrium temperature of a plate	$T = \left(\frac{\alpha S + Q_{int}/A}{\varepsilon \sigma} \right)^{1/4}$	α solar absorptance; one side lit, one radiating
Planetary equilibrium temperature	$T_{eq} = \left(\frac{S(1 - A_B)}{4\sigma} \right)^{1/4}$	A_B Bond albedo; Earth 255 K
Conduction	$Q = \frac{kA}{L} \Delta T$	k conductivity
Euler's equations	$I_x \dot{\omega}_x + (I_z - I_y)\omega_y \omega_z = T_x$ (cyclic)	principal axes; Chapter 6
Gravity-gradient torque	$T_{gg} = \frac{3\mu}{2r^3} \lvert I_z - I_y \rvert \sin 2\theta$	θ deviation from local vertical
Solar radiation pressure torque	$T_{srp} = \frac{S}{c} A_s (1 + q) \cos\phi \, (c_{ps} - c_g)$	q reflectance
Aerodynamic torque	$T_a = \frac{1}{2} \rho v^2 C_D A_s (c_{pa} - c_g)$	dominant below ≈ 500 km
Magnetic torque	$T_m = DB$, $B \approx B_0 (R/r)^3 \sqrt{1 + 3\sin^2 \lambda}$	D residual dipole (A m²)
Wheel momentum to store	$H = T_d \cdot \frac{T_{orbit}}{4} \cdot 0.707$	cyclic disturbance estimate
Slew time (bang-bang)	$t = 2\sqrt{\theta I / T}$	θ slew angle, T torque

B.5 Communication Links

TABLE B.5 Communication links.

Quantity	Equation	Symbols and notes
Free-space path loss	$L_{fs} = \left(\frac{4\pi d}{\lambda} \right)^2$; in dB: $92.45 + 20\log f_{GHz} + 20\log d_{km}$	Chapter 7, Appendix D
Antenna gain	$G = \eta \left(\frac{\pi D}{\lambda} \right)^2$	η ≈ 0.55–0.70 for dishes
Half-power beamwidth	$\theta_{3dB} \approx 70 \frac{\lambda}{D}$ degrees	
Effective isotropic radiated power	$\mathrm{EIRP} = P_t G_t / L_t$	dBW
Received carrier-to-noise density	$\frac{C}{N_0} = \mathrm{EIRP} - L_{fs} - L_a + \frac{G}{T} - 10\log k$	in dB; −10 log k = +228.6
Energy per bit to noise density	$\frac{E_b}{N_0} = \frac{C}{N_0} - 10\log R_b$	R_b bit rate
Link margin	$M = \left(\frac{E_b}{N_0} \right)_{received} - \left(\frac{E_b}{N_0} \right)_{required}$	typically ≥ 3 dB
System noise temperature	$T_{sys} = T_{ant} + (L - 1)T_0 + L T_{rx}$	T_0 = 290 K
Shannon capacity	$C = B \log_2 (1 + S/N)$	upper limit on bit rate
Doppler shift	$\Delta f = f \frac{v_r}{c}$	± 50 kHz at 2.2 GHz from LEO
Slant range to a satellite	$d = \sqrt{(R + h)^2 - R^2 \cos^2 \varepsilon} - R \sin \varepsilon$	ε elevation angle

B.6 Remote Sensing, Optics and Radar

TABLE B.6 Remote sensing, optics and radar.

Quantity	Equation	Symbols and notes
Planck's law	$B_\lambda(T) = \frac{2hc^2}{\lambda^5}\,\frac{1}{e^{hc/(\lambda kT)}-1}$	spectral radiance; Chapter 13
Wien's law	$\lambda_{max} = 2898\ \mu\mathrm{m\,K}/T$	Sun 0.50 µm, Earth 10 µm
Stefan–Boltzmann law	$M = \varepsilon\sigma T^4$	
Diffraction limit (Rayleigh)	$\theta = 1.22\,\lambda/D$	Chapters 14 and 38
Ground sample distance	$\mathrm{GSD} = \frac{H\,p}{f}$	p pixel pitch, f focal length, H altitude
Swath width (nadir)	$W \approx 2H\tan(\mathrm{FOV}/2)$	
Signal-to-noise ratio of a detector	$\mathrm{SNR} = \frac{S}{\sqrt{S + n_{pix}(B + Dt + R^2)}}$	S signal electrons, B background, D dark current, R read noise
NDVI	$\mathrm{NDVI} = \frac{\rho_{NIR} - \rho_{red}}{\rho_{NIR} + \rho_{red}}$	vegetation index
Radar range resolution	$\delta_r = \frac{c}{2B}$ (slant), ground: $\delta_r/\sin\theta_i$	B chirp bandwidth; Chapter 15
SAR azimuth resolution (stripmap)	$\delta_a \approx L_a/2$	L_a antenna length
Radar equation	$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4}$	σ radar cross-section
Altimeter range	$R = \frac{c\,\Delta t}{2}$	Δt echo delay

B.7 Space Environment and Astronomy

TABLE B.7 Space environment and astronomy.

Quantity	Equation	Symbols and notes
Dipole magnetic field	$B = B_0\left(\frac{R_E}{r}\right)^3\sqrt{1+3\sin^2\lambda}$	$B_0 \approx 30$ µT at Earth's equator; Chapter 23
Dipole field line (L-shell)	$r = L R_E \cos^2\lambda$	λ magnetic latitude
Gyrofrequency	$f_g = \frac{qB}{2\pi m}$	electrons: 28 GHz per tesla
Plasma frequency	$f_p \approx 8.98\sqrt{n_e}$ Hz	n_e in m^{-3}; sets ionospheric reflection
Ionospheric group delay	$\Delta\rho = \frac{40.3}{f^2}\,\mathrm{TEC}$	metres, TEC in electrons/m^2; Chapter 18
Dynamic pressure of the solar wind	$p = \rho_{sw} v_{sw}^2 = n m_p v^2$	≈ 1–2 nPa, sets magnetopause distance
Magnetopause stand-off	$\frac{r_{mp}}{R_E} \approx \left(\frac{2B_0^2}{\mu_0\, n m_p v^2}\right)^{1/6}$	≈ 10 R_E
Apparent magnitude	$m_1 - m_2 = -2.5\log_{10}\frac{F_1}{F_2}$	5 magnitudes = factor 100
Distance modulus	$m - M = 5\log_{10}\frac{d}{10\ \mathrm{pc}}$	

Quantity	Equation	Symbols and notes
Transit depth	$\delta = (R_p/R_\star)^2$	Earth–Sun: 84 ppm; Chapter 39
Hubble–Lemaître law	$v = H_0 d$	H_0 ≈ 67–73 km/s/Mpc
Redshift	$1 + z = \frac{\lambda_{obs}}{\lambda_{emit}}$	
Radiation dose	$D = E/m$ (gray), $H = w_R D$ (sievert)	Chapter 30

USING THESE EQUATIONS SAFELY

Every equation here rests on assumptions — two bodies and point masses for the conic equations, ideal gas and frozen chemistry for the nozzle equations, an exponential atmosphere for the entry formulas, isotropic antennas for path loss. Use them to size, estimate and check; use validated tools (high-fidelity propagators, chemical-equilibrium codes, ray-tracing and finite-element models) for design decisions, and always check that an answer has the right units and order of magnitude.

APPENDIX

C Solar System Data

Reference data for mission design and planetary science — physical and orbital properties of the Sun, the planets, the Moon and Pluto; gravitational parameters, spheres of influence and idealised transfer requirements from Earth; the major moons; the small bodies visited by spacecraft; Lagrange points; and the US Standard Atmosphere. Values follow the NASA Planetary Fact Sheets, JPL Solar System Dynamics and IAU working-group data, rounded for engineering use.

IN THIS CHAPTER

C.1 The Planets
C.2 Gravitational Parameters and Mission Design Data
C.3 Major Moons
C.4 Small Bodies Visited by Spacecraft
C.5 Lagrange Points
C.6 The US Standard Atmosphere 1976

The tables in this appendix give the values used in the chapters on orbital mechanics (Chapter 3), deep-space mission design (Chapter 24) and planetary exploration (Chapters 25–29). Masses and radii are known far more precisely than shown here; the values are rounded so that the tables stay readable. When designing a real trajectory, use the gravitational parameters and ephemerides distributed by JPL (the DE440 series) through the SPICE toolkit.

C.1 The Planets

TABLE C.1 Physical properties of the planets, the Moon and Pluto. A negative rotation period means retrograde rotation. Gravity and escape speed are at the equator of the 1-bar level for the giant planets. Source: NASA Planetary Fact Sheet.

Body	Mass (10^{24} kg)	Equatorial radius (km)	Mean density (kg/m^3)	Surface gravity (m/s^2)	Escape speed (km/s)	Sidereal rotation (h)	Obliquity (°)
Mercury	0.330	2 440	5 429	3.7	4.3	1 407.6	0.03
Venus	4.87	6 052	5 243	8.9	10.4	−5 832.5	177.4
Earth	5.97	6 378	5 514	9.8	11.2	23.93	23.4
Moon	0.073	1 738	3 344	1.6	2.4	655.7	6.7
Mars	0.642	3 396	3 934	3.7	5.0	24.62	25.2
Jupiter	1 898	71 492	1 326	23.1	59.5	9.93	3.1
Saturn	568	60 268	687	9.0	35.5	10.66	26.7
Uranus	86.8	25 559	1 270	8.7	21.3	−17.24	97.8
Neptune	102	24 764	1 638	11.0	23.5	16.11	28.3
Pluto	0.013 0	1 188	1 854	0.62	1.2	−153.3	122.5

TABLE C.2 Orbital properties of the planets and Pluto (epoch J2000). Moon counts change as surveys find more small irregular moons; the 2026 values include the Jupiter and Saturn moons announced by the Minor Planet Center in March 2026.

Body	Semi-major axis (au)	Eccentricity	Inclination to ecliptic (°)	Orbital period	Mean surface temperature	Surface pressure	Known moons (2026)
Mercury	0.387	0.206	7.0	88.0 d	167 °C (−180 to 430)	≈ 0	0
Venus	0.723	0.007	3.4	224.7 d	464 °C	92 bar	0
Earth	1.000	0.017	0.0	365.26 d	15 °C	1.014 bar	1
Mars	1.524	0.093	1.8	687.0 d	−65 °C	0.006 bar	2
Jupiter	5.203	0.048	1.3	11.86 yr	−110 °C (1 bar)	—	101
Saturn	9.555	0.054	2.5	29.45 yr	−140 °C (1 bar)	—	285
Uranus	19.22	0.047	0.8	84.0 yr	−195 °C (1 bar)	—	29
Neptune	30.11	0.009	1.8	164.8 yr	−200 °C (1 bar)	—	16
Pluto	39.48	0.249	17.2	248 yr	−225 °C	≈ 10 µbar	5

The giant planets have no solid surface; their "surface" values refer to the level where the pressure is 1 bar. Saturn's mean density is lower than that of water. Venus and Uranus rotate backwards; Uranus is tipped on its side, so each pole faces the Sun for 42 years at a time.

C.2 Gravitational Parameters and Mission Design Data

Table C.3 gives the gravitational parameter $\mu = GM$ of each body, its J_2 oblateness coefficient, and the radius of its *sphere of influence* — the region inside which the planet's gravity, rather than the Sun's, is the better reference for a patched-conic trajectory (Chapter 24). The sphere-of-influence radius is $r_{\mathrm{SOI}} = a\,(m/M_\odot)^{2/5}$, where a is the planet's orbital radius.

TABLE C.3 Gravitational parameters, oblateness and spheres of influence. Computed from JPL values; the Moon's sphere of influence is with respect to the Earth.

Body	$\mu = GM$ (km³/s²)	J_2 (× 10⁻⁶)	Sphere of influence (10^3 km)	in planet radii
Sun	132 712 440 041	0.2	—	—
Mercury	22 032	50	112	46
Venus	324 859	4.5	616	102
Earth	398 600	1 082.6	925	145
Moon	4 902.8	203	66 (about Earth)	38
Mars	42 828	1 960	577	170
Jupiter	126 686 534	14 736	48 200	674
Saturn	37 931 187	16 298	54 600	907
Uranus	5 793 939	3 343	51 800	2 028
Neptune	6 836 529	3 411	86 800	3 504
Pluto	869.6	—	3 150	2 648

Table C.4 gives the idealised *Hohmann transfer* from Earth to each planet, assuming circular, coplanar orbits. Real launch opportunities differ — Mars needs a C3 of 8–16 km²/s² depending on the year, and missions to the outer planets use gravity assists to cut both C3 and flight time (Chapter 24) — but the table shows the scale of each journey. *C3* is the square of the hyperbolic excess speed at departure; v_∞ at arrival is the speed relative to the planet before its gravity accelerates the spacecraft.

TABLE C.4 Idealised Hohmann transfers from Earth (circular coplanar orbits; Pluto at its mean distance). The synodic period is the interval between successive launch opportunities.

Destination	Departure C3 (km²/s²)	Δv from 400 km LEO (km/s)	Arrival v_∞ (km/s)	Transfer time	Synodic period (days)
Mercury	56.7	5.54	9.61	105 d	116
Venus	6.2	3.46	2.71	146 d	584
Mars	8.7	3.57	2.65	259 d	780
Jupiter	77.3	6.29	5.64	2.73 yr	399
Saturn	105.9	7.28	5.44	6.06 yr	378
Uranus	127.3	7.98	4.66	16.1 yr	370
Neptune	135.8	8.25	4.05	30.7 yr	367
Pluto	139.6	8.37	3.69	45.5 yr	367

C.3 Major Moons

TABLE C.5 Properties of the major moons. Source: JPL Solar System Dynamics planetary satellite physical and mean orbital parameters.

Moon	Planet	Mean radius (km)	μ (km³/s²)	Orbit radius (10^3 km)	Period (days)	Notes
Moon	Earth	1 737.4	4 902.8	384.4	27.32	Synchronous rotation; polar ice in shadowed craters
Phobos	Mars	11.1	7.1×10^{-4}	9.38	0.319	Target of JAXA's MMX sample return
Deimos	Mars	6.2	1.0×10^{-4}	23.46	1.263	
Io	Jupiter	1 821.6	5 959.9	421.8	1.769	Most volcanically active body known
Europa	Jupiter	1 560.8	3 202.7	671.1	3.551	Subsurface ocean; Europa Clipper arrives 2030
Ganymede	Jupiter	2 631.2	9 887.8	1 070.4	7.155	Largest moon; own magnetic field; JUICE target
Callisto	Jupiter	2 410.3	7 179.3	1 882.7	16.69	Heavily cratered

Moon	Planet	Mean radius (km)	μ (km³/s²)	Orbit radius (10^3 km)	Period (days)	Notes
Mimas	Saturn	198.2	2.50	185.5	0.942	Possible young internal ocean
Enceladus	Saturn	252.1	7.21	238.0	1.370	South-polar water plumes
Tethys	Saturn	531.1	41.2	294.7	1.888	
Dione	Saturn	561.4	73.1	377.4	2.737	
Rhea	Saturn	763.8	153.9	527.1	4.518	
Titan	Saturn	2 574.7	8 978.1	1 221.9	15.945	1.5 bar N_2 atmosphere; methane lakes; Dragonfly target
Iapetus	Saturn	734.5	120.5	3 560.8	79.33	Two-toned surface
Miranda	Uranus	235.8	4.3	129.9	1.413	Extreme cliffs
Ariel	Uranus	578.9	83.5	190.9	2.520	
Umbriel	Uranus	584.7	85.1	266.0	4.144	
Titania	Uranus	788.4	228.2	436.3	8.706	Largest moon of Uranus
Oberon	Uranus	761.4	192.4	583.5	13.46	
Triton	Neptune	1 353.4	1 427.6	354.8	5.877 (retrograde)	Captured Kuiper-belt object; nitrogen geysers
Charon	Pluto	606	106.1	19.6	6.387	Mutually locked with Pluto

C.4 Small Bodies Visited by Spacecraft

TABLE C.6 Small bodies visited, or soon to be visited, by spacecraft (Chapters 25–29).

Body	Type	Size	Orbit	Spacecraft (year of visit)
1 Ceres	Dwarf planet (C-type)	939 km diameter	2.77 au, 4.6 yr	Dawn orbiter (2015–18)
4 Vesta	Differentiated asteroid	525 km	2.36 au, 3.6 yr	Dawn orbiter (2011–12)
433 Eros	S-type NEA	34 × 11 × 11 km	1.46 au, 1.76 yr	NEAR Shoemaker; landed 2001
25143 Itokawa	S-type NEA	535 × 294 × 209 m	1.32 au	Hayabusa sample return (2005, returned 2010)
162173 Ryugu	C-type NEA	0.9 km	1.19 au	Hayabusa2 sample return (2018–19, returned 2020)
101955 Bennu	B-type NEA	0.49 km	1.13 au	OSIRIS-REx sample return (2018–21, returned 2023)

Body	Type	Size	Orbit	Spacecraft (year of visit)
65803 Didymos / Dimorphos	S-type binary NEA	0.78 km / 0.15 km	1.64 au	DART impact (2022); Hera (arrives November 2026)
152830 Dinkinesh	S-type main-belt	0.79 km, with contact-binary moon	2.19 au	Lucy flyby (2023)
52246 Donaldjohanson	C-type main-belt	≈ 8 km elongated	2.38 au	Lucy flyby (April 2025)
16 Psyche	M-type (metal-rich)	≈ 220 km	2.92 au	Psyche orbiter (arrival 2029)
67P/Churyumov–Gerasimenko	Jupiter-family comet	4.1 × 3.3 × 1.8 km (two lobes)	3.46 au, 6.4 yr	Rosetta and Philae (2014–16)
1P/Halley	Halley-type comet	15 × 8 km	17.8 au, 76 yr	Giotto, Vega 1/2, Suisei, Sakigake (1986)
486958 Arrokoth	Cold classical KBO	36 km contact binary	44.6 au	New Horizons flyby (2019)
99942 Apophis	S-type NEA	≈ 340 m	0.92 au	OSIRIS-APEX (arrives 2029, after the 13 April 2029 Earth flyby at ≈ 32 000 km altitude)

C.5 Lagrange Points

In the three-body problem a small spacecraft can stay fixed relative to two large bodies at five *Lagrange points* (Chapter 3). L1, L2 and L3 lie on the line through the two bodies and are unstable — a spacecraft must make small station-keeping burns, typically a few metres per second per year — while L4 and L5 lead and trail the smaller body by 60° and are stable.

TABLE C.7 Lagrange-point locations computed from the circular restricted three-body problem.

System	L1 (from smaller body)	L2 (from smaller body)	L3	L4 / L5	Spacecraft
Sun–Earth	1.498 million km sunward	1.508 million km anti-sunward	opposite side of the Sun	60° ahead / behind Earth	L1: SOHO, ACE, DSCOVR, Aditya-L1, SWFO-L1, IMAP; L2: JWST, Gaia, Euclid, Roman
Earth–Moon	58 000 km Earth-side of the Moon (326 400 km from Earth)	64 500 km beyond the Moon (448 900 km from Earth)	381 700 km from Earth, opposite the Moon	at lunar distance, 60° ahead / behind	L2: Queqiao-1 and -2 halo orbits; the Gateway's NRHO is related to the L1/L2 halo family

C.6 The US Standard Atmosphere 1976

The *US Standard Atmosphere 1976* is an idealised, steady-state model of the Earth's atmosphere for moderate solar activity. Below 86 km it is identical to the ICAO standard atmosphere used in aviation. Above about 150 km the real density can be ten times higher or lower than shown,

depending on solar and geomagnetic activity (Chapters 2 and 22) — use an empirical model such as NRLMSIS 2.1 or JB2008 for drag and lifetime work.

TABLE C.8 US Standard Atmosphere 1976 (geometric altitude).

Altitude (km)	Temperature (K)	Pressure (Pa)	Density (kg/m^3)	Layer
0	288.15	101 325	1.225	Troposphere
5	255.68	54 048	0.736	
11	216.77	22 700	0.365	Tropopause
20	216.65	5 529	0.088 9	Stratosphere
32	228.49	889.1	0.013 6	
47	269.68	115.9	1.50×10^{-3}	Stratopause region
51	270.65	70.46	9.07×10^{-4}	Mesosphere
71	216.82	4.480	7.20×10^{-5}	
86	186.87	0.373 4	6.96×10^{-6}	Mesopause
100	195.08	3.20×10^{-2}	5.60×10^{-7}	Kármán line, thermosphere
150	634.39	4.54×10^{-4}	2.08×10^{-9}	
200	854.56	8.47×10^{-5}	2.54×10^{-10}	
300	976.01	8.77×10^{-6}	1.92×10^{-11}	
400	995.83	1.45×10^{-6}	2.80×10^{-12}	ISS altitude
500	999.24	3.02×10^{-7}	5.22×10^{-13}	
600	999.85	8.21×10^{-8}	1.14×10^{-13}	
800	999.99	1.70×10^{-8}	1.14×10^{-14}	
1 000	1 000.00	7.51×10^{-9}	3.56×10^{-15}	Exosphere

TABLE C.9 Atmospheres relevant to entry, descent and aerial vehicles (Chapters 27 and 28).

Planet / moon	Main gases	Surface pressure	Scale height (km)	Notes for entry and flight
Venus	96.5 % CO_2, 3.5 % N_2	92 bar	15.9	Surface 464 °C; sulphuric-acid clouds 48–70 km
Earth	78 % N_2, 21 % O_2	1.014 bar	8.5	
Mars	95 % CO_2, 2.6 % N_2, 1.9 % Ar	≈ 6 mbar (varies with season and height)	11.1	Thin: parachutes alone cannot land heavy payloads
Jupiter	90 % H_2, 10 % He	—	27	Galileo probe entered at 47.4 km/s (1995)
Titan	≈ 95 % N_2, ≈ 5 % CH_4	1.5 bar	≈ 20–40	Dense and cold (94 K): ideal for flight — Dragonfly

KEY TAKEAWAYS

- Use μ = *GM* from JPL ephemerides for trajectory work; the tables here are rounded for estimates.
- The Hohmann table gives the scale of each journey: Mars and Venus need about 3.5 km/s from LEO, the outer planets 6–8 km/s, and Neptune more than 30 years without gravity assists.
- Launch opportunities repeat at the synodic period — every 26 months for Mars, every 19 months for Venus, about every 13 months for the outer planets.
- Atmospheric density above 150 km varies by a factor of ten with solar activity; standard atmospheres are only a starting point.

APPENDIX

D Radio-Frequency Bands and Spectrum

> *A quick reference to the radio spectrum as used in space — the ITU and IEEE band names, the frequencies allocated to telemetry and command, deep-space, Earth observation, satellite communication and navigation services, the bands used by radars and radiometers, how the atmosphere and rain attenuate each band, free-space path loss from LEO to the outer planets, optical-communication wavelengths, and the international rules that govern who may transmit where.*

IN THIS CHAPTER

D.1 Band Names
D.2 Telemetry, Command and Deep-Space Frequencies
D.3 Communication and Broadcasting Satellites
D.4 Navigation Signals
D.5 Remote-Sensing Frequencies
D.6 Propagation Through the Atmosphere
D.7 Free-Space Path Loss
D.8 Optical Communication Wavelengths
D.9 Who Decides: Spectrum Regulation

Every satellite depends on the radio spectrum — to be commanded, to send its data home, to navigate, or to sense the Earth. Spectrum is a finite natural resource shared by all nations, managed through the International Telecommunication Union (ITU) and its Radio Regulations. This appendix gathers the frequencies and rules referred to in Chapters 7, 13, 15, 16, 17, 18 and 24. Band edges are approximate and simplified; the exact allocation for any service depends on the ITU Region (Region 1: Europe, Africa, the Middle East and northern Asia; Region 2: the Americas; Region 3: the rest of Asia and Oceania) and on national rules.

D.1 Band Names

Two naming systems are in use. The ITU names bands by decade of frequency (Table D.1); the radar "letter bands" standardised by the IEEE (Table D.2) are what satellite engineers use day to day. The letters began as wartime secrecy codes, which is why they follow no logical order.

TABLE D.1 ITU frequency band designations.

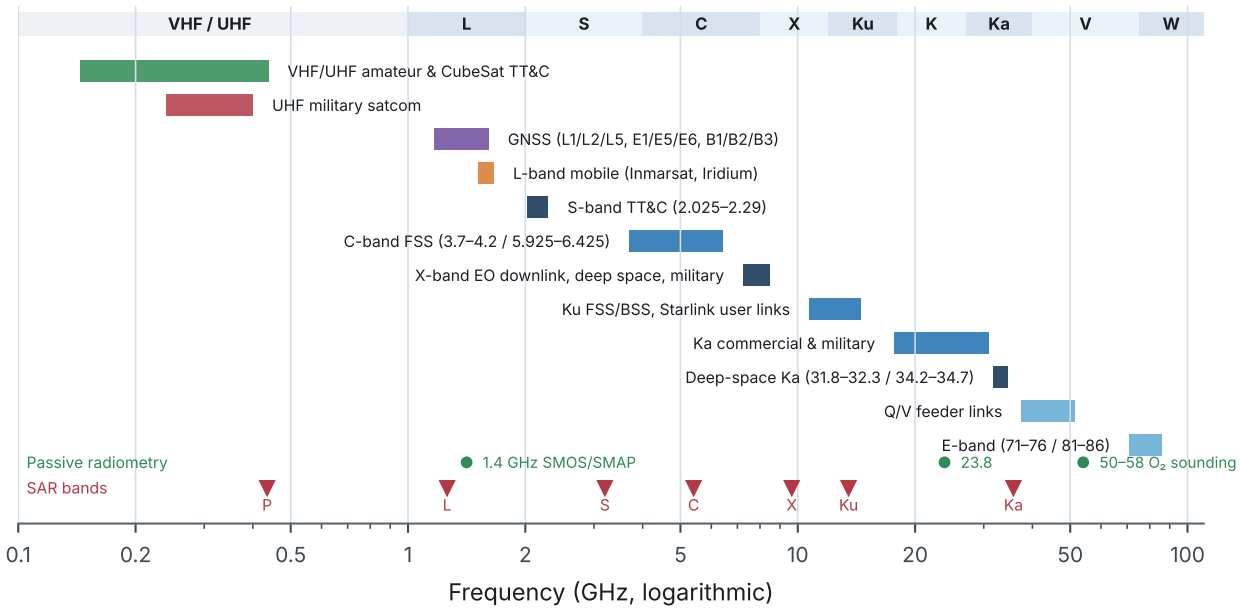
ITU band no.	Name	Frequency	Wavelength	Typical space uses
4	VLF (very low)	3–30 kHz	100–10 km	Whistlers and plasma waves measured by science payloads
5	LF (low)	30–300 kHz	10–1 km	Ground navigation (Loran), not used for space links
6	MF (medium)	0.3–3 MHz	1 000–100 m	Reflected by the ionosphere; radio astronomy impossible from the ground

ITU band no.	Name	Frequency	Wavelength	Typical space uses
7	HF (high)	3–30 MHz	100–10 m	Ionospheric sounding; amateur satellites (rare)
8	VHF (very high)	30–300 MHz	10–1 m	CubeSat and amateur links (145 MHz), weather APT (137 MHz), AIS (162 MHz)
9	UHF (ultra high)	300 MHz–3 GHz	1 m–10 cm	CubeSat TT&C (435–438 MHz), military narrowband satcom, GNSS, mobile satcom, S-band TT&C
10	SHF (super high)	3–30 GHz	10–1 cm	C, X, Ku and Ka bands: most satellite communication, SAR, EO downlinks
11	EHF (extremely high)	30–300 GHz	10–1 mm	Ka, Q/V and E bands; protected military satcom (44 GHz); radiometry
12	THF (tremendously high)	0.3–3 THz	1–0.1 mm	Sub-millimetre astronomy and limb sounders

TABLE D.2 IEEE radar letter bands (IEEE Std 521) and their main space uses. Note that the satellite industry often calls 17.7–21.2 GHz "Ka-band downlink" although it lies in the IEEE K band.

Letter band	Frequency (GHz)	Wavelength	Typical space uses
L	1–2	30–15 cm	GNSS, mobile satcom (Inmarsat, Iridium), L-band SAR, weather-satellite direct broadcast
S	2–4	15–7.5 cm	TT&C, deep space (legacy), mobile satcom, S-band SAR
C	4–8	7.5–3.75 cm	Fixed satellite service (TV distribution), C-band SAR (Sentinel-1)
X	8–12	3.75–2.5 cm	Deep space, EO data downlinks, military satcom, X-band SAR
Ku	12–18	2.5–1.67 cm	Direct-to-home TV, broadband (Starlink user links), altimeters, ISS Ku-band
K	18–27	1.67–1.11 cm	Ka-band downlinks (17.7–21.2 GHz), EESS data downlink (25.5–27 GHz)
Ka	27–40	11.1–7.5 mm	Uplinks of high-throughput satellites, deep-space Ka-band, SWOT radar interferometer
V	40–75	7.5–4 mm	Q/V-band feeder links, oxygen-band temperature sounding (50–60 GHz)
W	75–110	4–2.7 mm	E-band feeder links (71–86 GHz), cloud radars (94 GHz)

Figure D.1 places the main space services on a single logarithmic frequency axis.



Approximate band edges; exact allocations depend on ITU Region and service (ITU Radio Regulations, Article 5).

FIGURE D.1 The radio spectrum used in space, from VHF to E-band. Coloured bars show approximate allocations for the main satellite services; the triangles mark the centre frequencies of the radar bands used by imaging radars, and the dots mark some of the passive bands that radiometers use and that are protected from transmissions.

D.2 Telemetry, Command and Deep-Space Frequencies

Telemetry, tracking and command (TT&C) and science downlinks use bands allocated to the *space operation*, *space research* and *Earth exploration-satellite* (EESS) services. The ITU distinguishes *Category A* missions, within 2 million km of Earth, from *Category B* deep-space missions beyond that distance, which use separate, specially protected sub-bands because their signals arrive extremely weak (Table D.3).

TABLE D.3 Frequencies for telemetry, command and science data (simplified).

Use	Earth-to-space (uplink)	Space-to-Earth (downlink)	Examples
Amateur-satellite and CubeSats	144–146 MHz; 435–438 MHz	same bands	University CubeSats, AMSAT
Near-Earth S-band TT&C	2 025–2 110 MHz	2 200–2 290 MHz	Almost all LEO and GEO satellites, ISS, launchers
Near-Earth X-band	7 190–7 235 MHz	8 450–8 500 MHz	Lunar and L1/L2 missions
EESS X-band data downlink	—	8 025–8 400 MHz	Landsat, Sentinel, commercial EO
EESS / space research Ka-band	22.55–23.15 GHz	25.5–27.0 GHz	JPSS, NISAR, SWOT, Euclid, Roman, TDRS
Deep space S-band (Cat. B)	2 110–2 120 MHz	2 290–2 300 MHz	Legacy missions, Voyager (S/X)
Deep space X-band (Cat. B)	7 145–7 190 MHz	8 400–8 450 MHz	Workhorse band of the DSN, ESTRACK and CDSN
Deep space Ka-band (Cat. B)	34.2–34.7 GHz	31.8–32.3 GHz	Kepler; MRO demonstration; radio science on Cassini, Juno and BepiColombo
Mars and lunar proximity links	435–450 MHz (to surface)	390–405 MHz (from surface)	Electra relay radios on Mars orbiters, rovers and landers

D.3 Communication and Broadcasting Satellites

TABLE D.4 Main satellite communication bands (simplified).

Band	Downlink	Uplink	Services and users
UHF (military)	240–270 MHz; MUOS 360–380 MHz	290–320 MHz; MUOS 300–320 MHz	Narrowband tactical satcom (UFO legacy channels, MUOS)
L (mobile)	1 518–1 559 MHz	1 626.5–1 660.5 MHz	Inmarsat, Ligado/Viasat, Thuraya
L (Iridium)	1 616–1 626.5 MHz	same (time-division)	Iridium voice, data and Certus
L/S (Globalstar)	2 483.5–2 500 MHz	1 610–1 618.725 MHz	Globalstar; Apple emergency SOS
S (2 GHz MSS)	2 170–2 200 MHz	1 980–2 010 MHz	EchoStar, EU 2 GHz MSS; spectrum SpaceX agreed to buy in 2025 for direct-to-device
Cellular (direct-to-device)	600 MHz – 2.7 GHz	same	Terrestrial mobile bands reused from space: T-Mobile/Starlink, AST SpaceMobile
C	3.7–4.2 GHz	5.925–6.425 GHz	TV distribution, maritime, trunking; US 3.7–3.98 GHz cleared for 5G in 2020–23
X (military)	7.25–7.75 GHz	7.9–8.4 GHz	WGS, Skynet, Syracuse, SATCOMBw
Ku	10.7–12.75 GHz	12.75–13.25, 13.75–14.5 GHz	Direct-to-home TV, VSAT, Starlink and OneWeb user links, aero and maritime
Ka (commercial)	17.7–20.2 GHz	27.5–30.0 GHz	High-throughput satellites (ViaSat-3, Jupiter), gateways, ESIM
Ka (government)	20.2–21.2 GHz	30.0–31.0 GHz	WGS, military high-throughput payloads
Q/V	37.5–42.5 GHz	47.2–50.2, 50.4–51.4 GHz	Gateway (feeder) links for new constellations
EHF (protected)	20.2–21.2 GHz	43.5–45.5 GHz	AEHF and its successor ESS — anti-jam, nuclear-survivable
E	71–76 GHz	81–86 GHz	Experimental gateway links

D.4 Navigation Signals

Global navigation satellite systems share three frequency regions: 1 164–1 215 MHz (lower L-band), 1 215–1 300 MHz, and 1 559–1 610 MHz (upper L-band). Systems deliberately share centre frequencies (1 575.42 MHz and 1 176.45 MHz) so that one antenna and receiver front end can use satellites from several constellations (Chapter 18).

TABLE D.5 GNSS signals. GPS signals arrive at the ground with a minimum power of about −158.5 dBW — far below the noise.

System	Signal	Centre frequency (MHz)	Notes
GPS	L1 C/A, L1C	1 575.42	1.023 Mchip/s C/A code, used by every receiver
	L2C, P(Y)	1 227.60	Second civil frequency for ionospheric correction

System	Signal	Centre frequency (MHz)	Notes
	L5	1 176.45	Safety-of-life signal, 10.23 Mchip/s
Galileo	E1	1 575.42	Open Service; OSNMA authentication
	E5a / E5b	1 176.45 / 1 207.14	Wideband AltBOC signal
	E6	1 278.75	High Accuracy Service (≈ 20 cm)
GLONASS	L1OF / L2OF	1 602 + 0.5625 *k* / 1 246 + 0.437 5 *k*	FDMA: channel *k* from −7 to +6
	L3OC	1 202.025	New CDMA signal
BeiDou	B1I / B1C	1 561.098 / 1 575.42	
	B2a / B2b	1 176.45 / 1 207.14	B2b carries PPP corrections
	B3I	1 268.52	
QZSS	L1, L2, L5, L6	as GPS; L6 1 278.75	Centimetre-level augmentation (CLAS) on L6
NavIC	L5, S	1 176.45 / 2 492.028	L1 added on second-generation satellites
SBAS	L1, L5	1 575.42 / 1 176.45	WAAS, EGNOS, MSAS, GAGAN, SouthPAN

D.5 Remote-Sensing Frequencies

D.5.1 Active Sensors

Radar imagers and altimeters choose their band by trading resolution and antenna size against penetration of vegetation, soil and rain (Chapter 15). Longer wavelengths penetrate further into forest canopies and dry soil; shorter wavelengths give finer detail with smaller antennas.

TABLE D.6 Radar bands used by imaging radars, altimeters and cloud and rain radars.

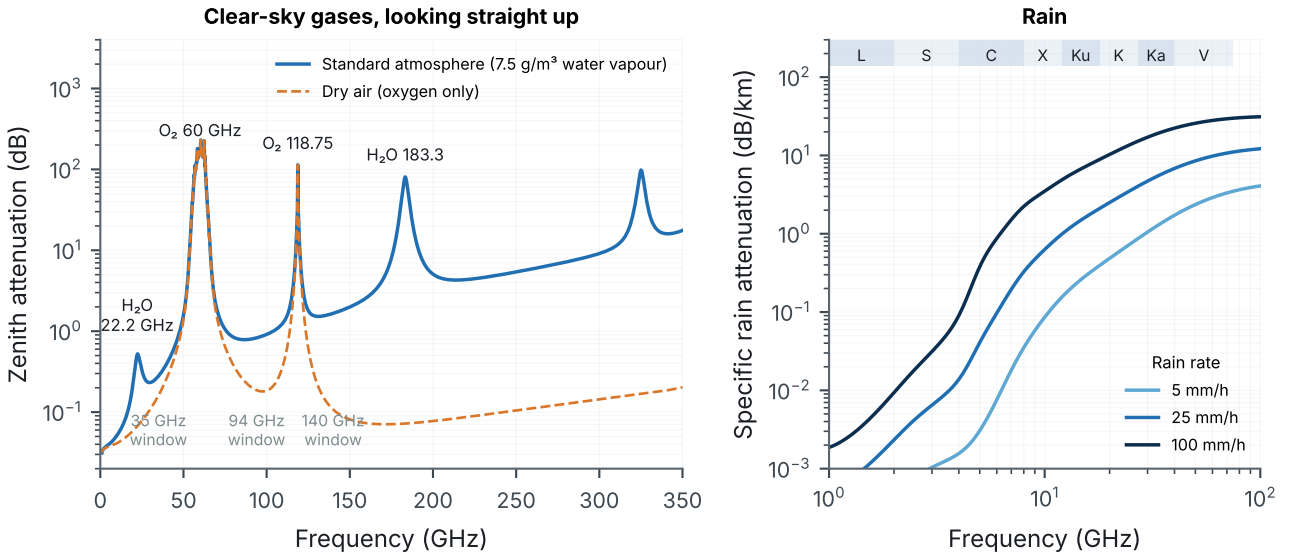
Band	Centre frequency	Wavelength	What it sees	Missions
P	435 MHz	69 cm	Penetrates forest canopy to trunks and branches	ESA BIOMASS (launched April 2025)
L	1.26 GHz	24 cm	Forests, soil moisture, ground deformation, ice	ALOS-2/ALOS-4, SAOCOM, NISAR (L)
S	3.2 GHz	9.4 cm	Crops, snow, sea ice	NISAR (S), NovaSAR-1
C	5.405 GHz	5.5 cm	All-round monitoring, sea ice, winds, floods	Sentinel-1, RADARSAT Constellation, Gaofen-3
X	9.65 GHz	3.1 cm	High-resolution imaging, change detection	TerraSAR-X/TanDEM-X, COSMO-SkyMed, ICEYE, Capella, Umbra
Ku	13.575 GHz	2.2 cm	Sea-surface height, winds, rain	Sentinel-3 and Sentinel-6 altimeters, GPM (13.6 GHz)
Ka	35.75 GHz	8.4 mm	Water-surface topography, snow, light rain	SWOT KaRIn, SARAL/AltiKa, GPM (35.5 GHz)
W	94.05 GHz	3.2 mm	Cloud droplets and ice	CloudSat, EarthCARE

D.5.2 Passive Sensors

Microwave radiometers measure the natural thermal emission of the Earth's surface and atmosphere. They rely on frequencies set by physics — the oxygen absorption complex near 50–60 GHz for temperature profiles, the 22.235 GHz and 183.31 GHz water-vapour lines for humidity, the 1.4 GHz "window" for soil moisture and ocean salinity. Because those frequencies cannot be moved, the Radio Regulations protect them: in bands such as 1 400–1 427 MHz, 23.6–24.0 GHz and 50.2–50.4 GHz *all emissions are prohibited*. Unwanted emissions from neighbouring services remain a concern: the decision at the 2019 World Radiocommunication Conference on 5G emissions next to the 23.8 GHz water-vapour band was criticised by meteorologists as too permissive.

D.6 Propagation Through the Atmosphere

The atmosphere is nearly transparent to radio below about 10 GHz, but above that gases and rain absorb increasingly (Figure D.2). Clear-sky absorption peaks at the oxygen lines (60 GHz and 118.75 GHz) and the water-vapour lines (22.2 GHz and 183.3 GHz); between them lie *windows* near 35, 94 and 140 GHz used by radars and radiometers. Rain is usually the larger problem for satellite links: at Ku-band a heavy shower (25 mm/h) attenuates about 1 dB per kilometre of rain path, at Ka-band (30 GHz) about 5 dB/km. Link designers therefore add a *rain margin* chosen from the statistics of rain at each ground station — for example, enough to keep the link closed for 99.9 % of an average year — and use adaptive coding and modulation, site diversity and uplink power control to ride out the worst storms. The ionosphere affects lower frequencies instead: it delays GNSS signals by up to tens of metres, rotates the polarisation of VHF and UHF signals (Faraday rotation) and causes scintillation near the equator and the poles (Chapter 23).



Gases: ITU-R P.676 line-by-line model, sea level, 1013 hPa, 15 °C. Rain: ITU-R P.838, 40° elevation, circular polarisation. Computed with ITU-Rpy.

FIGURE D.2 Atmospheric attenuation of radio waves. Left: clear-sky attenuation by oxygen and water vapour for a path looking straight up from sea level, computed with the ITU-R P.676 line-by-line model. Right: specific attenuation by rain for three rain rates, from ITU-R P.838; the total rain loss is this value multiplied by the length of the path through rain, typically a few kilometres.

D.7 Free-Space Path Loss

Most of the loss in any space link is simply geometric spreading. The *free-space path loss* between isotropic antennas is

$$L_{\text{fs}}\,(\text{dB}) = 92.45 + 20\log_{10} f_{\text{GHz}} + 20\log_{10} d_{\text{km}}$$

so every doubling of distance or frequency costs 6 dB. (The frequency dependence is offset by the higher gain of an antenna of a given size at higher frequency, which is why deep-space links moved from S- to X- to Ka-band.) Table D.7 gives typical values.

TABLE D.7 Free-space path loss (dB) for typical space links.

Link	Distance	437 MHz	2.2 GHz	8.4 GHz	20 GHz	32 GHz
LEO, overhead	550 km	140.1	154.1	165.7	173.3	177.4
LEO, low elevation	2 000 km	151.3	165.3	177.0	184.5	188.6
GEO	38 000 km	176.9	190.9	202.5	210.1	214.1
Moon	384 400 km	197.0	211.0	222.6	230.2	234.2
Sun–Earth L2	1.5 million km	208.8	222.8	234.5	242.0	246.1
Mars, closest	0.37 au	240.2	254.2	265.9	273.4	277.5
Mars, farthest	2.67 au	257.3	271.3	283.0	290.5	294.6
Jupiter	6.2 au	264.6	278.6	290.3	297.8	301.9
Voyager 1 (2026)	≈ 170 au	293.4	307.4	319.0	326.6	330.7

D.8 Optical Communication Wavelengths

Laser links avoid spectrum licensing for the beam itself and offer far higher data rates from small terminals (Chapters 7 and 17). Most terminals work at the telecommunication wavelength of 1 550 nm, where fibre-optic components such as erbium-doped amplifiers are mature; some use 1 064 nm (Nd:YAG lasers). Examples include NASA's LCRD relay (1.2 Gbit/s, 2021), the TBIRD CubeSat (200 Gbit/s bursts, 2022–23), the DSOC experiment on Psyche (267 Mbit/s from 33 million km in 2023, and links from beyond the distance of Mars until 2025), the European Data Relay System (1.8 Gbit/s at 1 064 nm), the inter-satellite lasers of Starlink, and the optical terminals specified by the US Space Development Agency for its Proliferated Warfighter Space Architecture. Clouds block optical links, so ground networks use several geographically separated stations.

D.9 Who Decides: Spectrum Regulation

The ITU Radio Regulations are an international treaty, revised every three to four years by a World Radiocommunication Conference (WRC). They allocate each band to one or more *services* (fixed-satellite, mobile-satellite, broadcasting-satellite, radionavigation-satellite, Earth exploration-satellite, space research, space operation, radio astronomy and others) with primary or secondary status. WRC-23 (Dubai, 2023) opened Ka-band to earth stations in motion

communicating with non-geostationary satellites and placed direct-to-device satellite connectivity in mobile bands and lunar-surface communications on the agenda of WRC-27, due in late 2027.

Filing and coordination. Before a satellite network can use a frequency, its national administration files it with the ITU Radiocommunication Bureau — first an advance publication or coordination request, then a notification for recording in the Master International Frequency Register. Coordination with earlier filings is negotiated between administrations; date priority ("first come, first served") governs most bands, while some broadcasting and fixed-satellite bands are governed by *plans* that guarantee each country access.

Non-geostationary constellations. Article 22 of the Radio Regulations sets *equivalent power flux-density* (EPFD) limits that protect geostationary networks from interference by non-geostationary constellations in shared Ku- and Ka-bands. Since WRC-19, a constellation must bring 10 % of its filed satellites into use within two years of the end of its seven-year regulatory period, 50 % within five years and 100 % within seven years, or its filing is reduced — a rule that discourages "paper satellites" (Chapter 17).

National licensing. Operators also need licences from their own regulator — in the United States the Federal Communications Commission (FCC) for commercial systems and the NTIA for federal ones; in the United Kingdom Ofcom — and landing rights in each country they serve.

KEY TAKEAWAYS

- Satellite engineers name bands by IEEE letters: L, S, C, X, Ku, K, Ka, V, W. The ITU allocates spectrum by service and Region.
- TT&C lives in S-band; deep space in X- and Ka-band sub-bands reserved for distances beyond 2 million km.
- GNSS signals cluster at 1 176, 1 207–1 278 and 1 575 MHz; shared centre frequencies enable multi-constellation receivers.
- Above 10 GHz rain fade dominates link margins; gas absorption peaks at 22, 60, 118 and 183 GHz.
- Free-space loss rises 6 dB per doubling of distance or frequency — from about 154 dB for LEO S-band to over 280 dB for Mars X-band.
- Spectrum is a treaty-governed shared resource: filings, coordination, EPFD limits and milestones shape every constellation.

APPENDIX

E Mission and Vehicle Databases

Reference tables of launch statistics, launch vehicles, launch sites and every orbital launch of 2025 and 2026, compiled from Jonathan McDowell's General Catalog of Artificial Space Objects (GCAT) and from the chapters of this manual.

IN THIS CHAPTER

E.1 Orbital Launch Attempts by Year
E.2 Most-Launched Vehicle Families
E.3 Busiest Launch Sites
E.4 Orbital Launch Log: 2025
E.5 Orbital Launch Log: 2026, to 13 October

This appendix collects reference data that would interrupt the flow of the chapters. The launch statistics and launch logs are generated directly from the **General Catalog of Artificial Space Objects (GCAT)** maintained by Jonathan McDowell (planet4589.org/space/gcat, licensed CC-BY 4.0), as updated on 13 October 2026. An orbital launch attempt is counted when a vehicle intended to reach orbit lifts off, whether or not it succeeds; suborbital test flights (including most Starship flights before September 2026) are not included. Small differences from other published counts arise from the treatment of such tests, of launches from one country by another's vehicle, and of partial failures.

E.1 Orbital Launch Attempts by Year

Orbital launch attempts (successful or not) by the country or region of the launching organisation, 1957 to 13 October 2026. Rocket Lab's launches from New Zealand count as US. Source: GCAT (J. McDowell, CC-BY 4.0).

TABLE E.1 Orbital launch attempts per year, 1957–2026 (* 2026 to 13 October).

Year	USA	USSR/Russia	China	Europe	Japan	India	Other	Total	Failures
1957	1	2	–	–	–	–	–	3	1
1958	23	5	–	–	–	–	–	28	20
1959	21	4	–	–	–	–	–	25	10
1960	29	9	–	–	–	–	–	38	19
1961	41	9	–	–	–	–	–	50	18
1962	59	22	–	–	–	–	–	81	14
1963	46	24	–	–	–	–	–	70	17
1964	64	36	–	–	–	–	–	100	15
1965	71	53	–	1	–	–	–	125	14
1966	78	51	–	1	2	–	–	132	17
1967	61	74	–	3	1	–	–	139	14

Year	USA	USSR/Russia	China	Europe	Japan	India	Other	Total	Failures
1968	48	80	–	1	–	–	–	129	10
1969	41	82	–	1	1	–	–	125	18
1970	29	87	1	5	2	–	–	124	11
1971	33	91	1	6	2	–	–	133	15
1972	32	79	–	1	1	–	–	113	8
1973	25	90	1	1	–	–	–	117	7
1974	23	85	2	2	1	–	–	113	8
1975	30	93	3	4	2	–	–	132	7
1976	26	100	3	–	2	–	–	131	5
1977	26	102	–	–	2	–	–	130	7
1978	33	91	1	–	3	–	–	128	5
1979	16	89	1	1	2	1	–	110	5
1980	15	90	–	1	2	1	–	109	5
1981	19	100	1	2	3	1	–	126	4
1982	18	108	1	1	1	–	–	129	10
1983	22	100	1	2	3	1	–	129	2
1984	23	97	3	4	3	–	–	130	3
1985	19	100	1	4	2	–	–	126	5
1986	9	94	2	3	2	–	–	110	8
1987	9	97	2	2	3	1	–	114	6
1988	11	94	4	8	2	1	1	121	6
1989	18	75	–	7	2	–	–	102	1
1990	27	79	5	6	3	–	1	121	7
1991	19	61	1	8	2	–	–	91	5
1992	29	55	4	7	1	1	–	97	3
1993	27	48	1	7	1	1	–	85	6
1994	28	49	5	8	2	2	–	94	4
1995	30	33	3	11	2	–	1	80	7
1996	33	27	4	11	1	1	–	77	6
1997	38	29	6	12	2	1	1	89	4
1998	36	25	6	11	2	–	2	82	6
1999	31	22	4	16	1	1	3	78	8
2000	29	32	5	16	1	–	2	85	4
2001	24	23	1	8	1	2	–	59	2
2002	18	25	5	12	3	1	1	65	4
2003	26	19	7	6	3	2	1	64	2
2004	19	22	8	3	–	1	1	54	3
2005	16	23	6	8	2	1	–	56	4
2006	23	23	6	7	6	1	–	66	4
2007	20	23	10	9	2	3	1	68	3

Year	USA	USSR/Russia	China	Europe	Japan	India	Other	Total	Failures
2008	20	26	11	7	1	3	1	69	3
2009	25	32	6	7	3	2	3	78	4
2010	15	30	15	7	2	3	2	74	4
2011	19	28	19	9	3	3	3	84	6
2012	16	23	19	11	2	2	5	78	5
2013	20	31	15	8	3	3	1	81	3
2014	24	32	16	11	4	4	1	92	3
2015	20	25	19	12	4	5	2	87	5
2016	23	17	22	11	4	7	2	86	2
2017	30	18	18	11	7	5	1	90	6
2018	34	17	39	11	6	7	–	114	2
2019	27	22	34	9	2	6	3	103	5
2020	44	12	39	10	4	2	3	114	10
2021	51	16	56	15	3	2	3	146	11
2022	87	21	64	6	1	5	2	186	7
2023	114	19	67	3	3	7	8	221	10
2024	157	17	68	3	7	5	5	262	6
2025	198	17	93	8	4	5	5	330	11
2026*	146	13	71	7	3	3	1	244	7

E.2 Most-Launched Vehicle Families

TABLE E.2 The thirty most-launched orbital vehicle families (to 13 October 2026).

Family	First orbital attempt	Attempts	Failures	Success rate
Soyuz	1966	1 181	31	97.4 %
Falcon 9	2010	695	2	99.7 %
Kosmos	1961	622	45	92.8 %
Proton	1967	425	43	89.9 %
Atlas	1958	327	40	87.8 %
Molniya	1960	320	33	89.7 %
Delta	1972	301	9	97.0 %
Voskhod	1963	299	13	95.7 %
Thor	1958	288	38	86.8 %
Long March 2	1974	235	4	98.3 %
Titan	1964	216	15	93.1 %
Long March 3	1984	177	8	95.5 %
Vostok	1960	156	8	94.9 %
Ariane	1979	144	7	95.1 %
Space Shuttle	1981	140	1	99.3 %
Tsiklon-3	1977	122	7	94.3 %

Family	First orbital attempt	Attempts	Failures	Success rate
Long March 4	1988	119	3	97.5 %
Ariane 5	1996	117	3	97.4 %
Atlas V	2002	110	0	100.0 %
Tsiklon-2	1969	105	1	99.0 %
Scout	1960	99	11	88.9 %
Electron	2017	88	4	95.5 %
PSLV	1993	64	4	93.8 %
H-IIA	2001	50	1	98.0 %
Pegasus	1990	46	4	91.3 %
Long March 6	2015	44	0	100.0 %
Thorad	1966	43	2	95.3 %
Kuaizhou	2013	40	3	92.5 %
Zenit-2	1985	36	7	80.6 %
Zenit-3SL	1999	36	3	91.7 %

E.3 Busiest Launch Sites

TABLE E.3 The twenty-five busiest orbital launch sites of all time (to 13 October 2026).

Launch site	Orbital launch attempts
Plesetsk	1 688
Baikonur	1 560
Cape Canaveral	1 112
Vandenberg	893
Kourou	336
Jiuquan	299
Kennedy Space Center	287
Xichang	242
Taiyuan	161
Sriharikota	103
Kapustin Yar	101
Tanegashima	99
Mahia (Rocket Lab LC-1)	84
Wenchang	51
Point Arguello	43
Pacific Ocean (Sea Launch)	36
Uchinoura (Kagoshima)	35
Wallops (MARS)	31
Wallops Island	28
Hainan commercial site (Wenchang)	24
Vostochny	21

Launch site	Orbital launch attempts
Yellow Sea (sea launch)	20
Semnan	16
Palmachim	13
Starbase	11

E.4 Orbital Launch Log: 2025

All 330 orbital launch attempts. × marks a failure to reach orbit. Times are UTC.

TABLE E.4 Orbital launch attempts in 2025.

No.	Date (UTC)	Vehicle	Site	Mission / payload	
1	Jan 4	Falcon 9	Canaveral	Thuraya 4-NGS	
2	Jan 6	Long March 3B	Xichang	Shi Jian 25	
3	Jan 6	Falcon 9	Canaveral	Starlink Group 6-71	
4	Jan 8	Falcon 9	Kennedy	Starlink Group 12-11	
5	Jan 10	Falcon 9	Vandenberg	NRO Starshield A6	
6	Jan 10	Falcon 9	Canaveral	Starlink Group 12-12	
7	Jan 13	Jielong-3	Yellow Sea	Weili Kongjian 01 zu	
8	Jan 13	Falcon 9	Canaveral	Starlink Group 12-4	
9	Jan 14	Falcon 9	Vandenberg	Transporter-12	
10	Jan 15	Falcon 9	Kennedy	BG-1/Hakuto-R M2	
11	Jan 16	New Glenn 7×2	Canaveral	DS-1	
12	Jan 16	Starship V2	Starbase	Ship 33	×
13	Jan 17	Long March 2D	Jiuquan	PRSC-EO1	
14	Jan 20	Gushenxing 1	Jiuquan	Yunyao-1 37-40	
15	Jan 21	Falcon 9	Kennedy	Starlink Group 13-1	
16	Jan 21	Falcon 9	Vandenberg	Starlink Group 11-8	
17	Jan 23	Long March 6A	Taiyuan	Qianfan Jigui 06	
18	Jan 23	Long March 3B	Xichang	TJS 14	
19	Jan 24	Falcon 9	Vandenberg	Starlink Group 11-6	
20	Jan 27	Falcon 9	Canaveral	Starlink Group 12-7	
21	Jan 29	GSLV Mk II	Sriharikota	IRNSS-1K?	
22	Jan 30	Falcon 9	Kennedy	Spainsat NG-1	
23	Feb 1	Falcon 9	Vandenberg	Starlink Group 11-4	
24	Feb 2	H3-22S	Tanegashima	QZS 6	
25	Feb 4	Falcon 9	Canaveral	Starlink Group 12-3	
26	Feb 4	Falcon 9	Kennedy	Maxar-3	
27	Feb 5	Soyuz-2-1V	Plesetsk	MKA-B 1 to 3	
28	Feb 8	Falcon 9	Canaveral	Starlink Group 12-9	
29	Feb 8	Electron	Mahia	Kineis 2A-2E	
30	Feb 11	Falcon 9	Vandenberg	Starlink Group 11-10	
31	Feb 11	Long March 8A	Wenchang	WHDW 02	
32	Feb 11	Falcon 9	Canaveral	Starlink Group 12-18	

No.	Date (UTC)	Vehicle	Site	Mission / payload	
33	Feb 15	Falcon 9	Canaveral	Starlink Group 12-8	
34	Feb 18	Electron	Mahia	BlackSky Gen3	
35	Feb 18	Falcon 9	Canaveral	Starlink Group 10-12	
36	Feb 21	Falcon 9	Canaveral	Starlink Group 12-14	
37	Feb 22	Long March 3B	Xichang	ZX-10R	
38	Feb 23	Falcon 9	Vandenberg	Starlink Group 15-1	
39	Feb 27	Falcon 9	Kennedy	IM-2/Odin/LTB/CGEO-1	
40	Feb 27	Falcon 9	Canaveral	Starlink Group 12-13	
41	Feb 27	Long March 2C	Jiuquan	Superview 1-03/04	
42	Feb 27	Soyuz-2-1A	Baikonur	Progress 7K-TGM No. 460	
43	Mar 1	Kuaizhou-1A	Jiuquan	Unknown	
44	Mar 2	Soyuz-2-1B	Plesetsk	Glonass-K2 No. 14L	
45	Mar 3	Falcon 9	Canaveral	Starlink Group 12-20	
46	Mar 6	Ariane 62	Kourou	CSO 3	
47	Mar 6	Starship V2	Starbase	Ship 34	×
48	Mar 9	Long March 3B	Xichang	TJS 15	
49	Mar 11	Long March 8	Hainan (comm.)	Qianfan	
50	Mar 12	Falcon 9	Vandenberg	SPHEREx	
51	Mar 13	Falcon 9	Canaveral	Starlink Group 12-21	
52	Mar 14	Falcon 9	Kennedy	Dragon C210.4	
53	Mar 15	Electron	Mahia	QPS-SAR-9	
54	Mar 15	Long March 2D	Jiuquan	Superview 3-02	
55	Mar 15	Falcon 9	Vandenberg	Transporter-13	
56	Mar 15	Falcon 9	Canaveral	Starlink Group 12-16	
57	Mar 16	Angara-1.2	Plesetsk	Rodnik Blok 17	
58	Mar 17	Gushenxing 1	Jiuquan	Yunyao-1 55-60	
59	Mar 18	Electron	Mahia	Kineis 4A-4E	
60	Mar 18	Falcon 9	Canaveral	Starlink Group 12-25	
61	Mar 21	Falcon 9	Vandenberg	NRO Starshield C1	
62	Mar 21	Gushenxing 1	Jiuquan	Yunyao-1 43-48	
63	Mar 24	Falcon 9	Canaveral	INTRUDER?	
64	Mar 26	Electron	Mahia	OTC Phase 1	
65	Mar 26	Long March 3B	Xichang	Tianlian 2-04	
66	Mar 26	Falcon 9	Vandenberg	Starlink Group 11-7	
67	Mar 29	Long March 7A	Wenchang	TJS 16	
68	Mar 30	Spectrum	Andoya	Going Full Spectrum	×
69	Mar 31	Falcon 9	Canaveral	Starlink Group 6-80	
70	Apr 1	Falcon 9	Kennedy	Dragon C207.4	
71	Apr 1	Long March 2D	Jiuquan	WHJSW 06	
72	Apr 3	Long March 6	Taiyuan	Tianping 3A-02	
73	Apr 4	Falcon 9	Vandenberg	Starlink Group 11-13	
74	Apr 6	Falcon 9	Canaveral	Starlink Group 6-72	

No.	Date (UTC)	Vehicle	Site	Mission / payload	
75	Apr 7	Falcon 9	Vandenberg	Starlink Group 11-11	
76	Apr 8	Soyuz-2-1A	Baikonur	Soyuz 11F732A48 No. 758	
77	Apr 10	Long March 3B	Xichang	TJS 17	
78	Apr 12	Falcon 9	Vandenberg	NRO Starshield A7	
79	Apr 13	Falcon 9	Kennedy	Starlink Group 12-17	
80	Apr 14	Falcon 9	Canaveral	Starlink Group 6-73	
81	Apr 16	Minotaur IV	Vandenberg	NROL-174 Asterion	
82	Apr 18	Long March 6A	Taiyuan	Shiyan 27 01-06	
83	Apr 20	Falcon 9	Vandenberg	NRO Starshield A8	
84	Apr 21	Falcon 9	Kennedy	Dragon C209.5	
85	Apr 22	Falcon 9	Canaveral	Bandwagon-3	
86	Apr 24	Long March 2F	Jiuquan	Shenzhou 20	
87	Apr 25	Falcon 9	Canaveral	Starlink Group 6-74	
88	Apr 27	Long March 3B	Xichang	Tianlian 2-05	
89	Apr 28	Falcon 9	Canaveral	Starlink Group 12-23	
90	Apr 28	Long March 5B/YZ-2	Wenchang	WHDW 03	
91	Apr 28	Falcon 9	Vandenberg	Starlink Group 11-9	
92	Apr 28	Atlas V 551	Canaveral	KA-01	
93	Apr 29	Falcon 9	Kennedy	Starlink Group 12-10	
94	Apr 29	Vega C	Kourou	Biomass	
95	Apr 29	Firefly Alpha	Vandenberg	LM-400 Tech Demo	×
96	May 2	Falcon 9	Canaveral	Starlink Group 6-75	
97	May 4	Falcon 9	Kennedy	Starlink Group 6-84	
98	May 7	Falcon 9	Canaveral	Starlink Group 6-93	
99	May 10	Falcon 9	Vandenberg	Starlink Group 15-3	
100	May 10	Falcon 9	Canaveral	Starlink Group 6-91	
101	May 11	Long March 6A	Taiyuan	Yaogan 40-02	
102	May 12	Long March 3C	Xichang	TJS 19	
103	May 13	Falcon 9	Vandenberg	Starlink Group 15-4	
104	May 13	Falcon 9	Kennedy	Starlink Group 6-83	
105	May 14	Long March 2D	Jiuquan	Taikong Jisuan WX	
106	May 14	Falcon 9	Canaveral	Starlink Group 6-67	
107	May 16	Falcon 9	Vandenberg	Starlink Group 15-5	
108	May 17	Zhuque-2E	Jiuquan	Tianyi	
109	May 17	Electron	Mahia	QPS-SAR-10	
110	May 18	PSLV-XL	Sriharikota	RISAT-1B	×
111	May 19	Gushenxing 1	Yellow Sea	Tianqi 16-18,20	
112	May 20	Long March 7A	Wenchang	ZX-3B	
113	May 21	Falcon 9	Canaveral	Starlink Group 12-15	
114	May 21	Lijian-1	Jiuquan	Taijing	
115	May 23	Soyuz-2-1B	Plesetsk	Nivelir-L No. 5?	
116	May 23	Falcon 9	Vandenberg	Starlink Group 11-16	

No.	Date (UTC)	Vehicle	Site	Mission / payload
117	May 24	Falcon 9	Canaveral	Starlink Group 12-22
118	May 27	Falcon 9	Vandenberg	Starlink Group 17-1
119	May 27	Starship V2	Starbase	Ship 35
120	May 28	Falcon 9	Kennedy	Starlink Group 10-32
121	May 28	Long March 3B	Xichang	Tianwen-2
122	May 29	Long March 4B	Jiuquan	SJ-26
123	May 30	Falcon 9	Canaveral	GPS III SV08
124	May 31	Falcon 9	Vandenberg	Starlink Group 11-18
125	Jun 2	Electron	Mahia	BlackSky Gen3-2
126	Jun 3	Falcon 9	Canaveral	Starlink Group 12-19
127	Jun 4	Falcon 9	Vandenberg	Starlink Group 11-22
128	Jun 5	Long March 6A	Taiyuan	WHDW 04
129	Jun 7	Falcon 9	Canaveral	SXM-10
130	Jun 8	Falcon 9	Vandenberg	Starlink Group 15-8
131	Jun 10	Falcon 9	Canaveral	Starlink Group 12-24
132	Jun 11	Electron	Mahia	QPS-SAR-11
133	Jun 13	Falcon 9	Vandenberg	Starlink Group 15-6
134	Jun 13	Falcon 9	Canaveral	Starlink Group 12-26
135	Jun 14	Long March 2D	Jiuquan	ZH1-02
136	Jun 17	Falcon 9	Vandenberg	Starlink Group 15-9
137	Jun 18	Falcon 9	Canaveral	Starlink Group 10-18
138	Jun 19	Angara A5	Plesetsk	14F166A No. 1
139	Jun 20	Long March 3B	Xichang	ZX 9C
140	Jun 23	Falcon 9	Canaveral	Starlink Group 10-23
141	Jun 23	Atlas V 551	Canaveral	KA-02
142	Jun 23	Falcon 9	Vandenberg	Transporter-14
143	Jun 25	Falcon 9	Kennedy	Dragon C213.1
144	Jun 25	Falcon 9	Canaveral	Starlink Group 10-16
145	Jun 26	Electron	Mahia	Hawk 12/Kestrel 0A
146	Jun 28	Falcon 9	Canaveral	Starlink Group 10-34
147	Jun 28	Electron	Mahia	Lyra 4
148	Jun 28	H-IIA 202	Tanegashima	GOSAT-GW
149	Jun 28	Falcon 9	Vandenberg	Starlink Group 15-7
150	Jul 1	Falcon 9	Kennedy	MTG-S1
151	Jul 2	Falcon 9	Canaveral	Starlink Group 10-25
152	Jul 3	Long March 4C	Xichang	SY28B-01
153	Jul 3	Soyuz-2-1A	Baikonur	Progress 7K-TGM No. 461
154	Jul 8	Falcon 9	Canaveral	Starlink Group 10-28
155	Jul 13	Falcon 9	Canaveral	Dror-1
156	Jul 14	Long March 7	Wenchang	Tianzhou-9
157	Jul 16	Falcon 9	Vandenberg	Starlink Group 15-2
158	Jul 16	Falcon 9	Canaveral	Kuiper KF-01

No.	Date (UTC)	Vehicle	Site	Mission / payload	
159	Jul 19	Falcon 9	Vandenberg	Starlink Group 17-3	
160	Jul 22	Falcon 9	Canaveral	O3b mPOWER 9/10	
161	Jul 23	Falcon 9	Vandenberg	TRACERS	
162	Jul 25	Soyuz-2-1B	Vostochny	Soyuz Rideshare 15	
163	Jul 26	Vega C	Kourou	CO3D/Microcarb	
164	Jul 26	Falcon 9	Canaveral	Starlink Group 10-26	
165	Jul 27	Falcon 9	Vandenberg	Starlink Group 17-2	
166	Jul 27	Long March 6A	Taiyuan	WHDW 05	
167	Jul 29	Shuang Quxian 1	Jiuquan	Enshi Xidou Shanquan	
168	Jul 29	Eris Block 1	Bowen	Test	×
169	Jul 30	Falcon 9	Canaveral	Starlink Group 10-29	
170	Jul 30	Long March 8A	Hainan (comm.)	WHDW 06	
171	Jul 30	GSLV Mk II	Sriharikota	NISAR	
172	Jul 31	Kuaizhou-1A	Xichang	PRSS	
173	Jul 31	Falcon 9	Vandenberg	Starlink Group 13-4	
174	Aug 1	Falcon 9	Kennedy	Dragon C206.6	
175	Aug 4	Falcon 9	Canaveral	Starlink Group 10-30	
176	Aug 4	Long March 12	Hainan (comm.)	WHDW 07	
177	Aug 5	Electron	Mahia	QPS-SAR-12	
178	Aug 8	Jielong-3	Yellow Sea	Gili 04 zu 01-11	
179	Aug 11	Falcon 9	Canaveral	Kuiper KF-02	
180	Aug 13	Ariane 62	Kourou	METOP-SG A1	
181	Aug 13	Vulcan Centaur VC4S	Canaveral	NTS-3	
182	Aug 13	Long March 5B/YZ-2	Wenchang	WHDW 08	
183	Aug 14	Falcon 9	Vandenberg	Starlink Group 17-4	
184	Aug 14	Falcon 9	Canaveral	Starlink Group 10-20	
185	Aug 15	Zhuque-2E	Jiuquan	WHJSW 07?	×
186	Aug 17	Long March 4C	Xichang	SY28B-02	
187	Aug 17	Long March 6A	Taiyuan	WHDW 09	
188	Aug 18	Falcon 9	Vandenberg	Starlink Group 17-5	
189	Aug 19	Lijian-1	Jiuquan	AIRSAT/DGS	
190	Aug 20	Soyuz-2-1B	Baikonur	Bion-M No. 2	
191	Aug 21	Angara-1.2	Plesetsk	OO MKA 3-6	
192	Aug 22	Falcon 9	Kennedy	X-37B No. 1/USSF-36	
193	Aug 22	Falcon 9	Vandenberg	Starlink Group 17-6	
194	Aug 23	Electron	Mahia	E-Space	
195	Aug 24	Falcon 9	Canaveral	Dragon C211.3	
196	Aug 25	Long March 8A	Hainan (comm.)	WHDW 10	
197	Aug 26	Falcon 9	Vandenberg	NAOS	
198	Aug 26	Starship V2	Starbase	Ship 37	
199	Aug 27	Falcon 9	Canaveral	Starlink Group 10-56	
200	Aug 28	Falcon 9	Kennedy	Starlink Group 10-11	

No.	Date (UTC)	Vehicle	Site	Mission / payload	
201	Aug 30	Falcon 9	Vandenberg	Starlink Group 17-7	
202	Aug 31	Falcon 9	Canaveral	Starlink Group 10-14	
203	Sep 2	Shavit 2	Palmachim	ʻOfeq-19	
204	Sep 3	Falcon 9	Vandenberg	Starlink Group 17-8	
205	Sep 3	Falcon 9	Canaveral	Starlink Group 10-22	
206	Sep 5	Long March 3C/YZ-1	Xichang	SY29	
207	Sep 5	Gushenxing 1	Jiuquan	Yunyao/Jiazhou	
208	Sep 5	Falcon 9	Kennedy	Starlink Group 10-57	
209	Sep 6	Long March 6A	Taiyuan	Yaogan 40-03	
210	Sep 6	Falcon 9	Vandenberg	Starlink Group 17-9	
211	Sep 8	Jielong-3	Yellow Sea	Gili 05 zu 01-11	
212	Sep 9	Long March 7A	Wenchang	Yaogan 45	
213	Sep 10	Falcon 9	Vandenberg	Tranche 1 TL-B	
214	Sep 11	Soyuz-2-1A	Baikonur	Progress 7K-TGM No. 462	
215	Sep 12	Falcon 9	Canaveral	Nusantara Lima	
216	Sep 13	Soyuz-2-1B	Plesetsk	Glonass/Mozhaets 6	
217	Sep 13	Falcon 9	Vandenberg	Starlink Group 17-10	
218	Sep 14	Falcon 9	Canaveral	Cygnus NG-23	
219	Sep 16	Long March 2C/YZ-1S	Jiuquan	WHJSW-07	
220	Sep 18	Falcon 9	Canaveral	Starlink Group 10-61	
221	Sep 18	Zoljanah	Semnan	Test?	×
222	Sep 19	Falcon 9	Vandenberg	Starlink Group 17-12	
223	Sep 21	Falcon 9	Canaveral	Starlink Group 10-27	
224	Sep 22	Falcon 9	Vandenberg	NRO Starshield C2	
225	Sep 24	Jielong-3	Yellow Sea	Gili 06 zu 01-11,SK1	
226	Sep 24	Falcon 9	Kennedy	IMAP	
227	Sep 25	Falcon 9	Canaveral	Starlink Group 10-15	
228	Sep 25	Atlas V 551	Canaveral	KA-03	
229	Sep 26	Falcon 9	Vandenberg	Starlink Group 17-11	
230	Sep 26	Long March 4C	Jiuquan	FY3-08	
231	Sep 27	Long March 6A	Taiyuan	WHDW 11	
232	Sep 29	Falcon 9	Vandenberg	Starlink Group 11-20	
233	Sep 29	Long March 2D	Xichang	Shiyan 30-01/02	
234	Oct 3	Falcon 9	Vandenberg	Starlink Group 11-39	
235	Oct 7	Falcon 9	Canaveral	Starlink Group 10-59	
236	Oct 8	Falcon 9	Vandenberg	Starlink Group 11-17	
237	Oct 11	Yinli-1	Yellow Sea	Jilin-1 Kuanfu 02B07	
238	Oct 13	Long March 2D	Jiuquan	Shiyan 31	
239	Oct 13	Starship V2	Starbase	Ship 38	
240	Oct 14	Falcon 9	Canaveral	Kuiper KF-03	
241	Oct 14	Electron	Mahia	Strix-5	
242	Oct 15	Falcon 9	Vandenberg	Tranche 1 TL-C	

No.	Date (UTC)	Vehicle	Site	Mission / payload	
243	Oct 16	Long March 8A	Hainan (comm.)	WHDW 12	
244	Oct 16	Falcon 9	Canaveral	Starlink Group 10-52	
245	Oct 17	Long March 6A	Taiyuan	Qianfan Jigui 18	
246	Oct 19	Lijian-1	Jiuquan	PRSC-HS1/AIRSAT	
247	Oct 19	Falcon 9	Canaveral	Starlink Group 10-17	
248	Oct 19	Falcon 9	Vandenberg	Starlink Group 11-19	
249	Oct 22	Falcon 9	Vandenberg	Starlink Group 11-5	
250	Oct 23	Long March 5	Wenchang	TJS 20	
251	Oct 24	Falcon 9	Canaveral	Spainsat NG-II	
252	Oct 25	Falcon 9	Vandenberg	Starlink Group 11-12	
253	Oct 26	H3-24W	Tanegashima	HTV-X1	
254	Oct 26	Long March 3B	Xichang	GF14-02	
255	Oct 26	Falcon 9	Canaveral	Starlink Group 10-21	
256	Oct 28	Falcon 9	Vandenberg	Starlink Group 11-21	
257	Oct 29	Falcon 9	Canaveral	Starlink Group 10-37	
258	Oct 31	Long March 2F	Jiuquan	Shenzhou 21	
259	Oct 31	Falcon 9	Vandenberg	Starlink Group 11-23	
260	Nov 2	Falcon 9	Canaveral	Bandwagon-4	
261	Nov 2	LVM3	Sriharikota	GSAT-7R	
262	Nov 3	Long March 7A	Wenchang	Yaogan 46	
263	Nov 4	Ariane 62	Kourou	Sentinel-1D	
264	Nov 5	Electron	Mahia	QPS-SAR-14	
265	Nov 6	Falcon 9	Canaveral	Starlink Group 6-81	
266	Nov 6	Falcon 9	Vandenberg	Starlink Group 11-14	
267	Nov 8	Long March 11	Yellow Sea	Shiyan 32 01-03	
268	Nov 9	Lijian-1	Jiuquan	Chutian	
269	Nov 9	Falcon 9	Kennedy	Starlink Group 10-51	
270	Nov 10	Long March 12	Hainan (comm.)	WHDW 13	
271	Nov 10	Gushenxing 1	Jiuquan	Jixing Gaofen 04C	×
272	Nov 11	Falcon 9	Canaveral	Starlink Group 6-87	
273	Nov 13	New Glenn 7×2	Canaveral	ESCAPADE	
274	Nov 14	Atlas V 551	Canaveral	Viasat-3 F2	
275	Nov 15	Falcon 9	Kennedy	Starlink Group 6-89	
276	Nov 15	Falcon 9	Canaveral	Starlink Group 6-85	
277	Nov 17	Falcon 9	Vandenberg	Sentinel-6B	
278	Nov 19	Falcon 9	Canaveral	Starlink Group 6-94	
279	Nov 19	Long March 2C	Jiuquan	Shi Jian 30A-C	
280	Nov 20	Electron	Mahia	BlackSky Gen3-3	
281	Nov 21	Falcon 9	Kennedy	Starlink Group 6-78	
282	Nov 21	Long March 3B	Xichang	TJS 21	
283	Nov 22	Falcon 9	Canaveral	Starlink Group 6-79	
284	Nov 23	Falcon 9	Vandenberg	Starlink Group 11-30	

No.	Date (UTC)	Vehicle	Site	Mission / payload	
285	Nov 25	Long March 2F	Jiuquan	Shenzhou 22	
286	Nov 25	Angara-1.2	Plesetsk	Rodnik Blok 18	
287	Nov 26	Nuri	Naro	CAS500-3	
288	Nov 27	Soyuz-2-1A	Baikonur	Soyuz 11F732A48 No. 753	
289	Nov 28	Falcon 9	Vandenberg	Transporter-15	
290	Nov 30	Long March 7A	Wenchang	Shijian 28	
291	Dec 1	Falcon 9	Kennedy	Starlink Group 6-86	
292	Dec 1	Vega C	Kourou	Kompsat-7	
293	Dec 2	Falcon 9	Vandenberg	Starlink Group 15-10	
294	Dec 2	Falcon 9	Canaveral	Starlink Group 6-95	
295	Dec 3	Zhuque-3	Jiuquan	Dummy payload	
296	Dec 4	Falcon 9	Vandenberg	Starlink Group 11-25	
297	Dec 5	Kuaizhou-1A	Jiuquan	Jiaotong VDES A/B	
298	Dec 6	Long March 8A	Hainan (comm.)	WHDW 14	
299	Dec 7	Falcon 9	Vandenberg	Starlink Group 11-15	
300	Dec 8	Long March 6A	Taiyuan	WHDW 15	
301	Dec 8	Falcon 9	Kennedy	Starlink Group 6-92	
302	Dec 9	Long March 4B	Jiuquan	Yaogan 47	
303	Dec 9	Long March 3B	Xichang	TJS 22	
304	Dec 9	Falcon 9	Canaveral	INTRUDER?	
305	Dec 10	Lijian-1	Jiuquan	Jilin-1/UAE-813	
306	Dec 10	Falcon 9	Vandenberg	Starlink Group 15-11	
307	Dec 11	Falcon 9	Canaveral	Starlink Group 6-90	
308	Dec 11	Long March 12	Hainan (comm.)	WHDW 16	
309	Dec 13	Kuaizhou-11	Jiuquan	Dier-5/Xiwang-5-2	
310	Dec 14	Electron	Mahia	RAISE-4	
311	Dec 14	Falcon 9	Vandenberg	Starlink Group 15-12	
312	Dec 15	Falcon 9	Canaveral	Starlink Group 6-82	
313	Dec 16	Long March 4B	Taiyuan	ZY3-04	
314	Dec 16	Atlas V 551	Canaveral	LA-04	
315	Dec 17	Ariane 62	Kourou	Galileosat 33-34	
316	Dec 17	Falcon 9	Kennedy	Starlink Group 6-99	
317	Dec 17	Falcon 9	Vandenberg	Starlink Group 15-13	
318	Dec 18	Electron	Wallops MARS	DiskSat 1-4	
319	Dec 20	Long March 5	Wenchang	TJS 23	
320	Dec 21	Electron	Mahia	QPS-SAR-15	
321	Dec 22	H3-22S	Tanegashima	QZS 5	×
322	Dec 23	HANBIT-NANO	Alcantara	Test	×
323	Dec 23	Long March 12A	Jiuquan	Dummy payload	
324	Dec 24	LVM3	Sriharikota	BlueBird Block-2 FM1	
325	Dec 25	Soyuz-2-1A	Plesetsk	Obzor-R No. 1	
326	Dec 25	Long March 8A	Hainan (comm.)	WHDW 17	

No.	Date (UTC)	Vehicle	Site	Mission / payload
327	Dec 26	Long March 3B	Xichang	FY4-03
328	Dec 28	Soyuz-2-1B	Vostochny	Soyuz Rideshare 16
329	Dec 30	Long March 4B	Jiuquan	Tianhui-7
330	Dec 30	Long March 7A	Wenchang	Shijian 29A/B

E.5 Orbital Launch Log: 2026, to 13 October

All 244 orbital launch attempts. × marks a failure to reach orbit. Times are UTC.

TABLE E.5 Orbital launch attempts in 2026, to 13 October.

No.	Date (UTC)	Vehicle	Site	Mission / payload	
1	Jan 3	Falcon 9	Vandenberg	COSMO-Skymed SG FM3	
2	Jan 4	Falcon 9	Canaveral	Starlink Group 6-88	
3	Jan 9	Falcon 9	Canaveral	Starlink Group 6-96	
4	Jan 11	Falcon 9	Vandenberg	Twilight	
5	Jan 12	PSLV-DL	Sriharikota	PSLV Rideshare 20	×
6	Jan 12	Falcon 9	Canaveral	Starlink Group 6-97	
7	Jan 13	Long March 6A	Taiyuan	YG50-01	
8	Jan 13	Long March 8A	Hainan (comm.)	WHDW 18	
9	Jan 14	Falcon 9	Canaveral	Starlink Group 6-98	
10	Jan 15	Long March 2C	Jiuquan	Alsat-3A	
11	Jan 15	Gushenxing 1	Yellow Sea	Tianqi 37-40	
12	Jan 16	Long March 3B	Xichang	SJ-32	×
13	Jan 17	Gushenxing-2	Jiuquan	Unk Sat 1-6	×
14	Jan 17	Falcon 9	Vandenberg	NRO Starshield C3	
15	Jan 18	Falcon 9	Canaveral	Starlink Group 6-100	
16	Jan 19	Long March 12	Hainan (comm.)	WHDW 19	
17	Jan 22	Falcon 9	Vandenberg	Starlink Group 17-30	
18	Jan 22	Electron	Mahia	Open Cosmos	
19	Jan 25	Falcon 9	Vandenberg	Starlink Group 17-20	
20	Jan 28	Falcon 9	Canaveral	GPS III SV09	
21	Jan 29	Falcon 9	Vandenberg	Starlink Group 17-19	
22	Jan 30	Electron	Mahia	NEONSAT-1A	
23	Jan 30	Falcon 9	Canaveral	Starlink Group 6-101	
24	Jan 31	Long March 2C	Jiuquan	Alsat-3B	
25	Feb 2	Falcon 9	Vandenberg	Starlink Group 17-32	
26	Feb 5	Soyuz-2-1B	Plesetsk	Unknown	
27	Feb 7	Long March 2F	Jiuquan	CSSHQ	
28	Feb 7	Falcon 9	Vandenberg	Starlink Group 17-33	
29	Feb 11	Falcon 9	Vandenberg	Starlink Group 17-34	
30	Feb 12	Jielong-3	Yangjiang	PRSC-E02	
31	Feb 12	Proton-M/DM-3	Baikonur	Elektro-L No. 5	
32	Feb 12	Vulcan Centaur VC4S	Canaveral	GSSAP	

No.	Date (UTC)	Vehicle	Site	Mission / payload	
33	Feb 12	Ariane 64	Kourou	Amazon LE-01	
34	Feb 13	Falcon 9	Canaveral	Dragon C212.5	
35	Feb 15	Falcon 9	Vandenberg	Starlink Group 17-13	
36	Feb 16	Falcon 9	Canaveral	Starlink Group 6-103	
37	Feb 20	Falcon 9	Canaveral	Starlink Group 10-36	
38	Feb 21	Falcon 9	Vandenberg	Starlink Group 17-25	
39	Feb 22	Falcon 9	Canaveral	Starlink Group 6-104	
40	Feb 24	Falcon 9	Canaveral	Starlink Group 6-110	
41	Feb 25	Falcon 9	Vandenberg	Starlink Group 17-26	
42	Feb 27	Falcon 9	Canaveral	Starlink Group 6-108	
43	Mar 1	Falcon 9	Vandenberg	Starlink Group 17-23	
44	Mar 2	Falcon 9	Canaveral	Starlink Group 10-41	
45	Mar 4	Falcon 9	Canaveral	Starlink Group 10-40	
46	Mar 5	Kairos	Kii	Tatara-1R	×
47	Mar 5	Electron	Mahia	BlackSky Gen3-4	
48	Mar 8	Falcon 9	Vandenberg	Starlink Group 17-18	
49	Mar 10	Falcon 9	Canaveral	Echostar 25	
50	Mar 12	Firefly Alpha	Vandenberg	LM-400 Tech Demo	
51	Mar 12	Long March 8A	Hainan (comm.)	WHDW 20	
52	Mar 12	Long March 2D	Xichang	Shiyan 30-03/04	
53	Mar 13	Falcon 9	Vandenberg	Starlink Group 17-31	
54	Mar 14	Falcon 9	Canaveral	Starlink Group 10-48	
55	Mar 15	Long March 6A	Taiyuan	YG50-02	
56	Mar 16	Kuaizhou-11	Jiuquan	Juntian-1-04A/Dongp	
57	Mar 17	Falcon 9	Vandenberg	Starlink Group 17-24	
58	Mar 17	Falcon 9	Canaveral	Starlink Group 10-46	
59	Mar 19	Falcon 9	Canaveral	Starlink Group 10-33	
60	Mar 20	Electron	Mahia	StriX-6	
61	Mar 20	Falcon 9	Vandenberg	Starlink Group 17-15	
62	Mar 22	Soyuz-2-1A	Baikonur	Progress 7K-TGM No. 463	
63	Mar 22	Falcon 9	Canaveral	Starlink Group 10-62	
64	Mar 22	Jielong-3	Yellow Sea	Weili Kongjian 02 zu	
65	Mar 23	Soyuz-2-1B	Plesetsk	Rassvet-3 G1	
66	Mar 25	Long March 2D	Taiyuan	Superview 2-05/06	
67	Mar 26	Falcon 9	Vandenberg	Starlink Group 17-17	
68	Mar 27	Long March 2C/YZ-1S	Jiuquan	Shiyan 33	
69	Mar 28	Electron	Mahia	Celeste	
70	Mar 30	Lijian-2	Jiuquan	XZC01/02	
71	Mar 30	Falcon 9	Vandenberg	Transporter-16	
72	Mar 30	Falcon 9	Canaveral	Starlink Group 10-44	
73	Apr 1	SLS Block 1	Kennedy	Integrity	
74	Apr 2	Falcon 9	Canaveral	Starlink Group 10-58	

No.	Date (UTC)	Vehicle	Site	Mission / payload	
75	Apr 3	Tianlong-3	Jiuquan	-	×
76	Apr 3	Soyuz-2-1A	Plesetsk	Meridian No. 21L?	
77	Apr 4	Atlas V 551	Canaveral	LA-05	
78	Apr 7	Falcon 9	Vandenberg	Starlink Group 17-35	
79	Apr 7	Minotaur IV	Vandenberg	STP-S29A	
80	Apr 7	Long March 8	Hainan (comm.)	Qianfan	
81	Apr 8	Long March 6A	Taiyuan	WHDW 21	
82	Apr 11	Falcon 9	Vandenberg	Starlink Group 17-21	
83	Apr 11	Jielong-3	Yangjiang	WHJSW-8	
84	Apr 11	Falcon 9	Canaveral	Cygnus NG-24	
85	Apr 14	Lijian-1	Jiuquan	Jilin-1 GF07	
86	Apr 14	Falcon 9	Canaveral	Starlink Group 10-24	
87	Apr 15	Falcon 9	Vandenberg	Starlink Group 17-27	
88	Apr 16	Soyuz-2-1B	Plesetsk	Unknown	
89	Apr 17	Long March 4C	Jiuquan	Daqi 2	
90	Apr 19	New Glenn 7×2	Canaveral	SpaceMobile-007	×
91	Apr 19	Falcon 9	Vandenberg	Starlink Group 17-22	
92	Apr 21	Falcon 9	Canaveral	GPS III SV10	
93	Apr 23	Electron	Mahia	JAXA ISTD-4	
94	Apr 23	Falcon 9	Vandenberg	Starlink Group 17-14	
95	Apr 23	Angara-1.2	Plesetsk	OO MKA 7-10	
96	Apr 24	Long March 2D	Xichang	WHJSW 9	
97	Apr 25	Long March 6	Taiyuan	PRSC-EO3	
98	Apr 25	Soyuz-2-1A	Baikonur	Progress 7K-TGM No. 464	
99	Apr 26	Falcon 9	Vandenberg	Starlink Group 17-16	
100	Apr 28	Atlas V 551	Canaveral	LA-06	
101	Apr 29	Falcon Heavy	Kennedy	Viasat 3F3	
102	Apr 30	Falcon 9	Vandenberg	Starlink Group 17-36	
103	Apr 30	Ariane 64	Kourou	Amazon LE-02	
104	May 1	Falcon 9	Canaveral	Starlink Group 10-38	
105	May 3	Falcon 9	Vandenberg	RideshareX-2026-100	
106	May 6	Falcon 9	Vandenberg	Starlink Group 17-29	
107	May 11	Long March 7	Wenchang	Tianzhou-10	
108	May 12	Falcon 9	Vandenberg	NRO Starshield A9	
109	May 12	Long March 6A	Taiyuan	Qianfan Jigui 09	
110	May 14	Zhuque-2E	Jiuquan	Dummy sat	
111	May 15	Lijian-1	Jiuquan	Taijing-3	
112	May 15	Falcon 9	Canaveral	Dragon C209.6	
113	May 17	Long March 8	Hainan (comm.)	Qianfan	
114	May 19	Vega C	Kourou	SMILE	
115	May 20	Falcon 9	Vandenberg	Starlink Group 17-42	
116	May 21	Falcon 9	Canaveral	Starlink Group 10-31	

No.	Date (UTC)	Vehicle	Site	Mission / payload
117	May 22	Electron	Mahia	StriX-7
118	May 22	Starship V3	Starbase	Ship 39
119	May 24	Long March 2F	Jiuquan	Shenzhou 23
120	May 25	Falcon 9	Canaveral	Starlink Group 10-47
121	May 26	Falcon 9	Vandenberg	Starlink Group 17-37
122	May 26	Long March 7A	Wenchang	TJS-24
123	May 29	New Glenn 7×2	Canaveral	-
124	May 29	Falcon 9	Canaveral	Starlink Group 10-53
125	May 29	Atlas V 551	Canaveral	LA-07
126	May 30	Falcon 9	Vandenberg	Starlink Group 17-41
127	May 30	Long March 2D	Xichang	WHJSW 10
128	Jun 1	Long March 12B	Jiuquan	Qianfan
129	Jun 3	Falcon 9	Vandenberg	Starlink Group 17-47
130	Jun 4	Falcon 9	Canaveral	Starlink Group 10-43
131	Jun 4	Long March 6A	Taiyuan	Qianfan Jigui 11
132	Jun 5	Long March 8	Hainan (comm.)	Qianfan Jigui 12
133	Jun 7	Falcon 9	Vandenberg	Starlink Group 17-43
134	Jun 8	Falcon 9	Canaveral	Starlink Group 10-35
135	Jun 9	Zhuque-2E	Jiuquan	Qianfan DTC1
136	Jun 11	Long March 5	Wenchang	TJS 25
137	Jun 11	HASTE	Wallops MARS	MACH-TB?
138	Jun 11	Falcon 9	Vandenberg	Starlink Group 17-44
139	Jun 12	H3-30S	Tanegashima	VEP-5
140	Jun 12	Falcon 9	Canaveral	Starlink Group 10-54
141	Jun 15	Lijian-1	Jiuquan	Jilin-1 GF07
142	Jun 15	Falcon 9	Vandenberg	Starlink Group 17-54
143	Jun 16	Long March 3B	Xichang	SJ-31 (TJS-26?)
144	Jun 17	Long March 12	Hainan (comm.)	WHDW 22
145	Jun 17	Kuaizhou-11	Jiuquan	Centispace 05
146	Jun 17	Falcon 9	Canaveral	Bluebird 8 to 10
147	Jun 17	Ariane 64	Kourou	Amazon LE-03
148	Jun 19	Falcon 9	Vandenberg	NRO Starshield C4
149	Jun 19	Electron	Mahia	VICTUS HAZE Puma
150	Jun 21	Falcon 9	Vandenberg	Starlink Group 17-28
151	Jun 23	Long March 7A	Wenchang	TJS-26A
152	Jun 23	Falcon 9	Canaveral	Starfall Demo
153	Jun 25	Falcon 9	Vandenberg	Starlink Group 17-45
154	Jun 26	Electron	Mahia	StriX-8
155	Jun 28	Falcon 9	Vandenberg	Starlink Group 17-40
156	Jun 29	Falcon 9	Canaveral	SXM-11
157	Jul 1	Long March 4B	Jiuquan	HY-2E
158	Jul 2	Falcon 9	Vandenberg	Starlink Group 17-46

No.	Date (UTC)	Vehicle	Site	Mission / payload	
159	Jul 2	Atlas V 551	Canaveral	LA-08	
160	Jul 3	Pegasus XL	Kwajalein	Swift Reboost	
161	Jul 4	Long March 6A	Taiyuan	Qianfan Jigui 13	
162	Jul 5	Falcon 9	Canaveral	Starlink Group 10-50	
163	Jul 5	Long March 8A	Hainan (comm.)	Qianfan Jigui 15	
164	Jul 7	Falcon 9	Vandenberg	Transporter-17	
165	Jul 9	Falcon 9	Canaveral	Starlink Group 10-42	
166	Jul 10	Long March 10B	Hainan (comm.)	CX-26	
167	Jul 11	Falcon 9	Vandenberg	Starlink Group 17-48	
168	Jul 14	Falcon 9	Vandenberg	Starlink Group 15-14	
169	Jul 14	Falcon 9	Canaveral	Starlink Group 10-45	
170	Jul 14	Soyuz-2-1A	Baikonur	Soyuz 11F732A48 No. 759	
171	Jul 16	Falcon 9	Vandenberg	Tranche 1 TL-E	
172	Jul 18	Vikram-1	Sriharikota	Solaras-S3/SCOPE	
173	Jul 19	Soyuz-2-1B	Plesetsk	Rassvet-3 G2	
174	Jul 21	Falcon 9	Vandenberg	Starlink Group 17-39	
175	Jul 21	Falcon 9	Canaveral	MRV/MEP	
176	Jul 22	Yinli-1	East China Sea	Dongpo/ZDG-3	
177	Jul 23	Long March 3B	Xichang	Tianlian 2-06	
178	Jul 23	Lijian-1	Jiuquan	Xiguang	
179	Jul 24	Starship V3	Starbase	Ship 40	
180	Jul 25	Falcon 9	Vandenberg	Starlink Group 17-51	
181	Jul 29	Long March 7A	Wenchang	Tianlian 3-01	
182	Jul 30	Long March 6A	Taiyuan	TJS-27A/B	
183	Jul 30	Falcon 9	Canaveral	INTRUDER?	
184	Aug 1	Falcon 9	Vandenberg	Starlink Group 17-52	
185	Aug 4	Long March 8A	Hainan (comm.)	WHDW 23	
186	Aug 4	Falcon 9	Vandenberg	Starlink Group 17-53	
187	Aug 5	Jielong-3	Yellow Sea	Lampung/Samarqand	
188	Aug 5	Falcon 9	Canaveral	Bluebird 11 to 13	
189	Aug 6	Electron	Mahia	QPS-SAR-13	
190	Aug 8	Falcon 9	Vandenberg	Starlink Group 17-38	
191	Aug 10	Long March 7A	Wenchang	Zhongxing-4B	×
192	Aug 10	H3-22S	Tanegashima	QZS 7	
193	Aug 11	Falcon 9	Canaveral	Starlink Group 10-19	
194	Aug 12	Falcon 9	Vandenberg	Starlink Group 17-49	
195	Aug 16	Falcon 9	Canaveral	Globalstar 2R-1	
196	Aug 16	Falcon 9	Vandenberg	SBST R-1	
197	Aug 16	Long March 12	Hainan (comm.)	WHDW 24	
198	Aug 17	Long March 2C	Taiyuan	SEO	
199	Aug 18	Zhuque-3	Jiuquan	Honghu-3	
200	Aug 19	Falcon 9	Vandenberg	Starlink Group 17-50	

No.	Date (UTC)	Vehicle	Site	Mission / payload
201	Aug 20	Electron	Mahia	QPS-SAR-18
202	Aug 21	Falcon 9	Canaveral	Starlink Group 10-39
203	Aug 22	Falcon 9	Vandenberg	Starlink Group 15-20
204	Aug 24	Soyuz-2-1B	Plesetsk	Glonass
205	Aug 25	Long March 6C	Taiyuan	ZK-14/15
206	Aug 25	Falcon 9	Canaveral	Starlink Group 10-49
207	Aug 26	Falcon 9	Vandenberg	Starlink Group 15-22
208	Aug 27	Ariane 62	Kourou	MTG-I2
209	Aug 30	Falcon Heavy	Kennedy	NGRST
210	Sep 1	Zhishenxing-1	Jiuquan	Pallas-1 Test
211	Sep 2	Falcon 9	Vandenberg	Starlink Group 15-23
212	Sep 2	Electron	Mahia	StriX-9
213	Sep 3	GSLV Mk II	Sriharikota	GISAT-1A
214	Sep 5	Spectrum	Andoya	Onward and Upward
215	Sep 6	Falcon 9	Vandenberg	Starlink Group 15-24
216	Sep 10	Long March 4B	Jiuquan	YG53/YG56
217	Sep 10	Falcon 9	Vandenberg	SBST R-2
218	Sep 11	Electron	Mahia	BlackSky Gen3-5
219	Sep 13	Falcon 9	Canaveral	O3b mPOWER 11/13
220	Sep 15	Vega C	Kourou	Sentinel-3C/Flex
221	Sep 15	Zhuque-2E	Jiuquan	Qianfan Digui 19
222	Sep 15	Yinli-1	East China Sea	Qianfan Digui 26
223	Sep 16	Soyuz-2-1B	Baikonur	Progress 7K-TGM No. 465
224	Sep 17	Long March 12	Hainan (comm.)	WHDW 25
225	Sep 17	Falcon 9	Vandenberg	SBST TH-1
226	Sep 17	Kuaizhou-11	Jiuquan	Tianyi-51/52
227	Sep 17	Soyuz-2-1B	Plesetsk	Glonass
228	Sep 19	Electron	Mahia	StriX-10
229	Sep 19	Long March 2D	Jiuquan	PIESAT 2-13 to 16
230	Sep 20	Falcon 9	Vandenberg	Starlink Group 15-27
231	Sep 20	Lijian-1	Jiuquan	Pengcheng
232	Sep 23	Long March 8A	Hainan (comm.)	WHDW 26
233	Sep 24	Long March 6A	Taiyuan	Yaogan 40-04
234	Sep 26	Electron	Mahia	StriX-12
235	Sep 26	Falcon 9	Vandenberg	SBST R-3
236	Sep 28	Starship V3	Starbase	Ship 41
237	Oct 1	Falcon 9	Canaveral	Dragon C213.2
238	Oct 1	Falcon 9	Vandenberg	Transporter-18
239	Oct 2	Falcon Heavy	Kennedy	NROL-97
240	Oct 7	Nuri	Naro	NeonSat
241	Oct 9	Long March 12	Hainan (comm.)	WHDW 27
242	Oct 10	Falcon 9	Vandenberg	Tranche 1 TL-A

No.	Date (UTC)	Vehicle	Site	Mission / payload
243	Oct 13	Falcon 9	Vandenberg	Starlink Group 15-25
244	Oct 13	Falcon 9	Canaveral	Dragon C211.4

APPENDIX

F Timeline of Space Exploration and Engineering

> *A chronology of the milestones described in this manual — from the theoretical foundations of rocketry and the first liquid-fuelled rocket, through the Space Race, the Shuttle and station era, and the rise of commercial spaceflight, to the events of 2025 and 2026 — followed by the missions planned for 2026–2035 as of October 2026.*

IN THIS CHAPTER

F.1 Foundations (1903–1956)
F.2 The Space Race (1957–1975)
F.3 Shuttles, Stations and Space Science (1976–1999)
F.4 The New Century (2000–2019)
F.5 The Commercial Era (2020–2026)
F.6 Planned Missions (as of October 2026)

Dates are in Coordinated Universal Time (UTC) where the precise day matters; a launch late in the evening in the Americas can fall on the next UTC day. "First" claims refer to the first *successful* achievement. Planned dates in the final table are those announced by the responsible agencies or companies as of October 2026; experience suggests that many will slip.

F.1 Foundations (1903–1956)

Date	Event
1903	Konstantin Tsiolkovsky publishes *Exploration of Outer Space by Means of Reaction Devices*, deriving the rocket equation
1919	Robert Goddard publishes *A Method of Reaching Extreme Altitudes*
1923	Hermann Oberth publishes *Die Rakete zu den Planetenräumen* (The Rocket into Planetary Space)
16 Mar 1926	Goddard launches the first liquid-propellant rocket, at Auburn, Massachusetts — a 2.5 s flight to 12.5 m
17 Aug 1933	GIRD-09, the first Soviet rocket with a liquid oxidiser, flies near Moscow
20 Jun 1944	A German V-2 (A4) test rocket reaches about 175 km — the first human-made object in space
24 Oct 1946	A camera on a captured V-2 launched from White Sands takes the first photographs of Earth from space (105 km)
24 Feb 1949	The two-stage Bumper-WAC reaches 393 km
Jul–Aug 1955	The United States and the Soviet Union announce plans to launch satellites for the International Geophysical Year

F.2 The Space Race (1957–1975)

Date	Event
21 Aug 1957	First successful flight of the R-7, the world's first intercontinental ballistic missile
4 Oct 1957	**Sputnik 1**, the first artificial satellite, launched by an R-7 from Baikonur

Date	Event
3 Nov 1957	Sputnik 2 carries the dog Laika, the first animal in orbit
1 Feb 1958	Explorer 1, the first US satellite; its Geiger counter discovers the Van Allen radiation belts
17 Mar 1958	Vanguard 1, the first solar-powered satellite — still in orbit, the oldest human-made object in space
1 Oct 1958	NASA begins operations
2 Jan 1959	Luna 1, the first spacecraft to escape Earth's gravity, flies past the Moon
14 Sep 1959	Luna 2 strikes the Moon, the first spacecraft to reach another world
7 Oct 1959	Luna 3 photographs the far side of the Moon
1 Apr 1960	TIROS-1, the first weather satellite
12 Aug 1960	Echo 1, a 30 m passive reflecting balloon, bounces telephone and television signals across the United States
18–19 Aug 1960	Discoverer 14 (Corona): the first film capsule recovered from orbit
12 Apr 1961	**Yuri Gagarin** orbits the Earth in Vostok 1 — the first human in space
5 May 1961	Alan Shepard's suborbital flight in Freedom 7
20 Feb 1962	John Glenn, the first American in orbit (Friendship 7)
10 Jul 1962	Telstar 1 relays the first live transatlantic television
14 Dec 1962	Mariner 2 flies past Venus — the first successful planetary encounter
16 Jun 1963	Valentina Tereshkova, the first woman in space (Vostok 6)
26 Jul 1963	Syncom 2, the first geosynchronous communication satellite (Syncom 3, geostationary, follows in 1964)
18 Mar 1965	Alexei Leonov makes the first spacewalk (Voskhod 2)
6 Apr 1965	Intelsat I "Early Bird", the first commercial communication satellite
15 Jul 1965	Mariner 4 returns the first close-up images of Mars
26 Nov 1965	France launches Astérix, becoming the third nation to orbit a satellite
3 Feb 1966	Luna 9 makes the first soft landing on the Moon
16 Mar 1966	Gemini 8 performs the first docking in orbit
27 Jan 1967	Apollo 1 fire kills three astronauts during a ground test
10 Oct 1967	The Outer Space Treaty enters into force
21–27 Dec 1968	Apollo 8: the first humans to orbit the Moon
20 Jul 1969	**Apollo 11**: Neil Armstrong and Buzz Aldrin land on the Moon
11 Feb 1970	Japan launches Ohsumi (fourth nation)
24 Apr 1970	China launches Dong Fang Hong 1 (fifth nation)
17 Nov 1970	Lunokhod 1, the first robotic rover on another world
15 Dec 1970	Venera 7 makes the first landing on Venus and transmits from the surface
19 Apr 1971	Salyut 1, the first space station
28 Oct 1971	The United Kingdom launches Prospero on Black Arrow
14 Nov 1971	Mariner 9, the first spacecraft to orbit another planet (Mars)
2 Dec 1971	Mars 3 makes the first soft landing on Mars; it transmits for about 20 seconds
23 Jul 1972	ERTS-1 (Landsat 1), the start of continuous civil land imaging
7–19 Dec 1972	Apollo 17, the last crewed lunar landing of the twentieth century
14 May 1973	Skylab, the first US space station
4 Dec 1973	Pioneer 10, the first spacecraft to fly past Jupiter
17 Jul 1975	Apollo and Soyuz dock in orbit

F.3 Shuttles, Stations and Space Science (1976–1999)

Date	Event
20 Jul 1976	Viking 1 lands on Mars; Viking 2 follows on 3 September
20 Aug / 5 Sep 1977	Voyager 2 and Voyager 1 launched
22 Feb 1978	First Navstar GPS satellite launched
27 Jun 1978	Seasat, the first civil imaging radar satellite
24 Dec 1979	First flight of Ariane 1
18 Jul 1980	India orbits Rohini with its SLV-3 rocket
12 Apr 1981	STS-1: first flight of the Space Shuttle Columbia
Apr 1984	STS-41C crew captures and repairs the Solar Maximum Mission satellite
28 Jan 1986	Challenger breaks up 73 s after launch; seven astronauts killed
20 Feb 1986	Launch of the Mir core module
Mar 1986	Giotto, Vega 1/2, Suisei and Sakigake fly past Halley's Comet
15 Nov 1988	Buran makes its only, uncrewed, orbital flight
19 Sep 1988	Israel launches Ofeq 1
25 Aug 1989	Voyager 2 flies past Neptune
18 Nov 1989	COBE launched; it maps the cosmic microwave background
24 Apr 1990	Hubble Space Telescope launched by STS-31
May 1992	STS-49 crew rescues Intelsat 603 by hand
Dec 1993	First Hubble servicing mission corrects its flawed optics
17 Jul 1995	GPS declared fully operational
2 Dec 1995	SOHO launched to the Sun–Earth L1 point
7 Dec 1995	The Galileo probe enters Jupiter's atmosphere; the orbiter begins eight years of study
4 Jul 1997	Mars Pathfinder lands; the Sojourner rover begins exploring
20 Nov 1998	Zarya, the first module of the International Space Station
Nov 1998	Iridium begins the first global satellite telephone service
23 Jul 1999	Chandra X-ray Observatory launched

F.4 The New Century (2000–2019)

Date	Event
2 Nov 2000	Expedition 1 arrives on the ISS — the start of continuous human presence in orbit
12 Feb 2001	NEAR Shoemaker lands on the asteroid Eros
1 Feb 2003	Columbia breaks up during re-entry; seven astronauts killed
15 Oct 2003	Yang Liwei flies Shenzhou 5; China becomes the third nation to launch a human
Jan 2004	Mars Exploration Rovers Spirit and Opportunity land; Opportunity drives for 14 years
21 Jun 2004	SpaceShipOne makes the first privately funded crewed spaceflight; wins the Ansari X Prize in October
1 Jul 2004	Cassini enters orbit around Saturn
14 Jan 2005	Huygens lands on Titan, the most distant landing yet
4 Jul 2005	Deep Impact strikes Comet Tempel 1
19 Jan 2006	New Horizons launched towards Pluto
11 Jan 2007	China's anti-satellite test destroys Fengyun-1C, creating more than 3 500 catalogued debris fragments

Date	Event
25 May 2008	Phoenix lands in Mars's northern arctic and touches water ice
28 Sep 2008	Falcon 1 becomes the first privately developed liquid-fuelled rocket to reach orbit
10 Feb 2009	Iridium 33 and Kosmos 2251 collide — the first accidental collision between intact satellites
7 Mar 2009	Kepler launched; it finds thousands of exoplanets
9 Oct 2009	LCROSS impact confirms water ice in a permanently shadowed lunar crater
4 Jun 2010	First flight of Falcon 9
13 Jun 2010	Hayabusa returns the first samples from an asteroid (Itokawa)
Jun 2010	IKAROS demonstrates solar-sail propulsion in interplanetary space
18 Mar 2011	MESSENGER, the first spacecraft to orbit Mercury
21 Jul 2011	Atlantis lands, ending the Space Shuttle programme after 135 flights
25 May 2012	SpaceX Dragon becomes the first commercial spacecraft to berth with the ISS
6 Aug 2012	Curiosity lands in Gale Crater using the sky crane
25 Aug 2012	Voyager 1 crosses the heliopause into interstellar space
14 Dec 2013	Chang'e 3 lands with the Yutu rover — the first lunar soft landing since 1976
19 Dec 2013	Gaia launched
6 Aug 2014	Rosetta arrives at Comet 67P; Philae lands on 12 November
24 Sep 2014	India's Mangalyaan enters Mars orbit on the country's first attempt
6 Mar 2015	Dawn enters orbit around Ceres after orbiting Vesta
14 Jul 2015	New Horizons flies past Pluto
22 Dec 2015	A Falcon 9 first stage lands after an orbital launch for the first time
5 Jul 2016	Juno enters orbit around Jupiter
15 Feb 2017	PSLV-C37 deploys 104 satellites in one launch
30 Mar 2017	First reflight of an orbital-class booster (Falcon 9, SES-10)
15 Sep 2017	Cassini plunges into Saturn
6 Feb 2018	First flight of Falcon Heavy
12 Aug 2018	Parker Solar Probe launched
1 Jan 2019	New Horizons flies past Arrokoth, the most distant object explored
3 Jan 2019	Chang'e 4 makes the first landing on the far side of the Moon
27 Mar 2019	India's anti-satellite test (Mission Shakti)
24 May 2019	First 60 operational-design Starlink satellites launched
9 Oct 2019	Mission Extension Vehicle 1 launched; it docks with Intelsat 901 on 25 Feb 2020

F.5 The Commercial Era (2020–2026)

Date	Event
30 May 2020	Crew Dragon Demo-2: the first crewed flight of a commercially built orbital spacecraft
31 Jul 2020	BeiDou-3 global navigation system commissioned
5 Dec 2020	Hayabusa2 capsule lands in Australia with samples of Ryugu
16 Dec 2020	Chang'e 5 returns 1.73 kg of lunar samples
18 Feb 2021	Perseverance lands in Jezero Crater
19 Apr 2021	Ingenuity makes the first powered, controlled flight on another planet
29 Apr 2021	Tianhe, the core module of China's Tiangong station

Date	Event
14 May 2021	Tianwen-1 lands the Zhurong rover; China becomes the second nation to operate a rover on Mars
Jul 2021	Virgin Galactic (11 July) and Blue Origin (20 July) fly their founders on suborbital flights
16 Sep 2021	Inspiration4, the first all-private orbital crew
15 Nov 2021	Russian anti-satellite test destroys Kosmos 1408, forcing the ISS crew to shelter
25 Dec 2021	James Webb Space Telescope launched by Ariane 5
12 Jul 2022	First full-colour images from JWST
26 Sep 2022	DART strikes Dimorphos and shortens its orbit by 33 minutes — the first planetary-defence test
16 Nov – 11 Dec 2022	Artemis I: uncrewed Orion flies around the Moon on the first SLS launch
20 Apr 2023	First integrated flight of Starship
23 Aug 2023	Chandrayaan-3 lands near the lunar south pole; India becomes the fourth nation to soft-land on the Moon
24 Sep 2023	OSIRIS-REx capsule returns samples of Bennu
19 Jan 2024	Japan's SLIM lands within 100 m of its target
22 Feb 2024	Intuitive Machines' Odysseus: the first commercial lunar landing (it tips over)
25 Jun 2024	Chang'e 6 returns the first samples from the lunar far side
9 Jul 2024	First flight of Ariane 6
12 Sep 2024	Polaris Dawn: the first commercial spacewalk
13 Oct 2024	Starship's Super Heavy booster is caught by the launch tower on its first attempt
14 Oct 2024	Europa Clipper launched
16 Jan 2025	First flight of New Glenn reaches orbit
2 Mar 2025	Firefly's Blue Ghost makes the first fully successful commercial lunar landing
18 Mar 2025	Crew-9 returns Butch Wilmore and Suni Williams after their Starliner test flight stretched to nine months
27 Mar 2025	Gaia passivated after mapping about two billion stars
1 Apr 2025	Fram2: the first crewed flight over the poles
29 Apr 2025	ESA's BIOMASS, the first P-band radar satellite
28 May 2025	China's Tianwen-2 launched to the near-Earth asteroid Kamoʻoalewa
30 Jul 2025	NISAR, the NASA–ISRO dual-band radar satellite, launched
Nov 2025	A crack in Shenzhou-20's window, probably from debris, forces its crew to return in Shenzhou-21; an uncrewed Shenzhou-22 is launched as a replacement
13 Nov 2025	New Glenn launches NASA's ESCAPADE and lands its booster
2025	Record 330 orbital launch attempts worldwide; more than 4 500 payloads launched
15 Jan 2026	Crew-11 returns a month early — the first medical evacuation from the ISS
1–10 Apr 2026	**Artemis II**: Wiseman, Glover, Koch and Hansen fly around the Moon — the first crewed lunar flight since 1972
19 May 2026	ESA–China SMILE magnetosphere mission launched on Vega-C
28 May 2026	A New Glenn rocket explodes during a static-fire test
Jun–Jul 2026	Tianwen-2 arrives at Kamoʻoalewa
30 Aug 2026	Nancy Grace Roman Space Telescope launched by Falcon Heavy, about eight months early
28 Sep 2026	Starship reaches orbit for the first time on Flight 14 and deploys Starlink V3 satellites
28 Sep 2026	JUICE flies past Earth on its way to Jupiter
8 Oct 2026	Crew-12 returns after about 235 days on the ISS

F.6 Planned Missions (as of October 2026)

The following events were scheduled as of mid-October 2026. They are listed to show where the field is heading; check current sources before relying on any date.

Planned date	Event
20 Oct 2026 (JST)	JAXA's Martian Moons eXploration (MMX) launch on H3, to return samples of Phobos around 2031
Nov 2026	ESA's Hera arrives at Didymos and Dimorphos to study the aftermath of DART
18 Nov 2026	Voyager 1 becomes the first spacecraft one light-day from Earth
21 Nov 2026	BepiColombo enters orbit around Mercury; its two orbiters separate in December
Late 2026	Gaganyaan G1, India's first uncrewed orbital test of its crew capsule
Late 2026	Starship Flight 15 with a planned ship catch; uncrewed Starliner-1 cargo flight
2027	Artemis III: crewed rendezvous and docking test with commercial lunar landers in Earth orbit
2027	Chang'e-7 lunar south-pole mission (postponed from 2026); Blue Moon MK1 lunar lander
2027	First crewed Gaganyaan flight (ISRO)
2027	ESCAPADE arrives at Mars; MMX arrives in Mars orbit
Nov 2027	Tianwen-2 sample capsule returns to Earth
2028	Ramses launched to Apophis (with DESTINY+ on H3); Space Rider first flight
Jul 2028	Dragonfly rotorcraft lander launched to Titan (arrival 2034)
13 Apr 2029	Asteroid Apophis passes about 32 000 km above the Earth's surface; OSIRIS-APEX and Ramses study it
2029	Psyche arrives at the metal-rich asteroid 16 Psyche
By 2030	China's planned first crewed lunar landing (Mengzhou spacecraft, Lanyue lander, Long March 10)
Apr 2030	Europa Clipper enters orbit around Jupiter
2030–31	ISS operations end; the US Deorbit Vehicle steers the station into the South Pacific
2031	JUICE enters orbit around Jupiter (July); MMX returns Phobos samples
2034	Dragonfly arrives at Titan
≈ 2035	ESA's LISA gravitational-wave observatory and NewAthena X-ray observatory (launches planned 2035–37)

KEY TAKEAWAYS

- It took 54 years from Tsiolkovsky's rocket equation (1903) to Sputnik (1957), and only 12 more to a human landing on the Moon.
- Most "firsts" of robotic planetary exploration came between 1959 and 1976; the decades since have brought orbiters, rovers and sample returns.
- Since 2015 reusable rockets and commercial spacecraft have transformed launch rates: 330 orbital launch attempts in 2025, compared with fewer than 100 a year in 2000–2017.
- 2026 brought the first crewed lunar flight in 54 years, the first orbital flight of Starship and the launch of the Roman Space Telescope.

APPENDIX

G Acronyms and Abbreviations

> *The acronyms and abbreviations used in this manual, with their meanings and the chapters in which they are defined or mainly used.*

IN THIS CHAPTER

G.1 A
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G.26 Z

Space engineering is notoriously rich in acronyms. This list gives every acronym that is spelled out in the chapters, together with common ones used without definition. The last column gives the chapters where each is defined or mainly used (letters denote appendices). Acronyms in other languages are followed by the English meaning where helpful. Units and chemical formulae are not listed; see Appendix A.

G.1 A

Acronym	Meaning	Chapters
ABFT	algorithm-based fault tolerance	45
ABI	Advanced Baseline Imager	16
ACC	anomaly correlation coefficient	47
ACE	Advanced Composition Explorer	
ACM	adaptive coding and modulation	17, 49
ACS	attitude control system	9
ACS3	Advanced Composite Solar Sail System	
ADCS	attitude determination and control subsystem	6
ADR	active debris removal	21
ADRAS-J	Active Debris Removal by Astroscale-Japan	
ADS-B	Automatic Dependent Surveillance–Broadcast (aircraft)	

Acronym	Meaning	Chapters
AEGIS	Autonomous Exploration for Gathering Increased Science	
AEPS	Advanced Electric Propulsion System	42
AFRL	Air Force Research Laboratory	53
AFSS	autonomous flight safety system	52
AGN	active galactic nuclei	51
AHI	Advanced Himawari Imager	16
AI	artificial intelligence	43, 52
AIAA	American Institute of Aeronautics and Astronautics	
AICON	DWD's AI-based global weather model (built with ECMWF's Anemoi software)	
AIFS	Artificial Intelligence Forecasting System	16, 47
AIGEFS	AI Global Ensemble Forecast System (NOAA)	
AIGFS	Artificial Intelligence Global Forecast System (NOAA)	
AIS	Automatic Identification System (ships)	46
AIT	assembly, integration and test	4
AMSR2	Advanced Microwave Scanning Radiometer 2	
AO	atomic oxygen	2
AOA	angle of arrival	19
AOD	aerosol optical depth	13
AOI	automated optical inspection	52
APCP	ammonium perchlorate composite propellant	35
API	application programming interface	14
APSK	amplitude-phase shift keying	7
AR	augmented reality	51
ARD	analysis-ready data	13, 45, 46
ARED	Advanced Resistive Exercise Device	30
ASAT	anti-satellite (weapon)	
ASCAT	Advanced Scatterometer	
ASE	Autonomous Sciencecraft Experiment	48
ASI	Agenzia Spaziale Italiana (Italian Space Agency)	
ASTER	Advanced Spaceborne Thermal Emission and Reflection Radiometer	
ATLID	atmospheric lidar	16
ATMS	Advanced Technology Microwave Sounder	
ATV	Automated Transfer Vehicle	
AUC	area under the (ROC) curve	
AVHRR	Advanced Very High Resolution Radiometer	

G.2 B

Acronym	Meaning	Chapters
B/H	base to the height	14
BAS	Bharatiya Antariksh Station	32
BATSE	Burst and Transient Source Experiment	39

Acronym	Meaning	Chapters
BEAM	Bigelow Expandable Activity Module	32
BLE	ballistic limit equation	2
BLS	box least squares	51
BO	bayesian optimisation	52
BOC	binary offset carrier	18
BOLE	Booster Obsolescence and Life Extension	35
BPSK	binary phase-shift keying	
BRDF	bidirectional reflectance distribution function	13
BSS	Broadcasting Satellite Service	17
BTU	British thermal unit	A
BVAD	Baseline Values and Assumptions Document	30

G.3 C

Acronym	Meaning	Chapters
C&DH	command and data handling	6
C/A	coarse/acquisition	18
CADRE	Cooperative Autonomous Distributed Robotic Exploration	
CAPSTONE	Cislunar Autonomous Positioning System Technology Operations and Navigation Experiment	
CAS	Chinese Academy of Sciences	
CASC	China Aerospace Science and Technology Corporation	
CASPER	Continuous Activity Scheduling Planning Execution and Replanning	
CBM	Common Berthing Mechanism	32
CCD	charge-coupled device	38
CCI	Climate Change Initiative	47
CCM	constant coding and modulation	49
CCSDS	Consultative Committee for Space Data Systems	7
CCW	Certain Conventional Weapons	50
CDM	conjunction data message	21, 50
CDMA	code division multiple access	18
CDR	Critical Design Review	4
CDRA	Carbon Dioxide Removal Assembly	30
CEMS	Copernicus Emergency Management Service	46
CEOS	Committee on Earth Observation Satellites	13
CEOS-ARD	CEOS Analysis Ready Data	13
CER	cost-estimating relationship	4
CERES	Clouds and the Earth's Radiant Energy System	
CFAR	constant false alarm rate	46
CFDP	CCSDS File Delivery Protocol	7, I
CFR	Code of Federal Regulations (USA)	
CFRP	carbon-fibre reinforced polymers	5
CG	centre of gravity	27

Acronym	Meaning	Chapters
CGMS	Coordination Group for Meteorological Satellites	16
CGRO	Compton Gamma Ray Observatory	39
CIMON	Crew Interactive MObile companioN (ISS assistant)	
CLA	coupled loads analysis	5
CLD	Commercial LEO Destinations	32
CLPS	Commercial Lunar Payload Services	27
CM	command module	31
CMB	cosmic microwave background	39
CME	coronal mass ejection	22, 47
CMG	control-moment gyroscope	6, 14, 32
CMOS	complementary metal–oxide–semiconductor	
CMSA	China Manned Space Agency	
CNES	Centre national d'études spatiales (French space agency)	
CNN	convolutional neural network	44, 46
CNSA	China National Space Administration	
CoDANN	Concepts of Design Assurance for Neural Networks	53
COG	Cloud-Optimized GeoTIFF	13, 45
COMINT	communications intelligence	
CONFERS	Consortium for Execution of Rendezvous and Servicing Operations	
COPUOS	Committee on the Peaceful Uses of Outer Space	21, 41
COPV	composite-overwrapped pressure vessel	34
CORS	continuously operating reference stations	18
COSPAR	Committee on Space Research	
COSTAR	Corrective Optics Space Telescope Axial Replacement	
COTS	commercial off-the-shelf	2
CPU	central processing unit	
CRPA	controlled-reception-pattern antenna	18
CRPS	continuous ranked probability score	
CSA	Canadian Space Agency	32
CSBF	Columbia Scientific Balloon Facility	8
CSDA	Commercial SmallSat Data Acquisition	13
CSI	critical success index	47
CSpO	Combined Space Operations	19
CT	computed tomography	52
CUSUM	cumulative sum	50
CVCM	collected volatile condensable material	2
CW	Clohessy–Wiltshire	
CZT	cadmium zinc telluride	38

G.4 D

Acronym	Meaning	Chapters
D2C	direct-to-cell	
DA-ASAT	direct-ascent anti-satellite (weapon)	
DAGGER	Deep leArninG Geomagnetic pErtuRbation (NASA geomagnetic-disturbance model)	
DAL	design assurance levels	53
DARPA	Defense Advanced Research Projects Agency	50
DART	Double Asteroid Redirection Test	26
DDD	displacement damage dose	2
DEM	digital elevation model	13, 46
DESI	Dark Energy Spectroscopic Instrument	
DET	direct energy transfer	5
DFMC	dual-frequency multi-constellation	18
DGMR	Deep Generative Model of Rain (nowcasting)	
DGNSS	Differential GNSS	18
DIAL	differential absorption lidar	12
DInSAR	Differential InSAR	15
DLR	Deutsches Zentrum für Luft- und Raumfahrt (German Aerospace Center)	
DMIPS	Dhrystone million instructions per second	
DMSP	Defense Meteorological Satellite Program	19
DNB	Day/Night Band	14
DO-178C	Software Considerations in Airborne Systems and Equipment Certification (RTCA)	
DoD	depth of discharge	5
DOP	dilution of precision	18
DR	Decommissioning Review	4
DRACO	Demonstration Rocket for Agile Cislunar Operations	
DRO	distant retrograde orbit	3, 24
DSCOVR	Deep Space Climate Observatory	
DSI	Deep Space Industries	41
DSM	digital surface model	14
DSN	Deep Space Network	7, 24, 49
DSOC	Deep Space Optical Communications	
DSP	Defense Support Program	20
DTN	delay- and disruption-tolerant networking	
DVB-S2	Digital Video Broadcasting – Satellite, second generation	
DVB-S2X	DVB-S2 extensions	

G.5 E

Acronym	Meaning	Chapters
EASA	European Union Aviation Safety Agency	
ECC	error-correcting codes	45

Acronym	Meaning	Chapters
ECEF	Earth-centred, Earth-fixed	3
ECI	earth-centred inertial	3
ECLSS	environmental control and life support systems	30
ECMWF	European Centre for Medium-Range Weather Forecasts	16, 47, 53
ECOSTRESS	ECOsystem Spaceborne Thermal Radiometer Experiment on Space Station	
ECSS	European Cooperation for Space Standardization	
ECV	Essential Climate Variable	13
ED-324	EUROCAE guidance on machine learning in aeronautical systems (with SAE ARP6983)	
EDAC	error detection and correction	2, 6
EDL	entry, descent and landing	27
EDRS	European Data Relay System	14, 17
EEE	electrical, electronic and electromechanical	4
EESS	Earth exploration-satellite service	
EGMS	European Ground Motion Service	15
EGNOS	European Geostationary Navigation Overlay Service	
EHF	extremely high frequency (30–300 GHz)	
EIRP	effective isotropic radiated power	7
ELA	Ensemble de Lancement Ariane (Ariane launch complex)	
ELINT	electronic intelligence	
EM	engineering model	4
EMC	electromagnetic compatibility	
EMIT	Earth Surface Mineral Dust Source Investigation	
EMU	Extravehicular Mobility Unit	33
ENL	equivalent number of looks (SAR speckle)	
ENS	ensemble	47
EO	earth observation	1, 13, 14, 46
EO-1	Earth Observing-1	
EO/IR	electro-optical / infrared	
EOL	end of life	5
EOSDIS	Earth Observing System Data and Information System	45
EP	electric propulsion	6
EPFD	equivalent power-flux-density	11, 17, D
EPS	electrical power subsystem	5
ERA5	ECMWF Reanalysis v5	
ESA	European Space Agency	32
ESA-ADB	ESA's anomaly-detection benchmark	49
ESCAPADE	Escape and Plasma Acceleration and Dynamics Explorers	
ESOC	European Space Operations Centre	49
ESTRACK	ESA Tracking Station Network	
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites	
EUSPA	EU Agency for the Space Programme	
EUV	extreme ultraviolet	

Acronym	Meaning	Chapters
EVA	extravehicular activity	
EVE	Earth Virtual Expert	44
EVI	Enhanced Vegetation Index	13

G.6 F

Acronym	Meaning	Chapters
FAA	Federal Aviation Administration	37
FAI	Fédération Aéronautique Internationale	1
FCC	Federal Communications Commission	1, D
FCI	Flexible Combined Imager	16
FCLS	fully constrained least squares	51
FDIR	fault detection, isolation and recovery	6, 48
FDL	Frontier Development Lab	45
FDOA	frequency difference of arrival	19
FEC	forward error correction	7
FEEP	field-emission electric propulsion	42
FEM	finite-element method	5
FFSC	full-flow staged combustion	35
FGS	Fine Guidance Sensor	38
FGSM	fast gradient sign method	53
FIRMS	Fire Information for Resource Management System	14
FISINT	foreign instrumentation signals intelligence	
FLOP	floating-point operations	44
FM	flight model	4
FMECA	failure modes, effects and criticality analysis	4
FN	false negative	43
FNO	fourier neural operator	44
FP	false positive	43
FPGA	field-programmable gate array	45
FRB	fast radio burst	51
FRP	fire radiative power	14
FRR	Flight Readiness Review	4
FSS	Fixed Satellite Service	17
FSW	flight software	
FTS	fourier-transform spectrometers	12

G.7 G

Acronym	Meaning	Chapters
G-FOLD	Guidance for Fuel-Optimal Large Diverts	
GAN	generative adversarial network	44

Acronym	Meaning	Chapters
GBAS	ground-based augmentation systems	18
GCAT	General Catalog of Artificial Space Objects (J. McDowell)	
GCOS	Global Climate Observing System	13
GCP	ground control point	13
GCR	galactic cosmic ray	2, 30
GCRF	Geocentric Celestial Reference Frame	3
GEDI	Global Ecosystem Dynamics Investigation	
GEM	Graphite Epoxy Motor (solid booster)	
GEO	Geostationary	1, 3, 16, 17
GEODSS	Ground-based Electro-Optical Deep Space Surveillance	21
GEVS	General Environmental Verification Standard	2
GFLOP	giga (10^9) floating-point operations	
GFM	Global Flood Monitoring	46
GFS	Global Forecast System (NOAA)	
GIC	geomagnetically induced current	22
GIQE	General Image Quality Equation	13
GLM	Geostationary Lightning Mapper	16
GLONASS	Globalnaya Navigatsionnaya Sputnikovaya Sistema (Russian GNSS)	
GNC	guidance, navigation and control	1, 6
GNN	graph neural network	44
GNSS	global navigation satellite systems	18
GNSS-R	GNSS reflectometry	12
GOCE	Gravity field and steady-state Ocean Circulation Explorer	
GOES	Geostationary Operational Environmental Satellite	
GOSAT	Greenhouse gases Observing Satellite (Japan)	
GP	gaussian process	43, 52
GPM	Global Precipitation Measurement	16
GPS	Global Positioning System	
GPU	graphics processing unit	
GRACE	Gravity Recovery and Climate Experiment	
GRACE-FO	GRACE Follow-On	
GRAIL	Gravity Recovery and Interior Laboratory	
GRD	Ground Range Detected	46
GRU	gated recurrent unit	44
GSaaS	Ground-station-as-a-service	49
GSD	ground sampling distance	11, 12, 13
GSLV	Geosynchronous Satellite Launch Vehicle	
GSN	Goal Structuring Notation	53
GSSAP	Geosynchronous Space Situational Awareness Program	19, 21
GTO	geostationary transfer orbit	34, 36

G.8 H

Acronym	Meaning	Chapters
HALEU	high-assay low-enriched uranium	42
HALO	Habitation and Logistics Outpost	32
HAND	height above nearest drainage	46
HAPS	high-altitude platform stations	1, 8
HAS	High Accuracy Service	18
HASDM	High Accuracy Satellite Drag Model	50
HBTSS	Hypersonic and Ballistic Tracking Space Sensor	20
HDRM	hold-down and release mechanism	5
HEK	Heliophysics Event Knowledgebase	51
HEO	highly elliptical orbits	1, 17, 39
HGA	high-gain antenna	7
HGEFS	Hybrid Global Ensemble Forecast System (NOAA)	
HGV	hypersonic glide vehicle	20
HiRISE	High Resolution Imaging Science Experiment	25
HLS	Harmonized Landsat and Sentinel-2	14
HMI	Helioseismic and Magnetic Imager	47
HPE	Hewlett Packard Enterprise	
HPSC	High-Performance Spaceflight Computing	6, 45
HRES	ECMWF high-resolution deterministic forecast	
HSS	heidke skill score	47
HTPB	hydroxyl-terminated polybutadiene (solid-propellant binder)	
HTS	high-throughput satellites	17
HTV-X	H-II Transfer Vehicle X	
HWO	Habitable Worlds Observatory	38, 39

G.9 I

Acronym	Meaning	Chapters
IADC	Inter-Agency Space Debris Coordination Committee	
IASI	Infrared Atmospheric Sounding Interferometer	
IAU	International Astronomical Union	
IBP	interval bound propagation	53
ICAO	International Civil Aviation Organization	
ICBM	intercontinental ballistic missile	
IDA	International Docking Adapter	32
IDSS	International Docking System Standard	31
IEEE	Institute of Electrical and Electronics Engineers	
IGRF	International Geomagnetic Reference Field	23
IGS	Information Gathering Satellites	19
IKAROS	Interplanetary Kite-craft Accelerated by Radiation Of the Sun	

Acronym	Meaning	Chapters
ILLUMA-T	Integrated LCRD Low Earth Orbit User Modem and Amplifier Terminal	
ILRS	International Lunar Research Station	1
IMAP	Interstellar Mapping and Acceleration Probe	
IME	integrated mass enhancement	46
IMS	Inductive Monitoring System	49
IMU	inertial measurement unit	34
INCO	Instrumentation and Communications Officer (Mission Control)	
INCOSE	International Council on Systems Engineering	4
IOS	in-orbit servicing	40
IoT	Internet of Things	17
IPCC	Intergovernmental Panel on Climate Change	16
IPO	initial public offering	52
ISAM	in-space servicing, assembly and manufacturing	40
ISLE	in-suit light exercise	33
ISO	International Organization for Standardization	
ISO/IEC	International Organization for Standardization / International Electrotechnical Commission	
ISR	intelligence, surveillance and reconnaissance	1, 8, 19
ISRO	Indian Space Research Organisation	
ISRU	in situ resource utilisation	
ISS	International Space Station	30, 31, 32, 33 …
ITRF	International Terrestrial Reference Frame	3
ITU	International Telecommunication Union	7, 17, D
ITU-R	ITU Radiocommunication Sector	
IV&V	independent verification and validation	4, 6, 53
IW	Interferometric Wide-swath	15
IXPE	Imaging X-ray Polarimetry Explorer	

G.10 J

Acronym	Meaning	Chapters
J2000	Standard astronomical epoch, 12:00 TT on 1 January 2000	
JAXA	Japan Aerospace Exploration Agency	
JB2008	Jacchia–Bowman 2008 atmospheric density model	
JD	Julian Date	A
JPL	Jet Propulsion Laboratory (NASA/Caltech)	
JPSS	Joint Polar Satellite System	
JUICE	JUpiter ICy moons Explorer	
JWST	James Webb Space Telescope	38, 51

G.11 K

Acronym	Meaning	Chapters
KARI	Korea Aerospace Research Institute	
KaRIn	ka-band radar interferometer	16

G.12 L

Acronym	Meaning	Chapters
L/D	lift-to-drag ratio	
LADEE	Lunar Atmosphere and Dust Environment Explorer	
LC	Launch Complex	
LCL	latching current limiter	5
LCRD	Laser Communications Relay Demonstration	17
LCROSS	Lunar Crater Observation and Sensing Satellite	
LCVG	liquid cooling and ventilation garment	33
LDB	long-duration balloon	8
LDPC	low-density parity-check	7
LDSD	Low-Density Supersonic Decelerator	8
LEO	low Earth orbit	1, 3, 16, 17 …
LEOP	launch and early orbit phase	11
LET	linear energy transfer	2
LGA	low-gain antenna	7
LI	Lightning Imager	16
LIGO	Laser Interferometer Gravitational-Wave Observatory	
LISA	Laser Interferometer Space Antenna	39
LLM	large language model	44, 49
LLO	low lunar orbit	41
LM	Lunar Module	27, 31
LMI	Lightning Mapping Imager	16
LOC	loss of crew	31
LOLA	Lunar Orbiter Laser Altimeter	12
LORRI	Long Range Reconnaissance Imager (New Horizons)	
LOX	liquid oxygen	37
LQR	linear-quadratic regulators	6
LRO	Lunar Reconnaissance Orbiter	25, 41
LRR	Launch Readiness Review	4
LRV	Lunar Roving Vehicle	28, 33
LSST	Legacy Survey of Space and Time	51
LST	land-surface temperature	14
LSTM	long short-term memory	44, 49
LTE	Long-Term Evolution (4G mobile standard)	
LuGRE	Lunar GNSS Receiver Experiment	18

Acronym	Meaning	Chapters
LVM3	Launch Vehicle Mark 3	
LVS	Lander Vision System	48
LWIR	long-wave infrared	

G.13 M

Acronym	Meaning	Chapters
MAE	mean absolute error	43
MARS	Methane Alert and Response System	46
MASINT	measurement and signature intelligence	19
MAST	Mikulski Archive for Space Telescopes	38
MAVEN	Mars Atmosphere and Volatile Evolution	
MBSE	model-based systems engineering	4
MC/DC	modified condition/decision coverage	
MCR	Mission Concept Review	4
MCS	mission control system	49
MDO	multidisciplinary design optimisation	52
MECO	main-engine cut-off	34
MEMS	microelectromechanical systems	
MEO	Medium Earth orbit	1, 3, 17
MEP	Mission Extension Pod	40
MESSENGER	MErcury Surface, Space ENvironment, GEochemistry and Ranging	
METOX	metal oxide	33
MEV	Mission Extension Vehicle	40
MILP	mixed-integer linear programming	48, 49, 53
MIPS	million instructions per second	48
MIRI	Mid-Infrared Instrument (JWST)	
MISSE	Materials International Space Station Experiment	2
ML	machine learning	43, 50, 51
MLI	multi-layer insulation	5
MMH	monomethylhydrazine	
MMOD	micrometeoroids and orbital debris	
MMRTG	multi-mission radioisotope thermoelectric generator	28
MMS	Magnetospheric Multiscale	
MMU	Manned Manoeuvring Unit	33
MMX	Martian Moons eXploration	
MOC	mission operations centre	7
MODIS	Moderate Resolution Imaging Spectroradiometer	
MOLA	Mars Orbiter Laser Altimeter	12
MOXIE	Mars Oxygen In-Situ Resource Utilization Experiment	
MPC	model predictive control	6
MPD	magnetoplasmadynamic (thruster)	

Acronym	Meaning	Chapters
MPE	maximum predicted environment	
MRE	molten regolith electrolysis	41
MRL	Manufacturing Readiness Levels	4
MRO	Mars Reconnaissance Orbiter	
MRV	Mission Robotic Vehicle	40
MSI	MultiSpectral Instrument (Sentinel-2)	
MSR	Mars Sample Return	29
MSS	Mobile Satellite Service	17
MTF	modulation transfer function	12, 13
MTG	Meteosat Third Generation	
MTTF	mean time to failure	
MUOS	Mobile User Objective System	
MWIR	mid-wave infrared	

G.14 N

Acronym	Meaning	Chapters
NAC	Narrow Angle Camera	51
NASA	National Aeronautics and Space Administration (USA)	
NBL	Neutral Buoyancy Laboratory	33
NBR	Normalised Burn Ratio	13
NDBI	Normalised Difference Built-up Index	13
NDMI	Normalised Difference Moisture Index	13
NDSI	Normalised Difference Snow Index	13
NDVI	Normalised Difference Vegetation Index	13
NDWI	Normalised Difference Water Index	13
NEA	near-Earth asteroid	41
NEO	near-Earth object	26
NEP	nuclear electric propulsion	42
NERVA	Nuclear Engine for Rocket Vehicle Application	
NESZ	noise-equivalent sigma-zero	15
NET	no earlier than	
NGA	National Geospatial-Intelligence Agency	50
NIIRS	National Imagery Interpretability Rating Scale	13
NIR	near-infrared	
NISAR	NASA–ISRO Synthetic Aperture Radar	
NIST	National Institute of Standards and Technology (USA)	53
NMF	non-negative matrix factorisation	51
NOAA	National Oceanic and Atmospheric Administration (USA)	47
NPA	Notice of Proposed Amendment	53
NPP	(Suomi) National Polar-orbiting Partnership	
NPR	NASA Procedural Requirements	I

Acronym	Meaning	Chapters
NPSH	net positive suction head	35
NRHO	near-rectilinear halo orbit	3, 24, 40
NRLMSIS	Naval Research Laboratory Mass Spectrometer and Incoherent Scatter radar model	
NRO	National Reconnaissance Office	19, 50
NSSL	National Security Space Launch	19
NTN	non-terrestrial networks	7, 17, 49
NTO	nitrogen tetroxide	
NTR	nuclear thermal rocket	42
NWP	numerical weather prediction	16, 47

G.15 O

Acronym	Meaning	Chapters
O/F	oxidiser-to-fuel mixture ratio	
OBC	on-board computer	
OBDH	on-board data handling	6
OBIA	object-based image analysis	13
OCI	ocean-colour instrument	16
OCO-2	Orbiting Carbon Observatory-2	
OD	orbit determination	3
ODD	operational design domain	53
OLCI	Ocean and Land Colour Instrument (Sentinel-3)	
OLI	Operational Land Imager (Landsat)	
OMM	Orbit Mean-elements Message	3
OPIR	Overhead Persistent Infrared	
OPS-SAT	ESA's flying laboratory satellite for operations experiments (2019–24)	
ORR	Operational Readiness Review	4
OSAM	on-orbit servicing, assembly and manufacturing	
OSIRIS-REx	Origins, Spectral Interpretation, Resource Identification, Security–Regolith Explorer	
OSNMA	Open Service Navigation Message Authentication	18
OTV	orbital transfer vehicle	11

G.16 P

Acronym	Meaning	Chapters
PA	product assurance	4
PACE	Plankton, Aerosol, Cloud, ocean Ecosystem	
PCA	principal component analysis	43
PCEC	Project Cost Estimating Capability	4
PDGS	payload data ground segment	7
PDOP	position dilution of precision	
PDR	Preliminary Design Review	4

Acronym	Meaning	Chapters
PDU	power distribution unit	5
PGD	projected gradient descent	53
PHM	prognostics and health management	49
PICA	phenolic impregnated carbon ablator	
PINN	physics-informed neural network	44
PIXL	Planetary Instrument for X-ray Lithochemistry (Perseverance)	
PLAR	Post-Launch Assessment Review	4
PLATO	PLAnetary Transits and Oscillations of stars	
PLSS	portable life support system	33
PMD	propellant management device	6
PNP	probability of no penetration	2
PnP	Perspective-n-Point	48
PNT	positioning, navigation and timing	1, 18
POD	probability-of-detection	52
POGO	longitudinal vibration from coupling of structure and propulsion (not an acronym)	
PPE	Power and Propulsion Element	32, 42
PPP	precise point positioning	18
PRA	probabilistic risk assessment	31
PRF	pulse repetition frequency	15
PRN	pseudo-random noise	18
PSI	persistent scatterer interferometry	15, 45
PSLV	Polar Satellite Launch Vehicle	
PSR	permanently shadowed region	41
PTQ	post-training quantisation	45
PUE	power usage effectiveness	53
PUNCH	Polarimeter to Unify the Corona and Heliosphere	
PWSA	Proliferated Warfighter Space Architecture	19

G.17 Q

Acronym	Meaning	Chapters
QAT	quantisation-aware training	45
QE	quantum efficiency	38
QM	qualification model	4
QPSK	quadrature phase-shift keying	
QWIP	quantum-well infrared photodetector	14
QZSS	Quasi-Zenith Satellite System (Japan)	

G.18 R

Acronym	Meaning	Chapters
R-CNN	region-based convolutional neural network	

Acronym	Meaning	Chapters
RAAN	right ascension of the ascending node	3
RAG	retrieval-augmented generation	44, 49, 52
RANSAC	random sample consensus	
RCA	rapid cycle amine	33
RCC	reinforced carbon-carbon	10
RDE	rotating detonation engine	42
RDM	radiation design margin	2
RDRE	rotating detonation rocket engine	42
REE	rare-earth elements	41
ReLU	rectified linear unit	44
RF	radio-frequency	1
RHU	radioisotope heater unit	5, 28
RL	reinforcement learning	43, 48
RLHF	reinforcement learning from human feedback	44
RLV	reusable launch vehicle	
RMSE	root-mean-square error	
RNN	recurrent neural network	44
RO	radio occultation	12, 16
ROC	receiver operating characteristic	43, 46
RORSAT	radar ocean-reconnaissance satellite	42
ROS	Russian Orbital Station	32
ROSA	Roll-Out Solar Array	5
ROSES	Research Opportunities in Space and Earth Sciences	9
RP-1	Rocket Propellant-1 (refined kerosene)	
RPC	rational polynomial coefficient	13
RPO	rendezvous and proximity operations	40
RPS	radioisotope power systems	5
RRM	Robotic Refuelling Mission	40
RSGS	Robotic Servicing of Geosynchronous Satellites	40
RTG	radioisotope thermoelectric generator	5, 24, 26
RTK	real-time kinematic	18
RTLS	return to launch site	34
RUL	remaining useful life	49

G.19 S

Acronym	Meaning	Chapters
SAA	South Atlantic Anomaly	2
SADA	solar array drive assembly	5
SAFER	Simplified Aid For EVA Rescue	33
SANS	spaceflight-associated neuro-ocular syndrome	30
SAR	synthetic aperture radar	12, 15

Acronym	Meaning	Chapters
SATCAT	Satellite catalogue	
SATCOM	satellite communications	19
SBAS	small baseline subset	15, 18
SBI	simulation-based inference	51
SBIRS	Space-Based Infrared System	
SDA	space domain awareness	11, 19, 21
SDC	silent data corruption	45
SDO	Solar Dynamics Observatory	47, 51
SEC-DED	single-error correction, double-error detection	
SECO	second-engine cut-off	34
SEE	single-event effect	2
SEFI	single-event functional interrupt	2, 45
SEL	single-event latch-up	2, 45
SEP	solar energetic particle	22
SEU	single-event upset	2, 45
SGD	stochastic gradient descent	43, 44
SGP4	Simplified General Perturbations model 4	
SIGINT	signals intelligence	19
SIR	System Integration Review	4
SLC	Space Launch Complex	
SLIM	Smart Lander for Investigating Moon	
SLS	Space Launch System	34, 36
SLSTR	Sea and Land Surface Temperature Radiometer (Sentinel-3)	
SMAP	Soil Moisture Active Passive	
SMILE	Solar wind Magnetosphere Ionosphere Link Explorer	
SMOS	Soil Moisture and Ocean Salinity	
SMT	satisfiability-modulo-theories	53
SNR	signal-to-noise ratio	12, 13, 38, 49
SOHO	Solar and Heliospheric Observatory	
SOI	sphere of influence	3
SOLAR-1	NOAA space-weather observatory at the Sun–Earth L1 point (formerly SWFO-L1)	
SPB	super-pressure balloon	8
SPE	solar particle event	2, 30
SPF	single points of failure	4
SPICE	Spacecraft, Planet, Instrument, C-matrix, Events (NAIF toolkit)	
SPIKE	Hubble Space Telescope long-range scheduling system	
SR-1	Space Reactor-1	42
SRP	solar radiation pressure	3
SRR	System Requirements Review	4
SRS	shock response spectrum	2
SRTM	Shuttle Radar Topography Mission	15
SSA	space situational awareness	21, 50

Acronym	Meaning	Chapters
SSLV	Small Satellite Launch Vehicle	
SSME	Space Shuttle main engine	52
SSO	sun-synchronous orbits	3
SST	sea-surface temperature	16, 21
SSTO	Single-stage-to-orbit	10
ST5	Space Technology 5	52
STIS	Space Telescope Imaging Spectrograph (Hubble)	
STM	structural-thermal model	4
STScI	Space Telescope Science Institute	39
SVM	support vector machine	43
SVOM	Space Variable Objects Monitor	
SWFO-L1	Space Weather Follow On–Lagrange 1	
SWIR	short-wave infrared	
SWME	spacesuit water membrane evaporator	33
SWOT	Surface Water and Ocean Topography (mission)	
SWPC	Space Weather Prediction Center	47

G.20 T

Acronym	Meaning	Chapters
T/W	thrust-to-weight ratio	
TAEM	Terminal Area Energy Management	10
TAG	touch-and-go	29
TAI	International Atomic Time	
TBIRD	TeraByte InfraRed Delivery	
TCE	threshold-crossing events”	51
TCN	temporal convolutional network	44
TCS	thermal control subsystem	5
TDB	Barycentric Dynamical Time	
TDI	time-delay integration	12, 14
TDOA	time difference of arrival	19
TDRS	Tracking and Data Relay Satellites	32
TDRSS	Tracking and Data Relay Satellite System	17
TEC	total electron content	18, 47
TEMPO	Tropospheric Emissions: Monitoring of Pollution	
TESS	Transiting Exoplanet Survey Satellite	39
TGO	Trace Gas Orbiter (ExoMars)	
THEMIS	Time History of Events and Macroscale Interactions during Substorms	
TID	total ionising dose	2, 45
TIR	thermal infrared	
TIRS	Thermal Infrared Sensor (Landsat)	
TMA	three-mirror anastigmat	12, 38

Acronym	Meaning	Chapters
TMG	thermal micrometeoroid garment	33
TML	total mass loss	2
TMR	triple modular redundancy	2, 45
TN	true negative	43
TOA	top-of-atmosphere	13
TOMS	Total Ozone Mapping Spectrometer	16
TOPS	tera (10^{12}) operations per second	
TP	true positive	43
TPS	thermal protection system	10
TPU	tensor processing unit	45
TraCSS	Traffic Coordination System for Space	21
TRL	Technology Readiness Level	4, 42
TRMM	Tropical Rainfall Measuring Mission	16
TRN	terrain-relative navigation	27, 48
TROPOMI	Tropospheric Monitoring Instrument (Sentinel-5P)	
TSS	true skill statistic	
TT&C	telemetry, tracking and command	1, 5, D
TVC	thrust vector control	34
TWTA	travelling-wave-tube amplifier	17

G.21 U

Acronym	Meaning	Chapters
UAE	United Arab Emirates	
UDMH	unsymmetrical dimethylhydrazine	
UERE	user equivalent range error	18
UHF	ultra high frequency (300 MHz–3 GHz)	
ULA	United Launch Alliance	36
ULE	ultra-low expansion (glass)	
USAF	United States Air Force	
USDV	US Deorbit Vehicle	32
USGS	United States Geological Survey	
USSR	Union of Soviet Socialist Republics	
UT1	Universal Time based on the Earth's rotation	
UTC	Coordinated Universal Time	
UTM	Universal Transverse Mercator	13

G.22 V

Acronym	Meaning	Chapters
VAE	variational autoencoder	
VASIMR	Variable Specific Impulse Magnetoplasma Rocket	42

Acronym	Meaning	Chapters
VCM	variable coding and modulation	7
VHF	very high frequency (30–300 MHz)	
VHR	very high resolution	14
VIIRS	Visible Infrared Imaging Radiometer Suite	14
VIPER	Volatiles Investigating Polar Exploration Rover	
ViT	vision transformer	44
VLEO	very low Earth orbit	2, 14
VNIR	visible and near-infrared	
VSAT	very small aperture terminal	

G.23 W

Acronym	Meaning	Chapters
WAAS	Wide Area Augmentation System	
WGS	Wideband Global SATCOM; also World Geodetic System	
WISE	Wide-field Infrared Survey Explorer	
WMAP	Wilkinson Microwave Anisotropy Probe	
WMO	World Meteorological Organization	16
WRC	World Radiocommunication Conference	D

G.24 X

Acronym	Meaning	Chapters
XMM	X-ray Multi-Mirror (XMM-Newton)	
XRISM	X-Ray Imaging and Spectroscopy Mission	

G.25 Y

Acronym	Meaning	Chapters
YAG	yttrium aluminium garnet (laser crystal)	
YOLO	“You Only Look Once” (single-stage object detector)	

G.26 Z

Acronym	Meaning	Chapters
ZBLAN	heavy-metal fluoride glass (ZrF_4–BaF_2–LaF_3–AlF_3–NaF)	

APPENDIX

H Glossary

Plain-language definitions of about 300 technical terms used in this manual, from ablator to zenith, covering orbital mechanics, spacecraft engineering, launch, remote sensing, communications, the space environment, exploration, human spaceflight and artificial intelligence, with pointers to the chapters where each is explained.

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Terms in *italics* within a definition have their own entry. The number in brackets is the chapter where the term is explained most fully; letters refer to appendices. Acronyms are listed in Appendix G.

H.1 A

Ablator. A heat-shield material that protects a vehicle during atmospheric entry by charring, melting or vaporising, carrying heat away with the lost material. Examples are PICA and Avcoat (10, 27).

Absorptance (α). The fraction of incident radiation, usually sunlight, that a surface absorbs. With *emittance* it sets a surface's equilibrium temperature (5).

Active debris removal. Deliberately capturing and de-orbiting or re-orbiting a defunct satellite or rocket body with another spacecraft (21, 40).

Adaptive coding and modulation. Changing a link's modulation and error-correcting code from moment to moment to match the received signal quality, so throughput stays high in clear weather and the link survives rain fades (7).

Aerobraking. Using repeated passes through the upper atmosphere of a planet to lower an orbit gradually, saving propellant (25).

Aerocapture. Using a single deep pass through an atmosphere to slow a spacecraft from a hyperbolic approach into orbit (25).

Airy disc. The bright central spot of the diffraction pattern produced by a circular aperture; its angular radius, $1.22\,\lambda/D$, sets the resolution limit of a telescope (14, 38).

Albedo. The fraction of incoming sunlight that a body or surface reflects. Earth's average is about 0.3 (2, 13).

Analysis-ready data. Satellite data already corrected and formatted (for example to surface reflectance on a standard grid) so that users can analyse it without further processing (13, 46).

Anomaly. (1) In orbital mechanics, an angle giving a body's position along its orbit — true, eccentric or mean anomaly (3). (2) In operations, any unexpected behaviour of a spacecraft (49).

Antenna gain. How strongly an antenna concentrates power in its main direction compared with an isotropic antenna, usually in dBi (7).

Aperture synthesis. Combining signals from separated antennas or positions to obtain the resolution of a much larger aperture — the principle behind *synthetic aperture radar* and radio interferometers (15).

Apoapsis / periapsis. The farthest and nearest points of an orbit from the central body; called apogee and perigee for Earth, aphelion and perihelion for the Sun (3).

Argument of periapsis (ω). The angle, in the orbital plane, from the *ascending node* to periapsis (3).

Artemis Accords. Non-binding principles for civil exploration of the Moon and beyond, led by the United States; 76 nations had signed by late September 2026 (41).

Artificial intelligence (AI). Computer systems that perform tasks normally requiring human intelligence, such as perception, planning or language. Most current space applications use *machine learning* (43).

Ascending node. The point where an orbit crosses the equatorial plane going north. Its longitude is the *right ascension of the ascending node* (3).

Astronomical unit (au). The mean Earth–Sun distance, fixed at exactly 149 597 870 700 m (A).

Atomic oxygen. Single oxygen atoms, the main constituent of the atmosphere at 200–600 km, which erode polymers and some coatings on spacecraft (2).

Attitude. The orientation of a spacecraft in space. The attitude determination and control subsystem measures and controls it (6).

Autonomy. The ability of a spacecraft or ground system to make decisions and act without human intervention. ECSS defines levels E1 (ground control) to E4 (goal-oriented on-board operations) (48).

H.2 B

Ballistic coefficient (β). Mass divided by the product of drag coefficient and reference area, $m/(C_D A)$. A high ballistic coefficient means the vehicle slows down less in the atmosphere (10, 21).

Ballistic missile. A missile that is powered only during the first part of its flight and then follows a free-fall trajectory, often reaching space (20).

Bandwidth. The range of frequencies a signal occupies or a system can pass, in hertz. It limits data rate and radar range resolution (7, 15).

Barycentre. The centre of mass of two or more bodies, around which they both orbit (3).

Beamforming. Steering and shaping an antenna's beam electronically by adjusting the phase and amplitude of many elements of a *phased array* (17).

Bipropellant. A rocket propellant system using separate fuel and oxidiser, such as hydrazine and nitrogen tetroxide or methane and liquid oxygen (6, 35).

Blackbody. An ideal object that absorbs all radiation falling on it and emits radiation according to *Planck's law* (13).

Body-mounted array. Solar cells fixed to a spacecraft's outer surfaces rather than on deployable wings (5).

Boil-off. The loss of cryogenic propellant by evaporation due to heat leaking into a tank (35, 40).

Bond albedo. The fraction of total incident solar power reflected by a planet in all directions and at all wavelengths (C).

Bus (satellite bus). The part of a satellite that provides structure, power, thermal control, attitude control, propulsion and communications to support the *payload* (12).

H.3 C

C3 (characteristic energy). Twice the specific orbital energy of a departure hyperbola, equal to the square of the *hyperbolic excess speed*; a launch vehicle's capability is quoted as payload mass versus C3 (24).

Cadence (revisit). How often a satellite or constellation can observe the same place (11, 14).

Capsule. A blunt, wingless re-entry vehicle, such as Soyuz, Dragon or Orion (31).

Catalogue (space object catalogue). The database of tracked objects in orbit with their orbital elements, maintained by the US Space Force and others (21).

Characteristic velocity (c*). A measure of combustion performance in a rocket chamber: chamber pressure times throat area divided by mass flow rate (35).

Chemical propulsion. Rocket propulsion in which energy comes from chemical reaction of the propellants (35).

Clean room. A room in which airborne particles are controlled to protect sensitive hardware, classified by particle count (ISO class) (4).

Coherent. Describing waves that keep a fixed phase relationship. Radar and interferometric processing depend on coherent signals (15).

Collision probability (Pc). The estimated probability that two objects will collide during a close approach, used to decide whether to manoeuvre (21).

Commissioning. The period after launch in which a spacecraft's systems and instruments are switched on, checked and calibrated before routine operations (4).

Conjunction. A predicted close approach between two space objects (21).

Constellation. A group of satellites working together, arranged in planes and phases to provide coverage (11).

Convolutional neural network (CNN). A *neural network* that applies learned filters across an image; the standard tool for image classification and detection (44).

Coronagraph. An instrument that blocks the light of a star inside a telescope so faint nearby objects, such as planets, can be seen (38).

Coronal mass ejection (CME). A huge cloud of magnetised plasma thrown out from the Sun, which can cause a *geomagnetic storm* when it reaches Earth (22).

Cosmic rays. Very energetic charged particles, mostly protons and heavier nuclei, from outside the Solar System (*galactic cosmic rays*) or from the Sun (2, 22).

Cross-range. How far sideways from its original ground track a re-entry vehicle can steer, which depends on its *lift-to-drag ratio* (10).

Cryocooler. A mechanical refrigerator that cools detectors or instruments to cryogenic temperatures, as on JWST's MIRI instrument (38).

Cryogenic propellant. A propellant stored as a very cold liquid — liquid hydrogen (20 K), liquid oxygen (90 K) or liquid methane (112 K) (35).

CubeSat. A standard small satellite built from 10 cm units (1U); 3U, 6U and 12U are common sizes (11).

H.4 D

Data assimilation. Combining observations with a numerical model to produce the best estimate of the state of a system, such as the atmosphere for a weather forecast (16, 47).

Decibel (dB). Ten times the base-10 logarithm of a power ratio (A, D).

Deep learning. *Machine learning* with neural networks of many layers (44).

Deep Space Network. NASA's three ground stations, at Goldstone, Madrid and Canberra, about 120° apart in longitude, which communicate with deep-space missions (7, 24).

Delta-v (Δv). The change in velocity a manoeuvre or mission requires; the currency of mission design (3).

Depth of discharge. The fraction of a battery's capacity used in a discharge cycle; lower depth gives longer life (5).

Diffraction limit. The finest angular detail an optical system can resolve, set by the wavelength divided by the aperture diameter (14, 38).

Digital twin. A continuously updated computer model of a physical system used to predict its behaviour and test operations (49, 52).

Direct-to-device. Satellite service to unmodified mobile phones, using terrestrial mobile spectrum (17).

Dispersion (launch). The spread of possible trajectories or impact points around the nominal, used in range safety (37).

Distribution shift. A change between the data on which an AI model was trained and the data it meets in operation, often degrading performance (43, 53).

Docking / berthing. Joining two spacecraft. In docking the active vehicle flies into contact; in berthing a robotic arm moves one vehicle onto the other's port (31, 32).

Doppler shift. The change in received frequency caused by relative motion between transmitter and receiver, used for navigation and radar (7, 18).

Dose (radiation). The energy deposited by radiation per kilogram of material (gray); weighted by biological effect it becomes equivalent dose (sievert) (2, 30).

Downlink / uplink. Transmission from space to ground / from ground to space (7).

Drag. The aerodynamic force opposing motion through an atmosphere; in low orbit it slowly lowers satellites (2, 21).

H.5 E

Earth observation. Gathering information about the Earth's surface, oceans and atmosphere from space (13).

Eccentricity (e). A number describing the shape of an orbit: 0 for a circle, between 0 and 1 for an ellipse, 1 for a parabola and above 1 for a hyperbola (3).

Ecliptic. The plane of the Earth's orbit around the Sun (3).

EIRP (effective isotropic radiated power). The transmitter power multiplied by the antenna gain — the power an isotropic antenna would need to give the same signal in the main beam (7).

Electric propulsion. Propulsion that accelerates propellant electrically — ion, Hall-effect, arcjet or electrospray thrusters — giving high *specific impulse* at low thrust (6, 42).

Electrolysis. Splitting a compound with electricity, for example water into hydrogen and oxygen for propellant or breathing (30, 41).

Emittance (ε). How effectively a surface radiates heat compared with a *blackbody* (5).

Engineering model. A representative version of a spacecraft or unit built to verify design and test procedures, but not intended to fly (4).

Entry, descent and landing (EDL). The sequence that takes a spacecraft from the top of an atmosphere to the surface (27).

Ephemeris. A table or model of the positions of a body over time (3).

Equatorial bulge (J2). The flattening of the Earth, which makes orbital planes precess; described by the J_2 coefficient (3).

Escape velocity. The speed needed at a given distance to leave a body's gravity without further propulsion, $\sqrt{(2\mu/r)}$ (3).

Exhaust velocity (effective). The speed at which a rocket's exhaust leaves relative to the vehicle, equal to *specific impulse* × g_0 (35).

Exoplanet. A planet orbiting a star other than the Sun (39).

Extravehicular activity (EVA). Any activity by an astronaut outside a spacecraft — a spacewalk or moonwalk (33).

H.6 F

Fairing (payload fairing). The aerodynamic nose cone that protects payloads during ascent and is jettisoned above the dense atmosphere (34).

Fault detection, isolation and recovery (FDIR). The on-board logic that detects faults, identifies the failed unit and switches to a safe configuration or a redundant unit (6, 48).

Field of regard / field of view. The region a sensor can be pointed at / the region it sees at one instant (13).

Flyby. A mission that passes a target body without orbiting or landing (26).

Flight-path angle (γ). The angle between a vehicle's velocity and the local horizontal (3, 10).

Footprint. The area on the ground covered by a satellite's antenna beam or sensor (11, 17).

Formation flying. Two or more spacecraft keeping a precise relative position, for example for interferometry (11).

Foundation model. A large *machine-learning* model trained on broad data, usually with *self-supervised learning*, that can be adapted to many tasks (44, 46).

Free-space path loss. The spreading loss of a radio signal with distance between isotropic antennas (7, D).

Frozen orbit. An orbit whose eccentricity and argument of perigee are chosen so that they stay nearly constant despite perturbations (11).

H.7 G

Gain-to-noise temperature ratio (G/T). The figure of merit of a receiving system: antenna gain divided by system noise temperature (7).

Gateway (feeder link). A ground station that connects a communication satellite to terrestrial networks (17).

Geomagnetic storm. A major disturbance of the Earth's magnetosphere caused by solar-wind structures, measured by indices such as Kp and Dst (22).

Geostationary orbit (GEO). A circular orbit 35 786 km above the equator with a period of one *sidereal day*, in which a satellite appears fixed in the sky (11).

Geosynchronous orbit. Any orbit with a period of one sidereal day; geostationary orbit is the equatorial, circular case (11).

Gimbal. A pivoted mount that lets an engine or instrument rotate about one or two axes; engine gimbals steer rockets (34).

Gravity assist (swing-by). Using a close pass of a planet to change a spacecraft's speed and direction relative to the Sun without propellant (24).

Gravity gradient. The variation of gravity across a body, which produces torques and tidal forces (6).

Gravity loss. The part of a rocket's Δv spent fighting gravity during ascent rather than gaining speed (34).

Ground sample distance (GSD). The distance on the ground between the centres of adjacent pixels in an image (13, 14).

Ground segment. All the ground facilities — antennas, control centres, data processing — that operate a space system (7).

Ground track. The path on the Earth's surface directly beneath an orbiting satellite (3).

H.8 H

Halo orbit. A periodic three-dimensional orbit around a *Lagrange point*, used by JWST and many solar observatories (3, 38).

Hall-effect thruster. An *electric propulsion* device that traps electrons in a magnetic field to ionise and accelerate propellant, typically xenon or krypton (6, 42).

Heat pipe. A sealed tube containing a working fluid that moves heat by evaporation and condensation with very small temperature drop (5).

Heliopause. The boundary where the solar wind meets the interstellar medium, crossed by Voyager 1 in 2012 and Voyager 2 in 2018 (26).

High-altitude platform station (HAPS). An aircraft or airship that stays in the stratosphere, typically at about 20 km, for weeks or months (8).

Highly elliptical orbit (HEO). An orbit with high eccentricity, such as a Molniya or Tundra orbit, giving long dwell time over high latitudes (11, 20).

Hohmann transfer. The two-impulse transfer between coplanar circular orbits with the least Δv (3, 24).

Hydrazine. N_2H_4, a storable propellant used alone over a catalyst (monopropellant) or with an oxidiser (6).

Hyperbolic excess speed (v_∞). The speed a spacecraft has relative to a planet far outside its gravity well (24).

Hypergolic. Describing propellants that ignite spontaneously on contact, such as hydrazines with nitrogen tetroxide (35).

Hyperspectral imaging. Imaging in many (often hundreds of) narrow contiguous spectral bands, giving a spectrum for each pixel (13).

H.9 I

Inclination (i). The angle between an orbital plane and the reference plane, usually the Earth's equator (3).

Inertial measurement unit (IMU). A package of gyroscopes and accelerometers that measures rotation and acceleration for navigation (6, 34).

In situ resource utilisation (ISRU). Producing useful materials, such as oxygen, water or propellant, from local resources on the Moon, Mars or asteroids (41).

Integration and test. Assembling a spacecraft and verifying it by functional, environmental and end-to-end tests (4).

Interferometric SAR (InSAR). Comparing the phase of two radar images to measure topography or surface movement to millimetres (15).

Interplanetary trajectory. A path between planets, usually designed with *patched conics* and refined numerically (24).

Ionosphere. The ionised upper atmosphere, from about 60 km to more than 1 000 km, which reflects, delays and scatters radio waves (23).

Isotropic radiator. An ideal antenna radiating equally in all directions; the reference for *antenna gain* (7).

H.10 J–K

Jitter. Small, rapid unwanted motions of a pointing system, which blur images (6, 38).

Kármán line. The altitude of 100 km often used as the boundary of space (FAI); the US military and FAA use 50 miles (80 km) (1, 9).

Kepler's laws. Three laws of planetary motion: orbits are ellipses with the central body at one focus; a line from the central body sweeps equal areas in equal times; the square of the period is proportional to the cube of the semi-major axis (3).

Kessler syndrome. A cascade in which collisions between debris objects create more debris, making further collisions more likely (21).

Kick stage. A small upper stage that delivers payloads to their final orbits after separation from the main launch vehicle (34).

H.11 L

Lagrange point. One of five points in a two-body system where a small object can remain stationary relative to the two bodies (3, C).

Lander. A spacecraft that makes a controlled landing on a surface (27).

Laser communication (optical communication). Using modulated laser light instead of radio waves to send data, allowing higher rates with smaller terminals (7).

Latency. The delay between sending and receiving a signal or a response — about 0.25 s each way via GEO, about 25–40 ms round trip via low-orbit broadband (7, 17).

Launch window. The period during which a launch can reach the desired orbit or trajectory (24, 34).

Lift-to-drag ratio (L/D). The ratio of aerodynamic lift to drag; it sets how much a re-entry vehicle can steer and how gently it decelerates (10).

Light-time. The time light takes to travel between two points — 1.3 s to the Moon, 3–22 minutes to Mars (7, 24).

Link budget. An accounting of all gains and losses between a transmitter and receiver, used to show that a communication link closes with margin (7).

Liquid rocket engine. An engine burning liquid propellants fed from tanks by pressure or pumps (35).

Long-duration balloon. A scientific balloon that flies for weeks, typically circling Antarctica in summer (8).

Low Earth orbit (LEO). Orbits from about 160 km to 2 000 km altitude (11).

Lunar south pole. The region around the Moon's south pole, of interest because permanently shadowed craters may hold water ice (27, 41).

H.12 M

Machine learning (ML). Methods by which computers learn patterns from data instead of following explicitly programmed rules (43).

Magnetometer. An instrument that measures magnetic field strength and direction, used for attitude sensing and space science (6, 23).

Magnetorquer. An electromagnet that produces attitude control torques by interacting with the Earth's magnetic field (6).

Magnetosphere. The region around a planet dominated by its magnetic field, shaped by the solar wind into a long tail (23).

Main-belt asteroid. An asteroid orbiting between Mars and Jupiter (26, 41).

Margin. The difference between what a design can do and what is required, kept to cover uncertainty (4).

Mass fraction (propellant mass fraction). The fraction of a stage's mass that is propellant (34).

Max-Q. The point in ascent at which *dynamic pressure* is greatest (34).

Medium Earth orbit (MEO). Orbits between LEO and GEO, notably those of GNSS satellites at about 19 000–23 000 km (11, 18).

Micrometeoroid. A natural particle smaller than about 1 mm travelling at tens of km/s (2).

Microgravity. The condition of apparent weightlessness in free fall, with residual accelerations around a millionth of g (9, 30).

Mission control centre. The facility from which a mission is commanded and monitored (7).

Model compression. Reducing an AI model's size and computational cost — by *quantisation*, pruning or distillation — so that it can run on limited hardware such as spacecraft processors (45).

Molniya orbit. A highly elliptical, 12-hour orbit with 63.4° inclination, giving long dwell over high northern latitudes (11).

Monopropellant. A single propellant that decomposes over a catalyst to produce thrust, usually hydrazine (6).

Multi-layer insulation (MLI). A blanket of many thin, reflective layers separated by spacers, which greatly reduces radiative heat transfer (5).

Multispectral imaging. Imaging in several (typically 4–15) spectral bands (13).

H.13 N

Near-Earth object (NEO). An asteroid or comet whose orbit brings it within 1.3 au of the Sun (26).

Near-rectilinear halo orbit (NRHO). A highly elongated *halo orbit* around the Moon, planned for the Gateway station (32).

Neural network. A machine-learning model made of layers of simple connected units (“neurons”) whose connection weights are learned from data (43, 44).

Nodal precession (regression of the nodes). The slow rotation of an orbital plane about the Earth’s axis caused by the *equatorial bulge* (3).

Noise temperature. A way of expressing noise power as the temperature of an equivalent resistor (7).

Nowcasting. Forecasting weather for the next few hours, largely by extrapolating the latest observations (16, 47).

Nuclear thermal propulsion. Heating hydrogen in a nuclear reactor and expanding it through a nozzle, giving roughly twice the specific impulse of chemical rockets (42).

H.14 O

Oberth effect. A burn performed at high speed, deep in a gravity well, produces a greater change in orbital energy than the same burn at low speed (3, 24).

Occultation. The passing of one body in front of another; radio and GNSS signals passing through an atmosphere during occultation are used to measure its profile (16, 26).

On-board processing. Processing data on the spacecraft, for example to filter clouds or detect events, rather than downlinking raw data (12, 45).

Optical depth. A measure of how much a medium attenuates light; an optical depth of 1 transmits 37 % (13).

Orbit determination. Estimating a spacecraft’s orbit from tracking measurements (3, 21).

Orbital elements. Six numbers that define an orbit and a position on it, such as semi-major axis, eccentricity, inclination, right ascension of the ascending node, argument of periapsis and true anomaly (3).

Orbital period. The time to complete one orbit (3).

Outgassing. The release of gas from materials in vacuum, which can contaminate optics and cold surfaces (2, 38).

Overfitting. When a machine-learning model learns the details and noise of its training data so closely that it performs poorly on new data (43).

H.15 P

Parking orbit. A temporary orbit in which a spacecraft waits before a departure burn (24).

Patched conics. Approximating a trajectory as a sequence of two-body conic sections, each inside one body’s *sphere of influence* (24).

Payload. The part of a spacecraft or launch vehicle that performs the mission — instruments, transponders or passengers (12, 34).

Perturbation. Any force other than the central body’s point-mass gravity that alters an orbit — J_2, drag, third bodies, solar radiation pressure (3).

Phased array. An antenna made of many elements whose relative phases are controlled to steer the beam electronically (17).

Physics-informed machine learning. Machine-learning models constrained or guided by known physical laws (44).

Pixel. One element of a digital image or detector array (13).

Planck’s law. The equation giving the spectrum of radiation emitted by a *blackbody* at a given temperature (13, B).

Planetary protection. Measures to prevent biological contamination of other worlds by Earth organisms, and of Earth by returned samples (24, 29).

Plasma. An ionised gas of free electrons and ions; most of the matter in space is plasma (23).

Polar orbit. An orbit with inclination near 90°, passing over both poles (11).

Pose estimation. Determining the position and orientation of an object, such as a target spacecraft, from sensor data (48).

Precession. A slow rotation of the axis of a spinning body or of an orbital plane (3, 6).

Pressure-fed / pump-fed. Engine feed systems in which propellant is pushed into the chamber by tank pressure or by turbopumps (35).

Primary mirror. The main light-collecting mirror of a reflecting telescope (38).

Proximity operations. Manoeuvring close to another spacecraft for inspection, docking or servicing (40).

Push-broom scanner. An imager with a line of detectors across the ground track that builds an image as the satellite moves forward (13).

H.16 Q–R

Quantisation (AI). Representing a model's weights and activations with fewer bits, for example 8-bit integers instead of 32-bit floating point (45).

Quaternion. A four-number representation of attitude that avoids the singularities of Euler angles (6).

Radar cross-section (RCS). The effective area that an object presents to a radar, in m^2 or dBsm (15, 21).

Radiation belts (Van Allen belts). Regions of trapped energetic electrons and protons in the Earth's magnetosphere (2, 23).

Radiator. A surface that rejects waste heat to space by thermal radiation (5).

Radioisotope thermoelectric generator (RTG). A power source that converts heat from radioactive decay of plutonium-238 into electricity (5, 26).

Reaction wheel. A spinning flywheel whose speed is changed by a motor to exchange angular momentum with the spacecraft and control its attitude (6).

Re-entry. The return of a vehicle into the atmosphere from space (10).

Regolith. The layer of loose, broken rock and dust covering a planetary surface (27, 41).

Reinforcement learning. Machine learning in which an agent learns a policy by trial and error, guided by rewards (43, 48).

Relay satellite. A satellite that forwards communications between other spacecraft and the ground, such as TDRS or Queqiao (7).

Remote sensing. Acquiring information about an object without physical contact, usually by measuring electromagnetic radiation (13).

Rendezvous. Bringing two spacecraft to the same orbit and position at the same time (31, 40).

Resolution. The ability to distinguish detail: spatial, spectral, radiometric or temporal (13).

Retrograde orbit. An orbit in the opposite direction to the central body's rotation (inclination above 90°) (3).

Reusability. Recovering and reflying launch vehicle stages, fairings or spacecraft (34, 36).

Revisit time. The time between successive observations of the same location (11).

Right ascension of the ascending node (RAAN, Ω). The angle from the vernal equinox direction to the *ascending node*, measured in the equatorial plane (3).

Rover. A mobile robotic vehicle that explores a planetary surface (28).

H.17 S

Safe mode. A minimal, stable spacecraft configuration entered automatically after a fault, waiting for ground instructions (6, 48).

Sample return. A mission that collects material from another body and brings it to Earth (29).

Scale height. The altitude change over which atmospheric pressure or density falls by a factor of e (2.718) (2).

Scintillation. Rapid fluctuations of a signal's amplitude and phase caused by irregularities in the medium it passes through, such as the ionosphere (23).

Scramjet. A supersonic-combustion ramjet, an air-breathing engine for hypersonic flight (42).

Segmented mirror. A large mirror made of smaller segments controlled to act as one surface, as on JWST (38).

Self-supervised learning. Machine learning in which a model learns from unlabelled data by predicting hidden parts of its input (44).

Semi-major axis (a). Half the longest diameter of an ellipse; it determines an orbit's period and energy (3).

Sensor fusion. Combining data from different sensors to obtain better estimates than any single sensor (6, 48).

Servicing (on-orbit). Inspecting, repairing, refuelling or upgrading a spacecraft in orbit (40).

Sidereal day. The Earth's rotation period relative to the stars, 23 h 56 min 4.1 s (3, A).

Signal-to-noise ratio (SNR). The ratio of signal power to noise power (7, 13).

Signals intelligence (SIGINT). Intelligence obtained by intercepting communications and electronic emissions (19).

Single-event effect. A malfunction caused by one energetic particle striking a sensitive part of a microcircuit — an upset (bit flip), latch-up or burnout (2).

Small satellite. A satellite of less than about 500 kg, including minisatellites, microsatellites, nanosatellites and *CubeSats* (11).

Solar array. An assembly of photovoltaic cells that converts sunlight into electricity (5).

Solar cycle. The roughly 11-year cycle of solar activity, measured by sunspot number (22).

Solar flare. A sudden release of energy in the Sun's atmosphere, producing X-rays and often energetic particles (22).

Solar sail. A large, thin reflective membrane propelled by the pressure of sunlight (42).

Solar wind. The continuous outflow of plasma from the Sun at 300–800 km/s (22).

Sounding rocket. A rocket that carries instruments on a suborbital flight, typically for 5–20 minutes above 100 km (9).

Space situational awareness (SSA) / space domain awareness. Knowledge of the location and behaviour of objects in space, and of the environment they operate in (21).

Space weather. Conditions in space, driven mainly by the Sun, that affect technology and human health (22).

Specific impulse (Isp). Thrust per unit weight flow of propellant, in seconds — a measure of propellant efficiency (35).

Sphere of influence. The region around a planet within which its gravity dominates for trajectory calculations (24, C).

Spin stabilisation. Keeping a spacecraft's attitude stable by spinning it (6).

Staging. Discarding empty tanks and engines during ascent so that the remaining vehicle is lighter (34).

Star tracker. A camera that identifies star patterns to determine a spacecraft's attitude to arc-second accuracy (6).

Starshade. A large, flower-shaped screen flown in front of a telescope to block a star's light for exoplanet imaging (38).

Station-keeping. Small manoeuvres that keep a satellite in its assigned orbit or slot (11).

Stratosphere. The atmospheric layer from about 11–15 km to 50 km, where temperature rises with height (2, 8).

Suborbital flight. A flight that reaches space but does not have enough speed to complete an orbit (9).

Sun-synchronous orbit. A near-polar orbit whose plane precesses once a year, so the satellite crosses each latitude at the same local solar time (11).

Supervised learning. Machine learning from examples labelled with the correct answer (43).

Surrogate model (emulator). A fast approximate model, often learned from data, that stands in for a slow physical simulation (44).

Swath. The width of the strip of ground imaged by a satellite on each pass (13).

Synodic period. The time between successive identical alignments of two orbiting bodies, such as Earth and Mars (24, C).

Synthetic aperture radar (SAR). An imaging radar that uses the motion of the platform to synthesise a long antenna, giving fine resolution day and night through clouds (15).

H.18 T

Telemetry. Measurements sent from a spacecraft to the ground describing its health and status (7).

Telecommand. Instructions sent from the ground to a spacecraft (7).

Terrain-relative navigation. Navigating during descent by comparing camera images with onboard maps, used by Perseverance and SLIM to land precisely (27, 48).

Thermal vacuum test. Testing a spacecraft in a chamber that reproduces space vacuum and temperature extremes (4).

Three-axis stabilisation. Controlling a spacecraft's attitude actively about all three axes (6).

Throttling. Varying a rocket engine's thrust (35).

Thrust-to-weight ratio. Thrust divided by weight; it must exceed 1 for a rocket to lift off (34).

Total electron content (TEC). The number of free electrons in a column through the ionosphere, which determines GNSS signal delay (18, 23).

Training data. The examples from which a machine-learning model learns (43).

Transfer orbit. An intermediate orbit between two others, such as a geostationary transfer orbit (3).

Transformer. A neural-network architecture based on attention, used in large language models and increasingly for images and weather (44).

Transit method. Detecting an exoplanet by the small dip in starlight as it passes in front of its star (39).

Transponder. A unit on a communication satellite that receives, amplifies, frequency-shifts and retransmits signals (17).

Troposphere. The lowest layer of the atmosphere, up to 8–18 km, where weather occurs (2, 16).

True anomaly (ν). The angle from periapsis to a body's current position, measured at the central body (3).

Turbopump. A turbine-driven pump that feeds propellant into a rocket engine at high pressure (35).

Two-line element set (TLE). A standard text format for publishing the orbits of catalogued objects (3, 21).

H.19 U–Z

Uncertainty quantification. Estimating how confident a model's predictions are (44).

Unsupervised learning. Machine learning that finds structure in unlabelled data, for example clustering (43).

Upper stage. The final stage(s) of a launch vehicle, which deliver the payload to orbit (34).

Validation / verification. Verification shows that a product meets its specified requirements ("built it right"); validation shows that it meets the user's real needs ("built the right thing") (4, 53).

Vernal equinox. The direction from the Earth to the Sun at the March equinox, used as the reference direction in celestial coordinates (3).

Vis-viva equation. The relation between speed, distance and semi-major axis in a two-body orbit (3, B).

Wavefront sensing and control. Measuring errors in the light waves reaching a telescope's focus and correcting them by moving mirror segments or deforming mirrors (38).

Wet mass / dry mass. A spacecraft's mass with and without propellant (4).

Whipple shield. A shield of a thin outer bumper spaced away from the main wall, which breaks up impacting particles (2).

Window (atmospheric). A wavelength band in which the atmosphere is relatively transparent (13, D).

Zenith / nadir. The point directly overhead / directly below (13).

APPENDIX

Standards, Data Sources and Further Reading

Where to go next — the engineering standards that govern spacecraft design, test, operations, debris mitigation and artificial intelligence; the open data services used to produce the figures and tables in this manual; and a selected reading list of textbooks, handbooks and reports for each Part of the book.

IN THIS CHAPTER

I.1 Engineering Standards
I.2 Open Data Sources
I.3 Further Reading

This manual explains principles and gives typical numbers. Real projects are built to standards and checked against primary data. This appendix lists the most widely used standards, the open data sources behind the computed figures and tables, and a short list of recommended books and reports. Standards are revised frequently; always use the issue specified in your contract or agency requirements.

I.1 Engineering Standards

I.1.1 Who Writes Space Standards

TABLE I.1 Main producers of space standards.

Organisation	What it publishes	Access
ECSS — European Cooperation for Space Standardization	More than 100 standards, plus handbooks, for project management (M), engineering (E), product assurance (Q) and sustainability (U), used by ESA and European industry	Free at ecss.nl
NASA	NASA Procedural Requirements (NPR), Standards (NASA-STD) and Handbooks (NASA-HDBK)	Free at standards.nasa.gov and nodis3.gsfc.nasa.gov
US Space Force / Aerospace Corporation	Space and Missile Systems Center standards (SMC-S series), mission-assurance guides	Mostly public
CCSDS — Consultative Committee for Space Data Systems	Data-system and communication protocols (“Blue Books”) adopted by more than 30 agencies	Free at public.ccsds.org
ISO TC 20/SC 14	International standards for space systems and operations (with ISO/IEC JTC 1/SC 42 for AI)	Purchase
ITU-R	Radio Regulations and Recommendations on propagation, interference and coordination	Free (Radio Regulations) at itu.int

Organisation	What it publishes	Access
IADC and UN COPUOS	Space debris mitigation and long-term sustainability guidelines	Free
AIAA, IEEE, SAE	Industry standards (e.g. AIAA S-series for pressure vessels and mass properties; IEEE 1149, SpaceWire under ECSS)	Purchase

I.1.2 Selected Standards by Topic

TABLE I.2 Selected standards and guidelines cited in this manual.

Topic	Standard	Title or subject	Chapter
Project management	NASA NPR 7120.5	NASA Space Flight Program and Project Management Requirements	4
Systems engineering	NASA NPR 7123.1; NASA/SP-2016-6105 Rev 2	Systems Engineering Processes and Requirements; NASA Systems Engineering Handbook	4
	ECSS-E-ST-10C	Space engineering — System engineering general requirements	4
	ISO/IEC/IEEE 15288	Systems and software engineering — System life cycle processes	4
Risk classification	NASA NPR 8705.4	Risk Classification for NASA Payloads (classes A–D)	4
Verification and test	ECSS-E-ST-10-03C	Space engineering — Testing	4, 5
	GSFC-STD-7000B (GEVS)	General Environmental Verification Standard for GSFC flight programs	4, 5
	SMC-S-016	Test Requirements for Launch, Upper-Stage and Space Vehicles	4, 34
Space environment	ECSS-E-ST-10-04C; ECSS-E-ST-10-12C	Space environment; methods for the calculation of radiation received and its effects	2, 22
Structures	NASA-STD-5001; ECSS-E-ST-32C	Structural design and test factors of safety; structural general requirements	5
Thermal control	ECSS-E-ST-31C	Thermal control general requirements	5
Mechanisms	ECSS-E-ST-33-01C	Mechanisms	5
Electrical and power	ECSS-E-ST-20C	Electrical and electronic	5
Propulsion	ECSS-E-ST-35C	Propulsion general requirements	6, 35
Attitude and orbit control	ECSS-E-ST-60-30C	Satellite attitude and orbit control system requirements	6
Software	ECSS-E-ST-40C; ECSS-Q-ST-80C; NASA NPR 7150.2; RTCA DO-178C	Software engineering; software product assurance; NASA software requirements; airborne software (often used as a reference)	6, 53
EEE components	ECSS-Q-ST-60C; NASA EEE-INST-002	Electrical, electronic and electromechanical components; parts selection and screening	5, 12

Topic	Standard	Title or subject	Chapter
Ground segment and operations	ECSS-E-ST-70C	Ground systems and operations	7
Space data links	CCSDS 131.0-B, 132.0-B, 231.0-B, 232.0-B, 732.0-B, 732.1-B	TM and TC synchronisation and coding; TM, TC, AOS and Unified Space Data Link protocols	7
Packets and files	CCSDS 133.0-B; 727.0-B; 734.2-B	Space Packet Protocol; CCSDS File Delivery Protocol (CFDP); Bundle Protocol (delay-tolerant networking)	7
RF and optical links	CCSDS 401.0-B; 141.0-B	Radio frequency and modulation systems; optical communications physical layer	7
Proximity links	CCSDS 211.0-B	Proximity-1 space link protocol (Mars relay)	7, 28
Data compression	CCSDS 121.0-B; 122.0-B; 123.0-B	Lossless data compression; image data compression; multispectral and hyperspectral compression	12, 13
Orbit and conjunction data	CCSDS 502.0-B; 508.0-B	Orbit Data Messages; Conjunction Data Message	3, 21
Radio spectrum	ITU Radio Regulations (2024 edition)	International allocation and coordination rules	7, 17, App. D
Propagation	ITU-R P.618, P.676, P.837, P.838	Earth–space propagation, gaseous attenuation, rain rate and rain attenuation	7, App. D
CubeSats	CubeSat Design Specification Rev. 14.1 (Cal Poly); ISO 17770	Mechanical and electrical interface requirements for CubeSats	11
Debris mitigation	ISO 24113:2023; ECSS-U-AS-10C; IADC-02-01; NASA-STD-8719.14; ESA ESSB-ST-U-007	Debris mitigation requirements, including post-mission disposal and casualty-risk limits	21
Debris policy	UN COPUOS Space Debris Mitigation Guidelines (2007); Guidelines for the Long-term Sustainability of Outer Space Activities (2019); FCC five-year rule (in force 2024)	International and national debris guidelines and rules	21
Launch and re-entry safety	FAA 14 CFR Part 450; DAFMAN 91-710	US commercial launch and re-entry licensing; range safety user requirements	34, 37
Human spaceflight	NASA-STD-3001 Vols 1 and 2; NASA Human Integration Design Handbook	Crew health and human-system design requirements	30–33
Planetary protection	COSPAR Policy on Planetary Protection; NASA NPR 8715.24	Contamination control for missions to other worlds	24, 29
Machine learning in space software	ECSS-E-HB-40-02A (15 November 2024)	Machine learning qualification handbook	45, 53
AI management and risk	ISO/IEC 22989:2022; ISO/IEC 23053:2022; ISO/IEC 23894:2023; ISO/IEC 42001:2023; NIST AI RMF 1.0 (2023)	AI concepts; ML framework; AI risk management; AI management systems	53
AI regulation and aviation guidance	Regulation (EU) 2024/1689 (EU AI Act); EASA AI Concept Paper Issue 2 (2024)	Risk-based regulation of AI systems; guidance for machine-learning applications in aviation	53

I.2 Open Data Sources

Most figures in this manual that show real data were computed from the following open sources, which readers can use to reproduce and extend them.

TABLE I.3 Open data sources.

Source	Content	Used in
GCAT — General Catalog of Artificial Space Objects (J. McDowell, planet4589.org)	Every launch and catalogued object since 1957, with orbits, masses and owners (CC-BY 4.0)	Chapters 1, 11, 36, 37; Appendix E
CelesTrak (celestrak.org) and Space-Track.org	Orbital elements (TLE/OMM), satellite catalogue, debris counts, space weather indices	Chapters 3, 11, 21
JPL Horizons and NAIF SPICE	Planetary, satellite and spacecraft ephemerides; mission geometry kernels	Chapters 3, 24–29; Appendix C
JPL Solar System Dynamics (ssd.jpl.nasa.gov)	Physical and orbital data for planets, moons, asteroids and comets; small-body database	Chapters 24–29, 41; Appendix C
NASA Planetary Fact Sheets (NSSDCA)	Planetary data and mission records	Appendix C
NOAA Space Weather Prediction Center; SILSO (Royal Observatory of Belgium)	Solar-wind data, geomagnetic indices, sunspot number	Chapters 22, 23
NASA Exoplanet Archive	Confirmed exoplanets and their properties	Chapter 39
Copernicus Data Space Ecosystem; USGS EarthExplorer	Sentinel and Landsat imagery and products	Chapters 13–16, 46
ECMWF (ERA5 reanalysis, open forecast data)	Weather and climate data for numerical and machine-learning models	Chapters 16, 47
ESA DISCOS and the annual ESA Space Environment Report	Debris and re-entry statistics	Chapter 21
UCS Satellite Database (to 2023)	Active satellites by purpose and owner	Chapters 1, 11
NASA Technical Reports Server (ntrs.nasa.gov)	NASA technical reports, conference papers and mission documents	All
eoPortal (ESA)	Mission and instrument descriptions	Chapters 12–16
ITU Space Network List; ITU-Rpy software	Satellite network filings; ITU-R propagation models in Python	Chapter 17; Appendix D
NASA Image and Video Library (images.nasa.gov)	Public-domain photographs used throughout this manual	All; Appendix J

I.3 Further Reading

I.3.1 Part I — Foundations

- J. R. Wertz, D. F. Everett and J. J. Puschell (eds.), *Space Mission Engineering: The New SMAD*, Microcosm Press, 2011 — the standard reference for mission and spacecraft sizing.
- P. Fortescue, G. Swinerd and J. Stark (eds.), *Spacecraft Systems Engineering*, 4th ed., Wiley, 2011.
- H. D. Curtis, *Orbital Mechanics for Engineering Students*, 4th ed., Butterworth-Heinemann, 2020.
- D. A. Vallado, *Fundamentals of Astrodynamics and Applications*, Microcosm Press (latest edition).
- R. R. Bate, D. D. Mueller and J. E. White, *Fundamentals of Astrodynamics*, Dover, 1971 (2nd ed. 2020).
- A. C. Tribble, *The Space Environment: Implications for Spacecraft Design*, revised ed., Princeton University Press, 2003.

- D. G. Gilmore (ed.), *Spacecraft Thermal Control Handbook, Vol. I*, 2nd ed., The Aerospace Press, 2002.
- F. L. Markley and J. L. Crassidis, *Fundamentals of Spacecraft Attitude Determination and Control*, Springer, 2014.
- NASA, *NASA Systems Engineering Handbook*, NASA/SP-2016-6105 Rev 2, 2016.
- NASA Ames Research Center, *State-of-the-Art of Small Spacecraft Technology* (updated annually).
- G. Maral, M. Bousquet and Z. Sun, *Satellite Communications Systems*, 6th ed., Wiley, 2020.

I.3.2 Parts II–III — Near Space, Suborbital Flight and Satellites

- M. D. Griffin and J. R. French, *Space Vehicle Design*, 2nd ed., AIAA, 2004.
- NASA Sounding Rockets Program Office, *Sounding Rocket Program Handbook* (NASA Wallops Flight Facility).
- Cal Poly CubeSat Laboratory, *CubeSat Design Specification*, Rev. 14.1, 2022.

I.3.3 Parts IV–V — Earth Observation, Communications and Navigation

- C. Elachi and J. van Zyl, *Introduction to the Physics and Techniques of Remote Sensing*, 3rd ed., Wiley, 2021.
- T. M. Lillesand, R. W. Kiefer and J. W. Chipman, *Remote Sensing and Image Interpretation*, 7th ed., Wiley, 2015.
- A. Moreira *et al.*, "A tutorial on synthetic aperture radar", *IEEE Geoscience and Remote Sensing Magazine*, 1(1), 2013.
- T. Pratt and J. E. Allnutt, *Satellite Communications*, 3rd ed., Wiley, 2019.
- P. Misra and P. Enge, *Global Positioning System: Signals, Measurements, and Performance*, 2nd ed., Ganga-Jamuna Press, 2006.
- P. J. G. Teunissen and O. Montenbruck (eds.), *Springer Handbook of Global Navigation Satellite Systems*, Springer, 2017.

I.3.4 Parts VI–VII — Security, Debris and the Space Environment

- H. Klinkrad, *Space Debris: Models and Risk Analysis*, Springer, 2006.
- ESA Space Debris Office, *ESA's Annual Space Environment Report* (published each year).
- Secure World Foundation, *Global Counterspace Capabilities: An Open Source Assessment* (updated annually).
- M. G. Kivelson and C. T. Russell (eds.), *Introduction to Space Physics*, Cambridge University Press, 1995.
- D. Hastings and H. Garrett, *Spacecraft–Environment Interactions*, Cambridge University Press, 1996.

I.3.5 Part VIII — Deep-Space Exploration

- S. Kemble, *Interplanetary Mission Analysis and Design*, Springer, 2006.
- A. J. Ball, J. R. C. Garry, R. D. Lorenz and V. V. Kerzhanovich, *Planetary Landers and Entry Probes*, Cambridge University Press, 2007.
- R. D. Braun and R. M. Manning, "Mars exploration entry, descent and landing challenges", *Journal of Spacecraft and Rockets*, 44(2), 2007.

- A. A. Siddiqi, *Beyond Earth: A Chronicle of Deep Space Exploration, 1958–2016*, NASA SP-2018-4041, 2018.

I.3.6 Parts IX–X — Human Spaceflight and Launch

- W. J. Larson and L. K. Pranke (eds.), *Human Spaceflight: Mission Analysis and Design*, McGraw-Hill, 2000.
- NASA, *Human Integration Design Handbook*, NASA/SP-2010-3407 (revised).
- G. P. Sutton and O. Biblarz, *Rocket Propulsion Elements*, 9th ed., Wiley, 2017.
- D. K. Huzel and D. H. Huang, *Modern Engineering for Design of Liquid-Propellant Rocket Engines*, AIAA, 1992.
- D. M. Goebel, I. Katz and I. G. Mikellides, *Fundamentals of Electric Propulsion*, 2nd ed., Wiley/JPL Space Science and Technology Series.
- R. W. Humble, G. N. Henry and W. J. Larson (eds.), *Space Propulsion Analysis and Design*, McGraw-Hill, 1995.

I.3.7 Parts XI–XII — Observatories and Emerging Frontiers

- P. Y. Bely (ed.), *The Design and Construction of Large Optical Telescopes*, Springer, 2003.
- J. P. Gardner *et al.*, "The James Webb Space Telescope Mission", *Publications of the Astronomical Society of the Pacific*, 135, 068001, 2023.
- K. F. Long, *Deep Space Propulsion: A Roadmap to Interstellar Flight*, Springer, 2012.
- V. Badescu (ed.), *Moon: Prospective Energy and Material Resources*, Springer, 2012.
- F. von der Dunk and F. Tronchetti (eds.), *Handbook of Space Law*, Edward Elgar, 2015.

I.3.8 Part XIII — Artificial Intelligence

- S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed., Pearson, 2020.
- I. Goodfellow, Y. Bengio and A. Courville, *Deep Learning*, MIT Press, 2016.
- C. M. Bishop and H. Bishop, *Deep Learning: Foundations and Concepts*, Springer, 2024.
- G. Camps-Valls, D. Tuia, X. X. Zhu and M. Reichstein (eds.), *Deep Learning for the Earth Sciences*, Wiley, 2021.
- ECSS, *ECSS-E-HB-40-02A Machine Learning Handbook*, 15 November 2024.

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>
> Space changes quickly. For current events, the most reliable sources are the agencies themselves (NASA, ESA, JAXA, CNSA, ISRO, Roscosmos press releases), company announcements checked against independent trackers, Jonathan McDowell's *Jonathan's Space Report*, Gunter's Space Page, the trade press (SpaceNews, SpacePolicyOnline, NASASpaceflight, Spaceflight Now) and peer-reviewed journals (*Acta Astronautica*, *Journal of Spacecraft and Rockets*, *Advances in Space Research*, *Remote Sensing of Environment*). Cross-check any number that matters against a primary source.

APPENDIX

J Image Credits

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TABLE J.1 Part opening images.

Part	NASA ID	Title	Date
I	iss028e007274	Earth Observation	2011-06-11
II	GSFC_20171208_Archive_e000153	Balloons on Ice: Launch # 2 takes flight in Antarctica	2017-12-08
III	KSC-20210811-PH-RNB01_0028	Landsat 9 Spacecraft Mate to EPSA	2021-08-11
IV	GSFC_20171208_Archive_e001386	Most Amazing High Definition Image of Earth - Blue Marble 2012	2017-12-08
V	KSC-2013-1231	KSC-2013-1231	2013-01-29
VI	GSFC_20171208_Archive_e002131	NASA Blue Marble 2007 West	2017-12-08
VII	GSFC_20171208_Archive_e001050	Two X Flares in Quick Succession	2017-12-08
VIII	PIA21381	Jupiter From Below (Enhanced Color)	2017-02-08
IX	iss065e145170	iss065e145170	2021-06-25
X	NHQ202211160030	Artemis I Launch	2022-11-16
XI	carina_nebula	James Webb Space Telescope NIRCam Image of the “Cosmic Cliffs” in Cari	2022-07-12
XII	iss043e124426	SpaceX (SpX)-6 Dragon spacecraft move to berth sequence	2015-04-17
XIII	ACD15-0062-004	ACD15-0062-004	2015-04-10
XIV	GSFC_20171208_Archive_e001282	NASA Google+ Hangout: ‘Earthrise’ A New Visualization - 45th Anniversa	2017-12-08

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Image	NASA ID	Title	Date
Chapter opening	s134e010592	View of ISS taken during STS-134 Flyaround	2011-05-30

J.3 Chapter 2: The Space Environment

TABLE J.3 Images in Chapter 2.

Image	NASA ID	Title	Date
Chapter opening	iss066e146457	iss066e146457	2022-02-18
misse	0201585	International Space Station (ISS)	2001-08-17
ldef	s32-541-008	LDEF grappled by remote manipulator system (RMS) during STS-32 retriev	1990-01-20
tvac	PIA13936	Juno Emerges from Thermal-Vac Testing	2011-03-29

J.4 Chapter 3: Orbital Mechanics and Astrodynamics

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Image	NASA ID	Title	Date
Chapter opening	art001e001712	Flight Day 14: Orion's Solar Array Divides Earth and Moon	2022-11-29

J.5 Chapter 4: Spacecraft Systems Engineering

TABLE J.5 Images in Chapter 4.

Image	NASA ID	Title	Date
Chapter opening	GSFC_20171208_Archive_e000495	James Webb Space Telescope Mirror Halfway Complete	2017-12-08
highbay	PIA23519	High Bay 1 in JPL's Spacecraft Assembly Facility	2019-12-23
clipper	PIA25958	Powerful Antenna Installed On Europa Clipper	2023-08-15
shake	GRC-2024-C-01582	Sierra Space Dream Chaser Spaceplane at the Space Environments Complex	2024-03-07

J.6 Chapter 5: Spacecraft Subsystems I: Structures, Thermal Control, Power and Mechanisms

TABLE J.6 Images in Chapter 5.

Image	NASA ID	Title	Date
Chapter opening	PIA13925	Juno Solar Panel Deployment Test	2009-10-13

Image	NASA ID	Title	Date
maven	GSFC_20171208_Archive_e001847	MAVEN Mission Primary Structure Complete	2017-12-08
radiator	s114e5283	P6 truss and radiator panels	2005-07-28
mli	iss015e10854	Survey view of damage to a MLI Blanket on the exterior of the FGB take	2007-06-06
irosa	iss068e026401	Expedition 68 EVA 82	2022-12-03
mmrtg	PIA23306	Mars 2020's MMRTG	2019-07-24

J.7 Chapter 6: Spacecraft Subsystems II: Attitude Control, Propulsion, Data Handling and Flight Software

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Image	NASA ID	Title	Date
Chapter opening	GRC-2003-C-00585	GRC-2003-C-00585	2004-05-01
startrackers	PIA26562	Star Trackers for Europa Clipper	2025-02-04
wheels	PIA25495	NASA's Europa Clipper Reaction Wheels Installed	2022-11-23
wheel_eva	S109E5396	STS-109 MS Massimino and Newman replace Reaction Wheel assembly during	2002-03-05

J.8 Chapter 7: Space Communications and the Ground Segment

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Image	NASA ID	Title	Date
Chapter opening	PIA13277	Tracks of a Giant	2010-08-25
dsoc_ftl	PIA24569	DSOC's Flight Laser Transceiver	2022-04-06
illumat	jsc2023e064875	ILLUMA-T Investigation Preflight Imagery	2023-10-19
madrid	PIA26147	Six Deep Space Network Antennas in Madrid Arrayed For the First Time	2024-04-26
dss53	PIA25137	The Deep Space Network's New DSS-53 at Night	2022-03-16

J.9 Chapter 8: High-Altitude Platforms and Space Balloons

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Image	NASA ID	Title	Date
Chapter opening	GSFC_20171208_Archive_e000115	Alien crop circle? No, that's just NASA's newest balloon launch pad	2017-12-08
ldb	GSFC_20171208_Archive_e000799	Balloon Program Wraps up in Antarctica, Heading to New Zealand	2017-12-08

Image	NASA ID	Title	Date
part2	GSFC_20171208_Archive_e000153	Balloons on Ice: Launch # 2 takes flight in Antarctica	2017-12-08
helios	ED01-0230-1	ED01-0230-1	2001-08-13
er2	EC01-0232-2	Lockheed ER-2 #809 high altitude research aircraft in flight	2001-08-01

J.10 Chapter 9: Sounding Rockets and Suborbital Vehicles

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Image	NASA ID	Title	Date
Chapter opening	GSFC_20171208_Archive_e000703	NASA Successfully Launches Suborbital Rocket from Wallops with Student	2017-12-08
mtex	GSFC_20171208_Archive_e000831	M-TeX and MIST Experiments Launched from Alaska	2017-12-08
evex	GSFC_20171208_Archive_e001503	NASA EVEX Experiment Launches from the Marshall Islands	2017-12-08
clouds	GSFC_20171208_Archive_e000130	NASA rocket to display artificial clouds in space	2017-12-08
x43	ED04-0261-1	NASA's B-52B launch aircraft takes off carrying the third X-43A hypers	2004-09-27

J.11 Chapter 10: Spaceplanes, Atmospheric Re-entry and Reusable Vehicles

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Image	NASA ID	Title	Date
Chapter opening	sts135-s-273	Space Shuttle Atlantis after its Final Landing	2011-06-21
orionhs	KSC-20240203-EM1_CM_320_0-0001 1	Artemis I Orion Heat Shield Inspection at Kennedy Space Center	2023-02-03
arcjet	jsc2024e078233	Arc Jet Testing of Orion Heat Shield Avcoat	2023-12-05
sts1	8111969	Space Shuttle Projects	1981-04-12
tiles	KSC-03pd3213	KENNEDY SPACE CENTER, FLA. - In the Orbiter Processing Facility, KSC	2003-12-09
dreamchaser	AFRC2017-0016-016	Dream Chaser Spacecraft Arrives at NASA Armstrong	2017-01-25
sts53	sts053-s-080	STS-53 Discovery, Orbiter Vehicle (OV) 103, lands on runway 22 at EAFB	1992-12-09

J.12 Chapter 11: Earth Orbits, Constellations and Small Satellites

TABLE J.12 Images in Chapter 11.

Image	NASA ID	Title	Date
Chapter opening	iss050e031340	J-SSOD-6 Deployment	2017-01-16
marco	PIA20342	MarCO CubeSat Engineers 2	2016-01-20
nanoracks	iss069e004397	Astronaut Frank Rubio installs the NanoRacks CubeSat Deployer	2023-04-20
starling	ACD23-0077-003	Starling's four CubeSats in space (illustration)	2023-07-12

J.13 Chapter 12: Satellite Buses and Payload Instruments

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Image	NASA ID	Title	Date
Chapter opening	KSC-20171205-PH_MAS01_0031	GOES-S Rotate to Vertical	2017-12-05
psp_mag	KSC-20180523-PH_LCH01_0084	Parker Solar Probe Magnetometer Boom Deployment	2018-05-23
clippermag	PIA24786	Europa Clipper Magnetometer (ECM) Boom	2021-08-05
goesfull	GSFC_20171208_Archive_e000007	Eclipse from Space - GOES Full Disk	2017-12-08

J.14 Chapter 13: Remote Sensing Fundamentals

TABLE J.14 Images in Chapter 13.

Image	NASA ID	Title	Date
Chapter opening	GSFC_20171208_Archive_e002144	Guinea-Bissau	2017-12-08
aljowf	PIA23955	Al Jowf, Saudi Arabia	2020-07-06
sf	PIA02606	ASTER Images San Francisco Bay Area	2000-04-26
gardencity	GSFC_20171208_Archive_e001881	Garden City, Kansas	2017-12-08
rimfire	GSFC_20171208_Archive_e001383	Satellite View of Rim Fire On August 25, 2013	2017-12-08
lakemead	PIA19731	Lake Mead and Drought	2015-07-20

J.15 Chapter 14: Electro-Optical and Infrared Imaging Satellites

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Image	NASA ID	Title	Date
Chapter opening	KSC-20210720-PH-RNB01_0022	Landsat 9 Spacecraft (Observatory) Move from High Bay into Cell	2021-07-20
laheat	PIA23786	Los Angeles Heat Wave	2020-08-19
applefire	PIA23798	ECOSTRESS Gauges the Apple Fire's Temperature From Space	2020-08-03
nile	GSFC_20171208_Archive_e001586	City Lights Illuminate the Nile	2017-12-08
methane	PIA26113	EMIT Identifying Methane Plumes Around the Globe	2023-11-17

Image	NASA ID	Title	Date
eyja	PIA13098	Iceland Volcano Puffs Ash as Lava Flow Cuts Through Eyjafjallajökull I	2010-05-05

J.16 Chapter 15: Synthetic Aperture Radar

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Image	NASA ID	Title	Date
Chapter opening	PIA26612	NISAR Team Deploys Radar Antenna Reflector	2025-08-15
seasat	PIA18137	NASA Seasat Satellite Shows Massachusetts Coast	2014-03-18
sirc	s93-48551	STS-59 payload SIR-C/X-SAR antenna view	1993-10-30
nisarfirst	PIA26616	NASA-ISRO Radar Satellite Captures First Image of Mount Desert Island	2025-09-25
ndakota	PIA26615	NASA-ISRO Radar Satellite Captures First Image of North Dakota Wetland	2025-09-25
srtm	PIA02723	Radar Image, Wrapped Color as Height, Lanai and West Maui, Hawaii	2000-02-19
izmit	PIA00557	Izmit, Turkey 1999 Earthquake Interferogram	2001-03-30
kilauea	PIA01710	Space Radar Image of Kilauea, Hawaii	1999-01-27

J.17 Chapter 16: Weather and Environmental Satellites

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Image	NASA ID	Title	Date
Chapter opening	iss072e001649	Hurricane Helene pictured from the space station	2024-09-25
goesu	KSC-20240624-PH-SPX01_0003	GOES-U Rollout for Launch	2024-06-24
ozone	GSFC_20171208_Archive_e001962	Snapshot of the Antarctic Ozone Hole 2010	2017-12-08
gpm	GSFC_20171208_Archive_e001331	GPM Core Observatory	2017-12-08
pace	KSC-20231204-PH-KLS01_0024	PACE Spacecraft Processing	2023-12-04
swot	PIA25772	SWOT Satellite's Sea Level 'First Light'	2023-03-24

J.18 Chapter 17: Communication Satellites

TABLE J.18 Images in Chapter 17.

Image	NASA ID	Title	Date
Chapter opening	KSC-20170808-PH_KLS01_0532	TDRS-M Spacecraft Lift to Transport Vehicle	2017-08-08
echo	LRC-1958-B701_P-03603	Inflation Tests of the Echo 1 Satellite in Weeksville, N.C.	1958-08-13

Image	NASA ID	Title	Date
syncom	41d-32-0067	View of the Syncom IV satellite in orbit over the earth	1984-08-31
intelsat	9301572	Space Shuttle Projects	1992-05-13
illumat	jsc2023e064876	ILLUMA-T Investigation Preflight Imagery	2023-10-19

J.19 Chapter 18: Global Navigation Satellite Systems

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Image	NASA ID	Title	Date
Chapter opening	PIA24573	Deep Space Atomic Clock Hardware	2021-06-30
navstar	9304431	Launch Vehicles	1993-03-29
tonga	PIA24905	Tonga Eruption Atmospheric Wave	2022-01-27
blueghost	KSC-20250114-PH-KLS01_0039	CLPS Firefly Blue Ghost Mission 1	2025-01-14
delta	0400536	Space Science	2004-01-01

J.20 Chapter 19: National Security Space and Signals Intelligence

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Image	NASA ID	Title	Date
Chapter opening	iss071e439624	An orbital sunrise colorfully illuminates the Earth's atmosphere	2024-08-06

J.21 Chapter 20: Missile Early Warning Satellites

TABLE J.21 Images in Chapter 20.

Image	NASA ID	Title	Date
Chapter opening	iss074e0431736	The exhaust plume left behind by Artemis II is pictured from the Inter	2026-04-01
dspbay	S44-71-011	STS-44 DSP / IUS spacecraft tilted to predeployment position in OV-104	1991-11-24
dsp	s91-50773	STS-44 DSP satellite and IUS during preflight processing at Cape Canav	1991-10-19
starship6	iss072e220043	The launch of the SpaceX Starship 6 rocket seen from the space station	2024-11-19
crs35	NHQ20261013_admin_0004	NASA's SpaceX CRS-35 Launch	2026-10-13

J.22 Chapter 21: Space Surveillance, Orbital Debris and Space Situational Awareness

TABLE J.22 Images in Chapter 21.

Image	NASA ID	Title	Date
Chapter opening	iss074e0540224	A piece of unidentified space debris is pictured from the Internationa	2026-04-27
hvi	wstf0507e04048	WSTF Hypervelocity Impact Test - CEV	2007-05-31
sts125	jsc2010e054445	STS-125 Orbital Debris	2009-07-02

J.23 Chapter 22: Space Weather

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Image	NASA ID	Title	Date
Chapter opening	GSFC_20171208_Archive_e002109	Blasting CME	2017-12-08
sunspot	GSFC_20171208_Archive_e001978	C3-class Solar Flare Erupts on Sept. 8, 2010 [Full Disk]	2017-12-08
xflare	GSFC_20171208_Archive_e000934	NASA's SDO Observes an X-class Solar Flare	2017-12-08
cme	cme-blow-out_18529951894_o	CME Blow Out	2015-06-25
parker	KSC-20180419-PH_KLS01_0009	Parker Solar Probe Antenna Deployment	2018-04-19
punch	KSC-20250121-PH-ANV01_0028	PUNCH Satellites Solar Array Deployment Test	2025-01-21
maven	PIA26304	MAVEN Detects Auroras During Solar Storm in 2024	2024-06-10

J.24 Chapter 23: Magnetospheric and Ionospheric Physics

TABLE J.24 Images in Chapter 23.

Image	NASA ID	Title	Date
Chapter opening	iss072e159172	The aurora borealis blankets the Earth	2024-10-30
belt3	GSFC_20171208_Archive_e001541	NASA's Van Allen Probes Discover a Surprise Circling Earth	2017-12-08
mms	GSFC_20171208_Archive_e001111	Magnetospheric Multiscale (MMS)	2017-12-08
aurora	iss073e0247726	The aurora australis arcs above a partly cloudy Indian Ocean	2025-06-12

J.25 Chapter 24: Deep-Space Mission Design

TABLE J.25 Images in Chapter 24.

Image	NASA ID	Title	Date

Image	NASA ID	Title	Date
Chapter opening	PIA22915	Voyager 2 Spacecraft Instruments	2018-12-10
dawn	PIA18922	Dawn Blue Glow Artist Concept	2015-03-02
dsn	PIA23682	NASA's Deep Space Antenna Upgrade to Affect Voyager	2020-03-04
clipper	KSC-20240528-PH-KLS01_0036	Europa Clipper Spacecraft Unboxing	2024-05-28
lucy	KSC-20210819-PH-GEB01_0026	Lucy Solar Array Pop & Catch	2021-08-19

J.26 Chapter 25: Planetary Orbiters

TABLE J.26 Images in Chapter 25.

Image	NASA ID	Title	Date
Chapter opening	PIA21771	Juno Over Jupiter's South Pole (Illustration)	2017-07-11
finale	PIA21439	Cassini Grand Finale Dive Illustration	2017-04-04
earthrise	GSFC_20171208_Archive_e000496	NASA Releases New High-Resolution Earthrise Image	2017-12-08
mro	PIA07245	Mars Reconnaissance Orbiter, Front View Artist Concept	2005-06-03
venus	PIA00478	Venus - Global View Centered at 180 degrees	1996-11-26
mercury	PIA13823	MESSENGER Explores Mercury - In Color	2011-01-25
grs	PIA21773	Jupiter's Great Red Spot (Enhanced Color)	2017-07-12
enceladus	PIA14599	Dark Moon, Dramatic Plume	2012-03-12

J.27 Chapter 26: Space Probes and Flyby Missions

TABLE J.27 Images in Chapter 26.

Image	NASA ID	Title	Date
Chapter opening	PIA19956	Pluto in Extended Color	2015-09-24
pioneer	ARC-1972-AC72-2135	ARC-1972-AC72-2135	1972-02-26
voyager	PIA17049	Voyager in Space Artist Concept	2013-09-12
neptune	PIA01492	PIA01492	1998-10-30
paleblue	PIA23645	Pale Blue Dot Revisited	2020-02-12
charon	PIA19968	Charon in Enhanced Color	2015-10-01
tempel	PIA02137	Tempel Alive with Light	2005-07-05
dart	PIA25329	NASA's DART at the Didymos System	2022-09-22

J.28 Chapter 27: Landers and Entry, Descent & Landing

TABLE J.28 Images in Chapter 27.

Image	NASA ID	Title	Date

Image	NASA ID	Title	Date
Chapter opening	PIA25287	InSight's Final Selfie	2022-05-23
aldrin	as11-40-5902	Astronaut Edwin Aldrin walks on lunar surface near leg of Lunar Module	1969-07-20
surveyor	AS12-48-7134	Apollo 12 Mission image - Astronaut Alan L. Bean,lunar module pilot,an	1969-11-20
pdv	KSC-20200422-PH-JPL01_0005	Mars 2020 Perseverance Powered Descent Vehicle (PDV) Stack Operations	2020-04-22
hirisedesc	PIA24270	HiRISE Captured Perseverance During Descent to Mars	2021-02-19
skycrane	PIA24428	High-Resolution Still Image of Perseverance's Landing	2021-02-19
viking	PIA00396	Viking Lander 2 First Picture On The Surface Of Mars	1996-12-12
phoenix	PIA10910	Ice on Mars - Now You See It	2008-06-23
seis	PIA22977	Putting SEIS on the Ground	2018-12-20
huygens	PIA07232	First Color View of Titan Surface	2005-01-15

J.29 Chapter 28: Planetary Rovers and Aerial Explorers

TABLE J.29 Images in Chapter 28.

Image	NASA ID	Title	Date
Chapter opening	PIA19807	Curiosity Low-Angle Self-Portrait at Buckskin Drilling Site on Mount S	2015-08-19
lunokhod	PIA12967	Soviet Union Lunar Rovers	2010-03-17
lrv	as15-86-11603	Astronaut James Irwin works at Lunar Roving Vehicle during Apollo 15 E	1971-07-31
rovers3	PIA11436	Newest is Biggest: Three Generations of NASA Mars Rovers	2008-11-19
wheel	PIA17751	Left-Front Wheel of Curiosity Rover, Approaching Three Miles	2013-12-20
opptracks	PIA14129	Autonomous Hazard Checks Leave Patterned Rover Tracks on Mars	2011-05-18
leopard	PIA26368	Perseverance Finds a Rock With 'Leopard Spots'	2024-07-25
ingenuity	PIA24542	Perseverance's Selfie with Ingenuity	2021-04-07
blade	PIA26244	Ingenuity's Navcam Reveals a Missing Rotor Blade	2024-02-27
viper	GRC-2019-C-09947	Virtual Intelligent Planetary Exploration Rover, VIPER Mobili...	2019-10-22

J.30 Chapter 29: Sample-Return Missions

TABLE J.30 Images in Chapter 29.

Image	NASA ID	Title	Date
Chapter opening	NHQ202308300001	OSIRIS-REx Sample Return Training	2023-08-30
schmitt	as17-140-21496	Astronaut Harrison Schmitt standing next to boulder during third EVA	1972-12-13
collector	PIA03188	Dust Collector with Aerogel	2005-11-03
aerogel	PIA03186	Particle Tracks in Aerogel	2005-11-03
bennu	2019-02-25_regolith_image_compilation	A Region of Bennu's Northern Hemisphere Close Up	2019-03-14
capsule	NHQ202309240002	OSIRIS-REx Sample Return	2023-09-24
tagsam	jsc2024e006057	OSIRIS-REx Asteroid Sample in Touch-and-Go-Sample-Acquisition-Mechanis	2024-01-17
ryugu	jsc2021e062383	jsc2021e062383	2021-11-30
ees	PIA25986	Artist's Concept of the Earth Entry System for Mars Sample Return	2023-08-03

J.31 Chapter 30: Human Spaceflight Fundamentals and Life Support

TABLE J.31 Images in Chapter 30.

Image	NASA ID	Title	Date
Chapter opening	iss074e0381623	ESA astronaut Sophie Adenot poses for a portrait inside the Internatio	2026-03-12
ared	iss072e126509	Astronaut Nick Hague exercises on the advanced resistive exercise devi	2024-10-29
eyeus	iss073e0026012	iss073e0026012	2025-05-06
twins	iss045e028258	Kelly during Twins Study Experiment Operations	2015-09-24
phantoms	KSC-20220808-PH-FMX01_0031	Artemis I MARES Payloads - Install into Crew Module	2022-08-08
mailbox	as13-62-8929	Interior view of "mail box" for purging carbon dioxide from Lunar Modu	1970-04-14
ogs	iss073e0118757	NASA astronaut Nichole Ayers cleans and services life support componen	2025-05-29
wrs	iss067e099373	iss067e099373	2022-06-01
peppers	iss066e083264	Chile Peppers growing inside the Advanced Plant Habitat in the Columbu	2021-11-26
meal	iss031e041675	Expedition 31 Crewmembers Share a Meal in the SM	2012-05-18
chapea	jsc2026e403153	CHAPEA Mission 2 Crew Poses in Habitat	2025-11-24

J.32 Chapter 31: Crewed Spacecraft

TABLE J.32 Images in Chapter 31.

Image	NASA ID	Title	Date
Chapter opening	iss073e0505071	The SpaceX Dragon crew spacecraft approaches the International Space S	2025-08-02
gemini	S65-63220	View of the Gemini 6 and Gemini 7 rendezvous	1965-12-15
lmA	as11-44-6642	KSC-as11-44-6642	1969-07-21
starliner	iss071e170979	Boeing's Starliner Spacecraft Approaches the International Space Stati	2024-06-06
orionselfie	art002e004357	Orion Snaps a Selfie During External Inspection	2026-04-03
ceit2	KSC-20250217-PH-SPX01_0002	SpaceX Crew-10 CEIT	2025-02-17
aa2D	KSC-20190702-PH_AWG07_0011	Ascent Abort-2 Liftoff	2019-07-02
spaA	KSC-20191104-PH-JSC01_0002	Boeing CST-100 Starliner Pad Abort Test	2019-11-04
a2splash	KSC-20260410-PH-JNV01_0002	Artemis II Splashdown	2026-04-10
soyuzC	NHQ201810040001	Expedition 56 Soyuz MS-08 Landing	2018-10-04
dsplash	NHQ202601150016	NASA's SpaceX Crew-11 Splashdown	2026-01-15

J.33 Chapter 32: Space Stations and Habitats

TABLE J.33 Images in Chapter 32.

Image	NASA ID	Title	Date
Chapter opening	iss01-389-023	Expedition One crew flyaround of ISS	2001-02-24
skylab	sl4-143-4706	View of Skylab space station cluster in Earth orbit from CSM	2008-08-18
mir	sts071-744-017	Mir space station	1995-06-29
iss2021	iss066e080762	iss066e080762	2021-11-08
irosa	iss065e116519	iss065e116519	2021-06-15
dextre	iss073e0546366	The Canadarm2 with its fine-tuned robotic hand, Dextre	2025-08-27
cupola	iss074e0604606	Astronauts Sophie Adenot and Jack Hathaway look out a window in the cu	2026-05-17
htvx	iss073e0988464	The new HTV-X1 in the grips of the Canadarm2	2025-10-29
cygnusxl	iss074e0458241	The Cygnus XL cargo spacecraft at its capture point 12 meters from the	2026-04-13
beamx2	iss067e214089	iss067e214089	2022-08-02
astrobee	iss060e007162	Mobility Test of the Free-Flying Astrobee Robotic Assistant	2019-07-12

J.34 Chapter 33: Extravehicular Activity Systems

TABLE J.34 Images in Chapter 33.

Image	NASA ID	Title	Date
Chapter opening	iss063e052528	iss063e052528	2020-07-21
white	s65-30427	ASTRONAUT EDWARD H. WHITE II - GEMINI-TITAN (GT)-IV - ZERO GRAVITY - O	2015-03-20
mmu	s84-27017	INFLIGHT (CREW ACTIVITY) - STS-11/41B	1984-02-11
hubble	s125e007221	STS-125 MS5 Feustel and MS3 Grunsfeld during EVA1	2009-05-14
orlan	iss073e0988196	Cosmonaut Alexey Zubritsky is pictured in his Orlan spacesuit conducti	2025-10-28
gloves	s123e006438	Whitson takes photo of EMU gloves after EVA 2 in the A/L during Joint	2008-03-16
safer	9412846	Space Shuttle Projects	1994-09-16
iss2026	iss074e0788297	NASA astronaut Chris Williams works outside the International Space St	2026-06-30
axnbl	jsc2025e077817	NASA Astronauts Complete First Dual Suit Run Wearing AxEMU	2025-09-24
cernan	AS17-145-22224	Astronaut Eugene Cernan inside the lunar module on lunar surface after	1972-12-12
axemu	axemu-prada-spacesuit-mockup-20241007-423	Artemis III Lunar Spacesuit	2024-10-07

J.35 Chapter 34: Launch Vehicle Engineering

TABLE J.35 Images in Chapter 34.

Image	NASA ID	Title	Date
Chapter opening	KSC-20240928-PH-KLS03_0026	NASA's SpaceX Crew-9 Liftoff	2024-09-28
sii	6758331	Saturn Apollo Program	1967-01-01
etsep	s29-72-059	STS-29 Discovery, OV-103, external tank (ET) separation	1989-03-13
barrel2	MAF_20250210_CS5_LH2_Barrel_out_VWC-04	Artemis V Moon Rocket in Early Phase of Production	2025-02-10
lh2tank	MSFC_011519_SLS_LH2_Lift-7	Space Launch System Liquid Hydrogen Tank Test Article Positioned in Te	2019-03-05
fairingtest	GRC-2012-C-01689	Ariane 5 Payload Fairing Test	2012-04-30
windtunnel	ACD25-0156-012	NASA, SpaceX Advance Wind Tunnel Tests at NASA Ames	2025-07-31
encap	GSFC_20240130_PACE_001980	PACE Encapsulation	2024-01-30

Image	NASA ID	Title	Date
acoustic	GRC-2024-C-09957	Orion Environmental Test Article in the Vibro Acoustic Lab at Armstron	2024-09-11
fhland2	KSC-20240625-PH-AJN01_0191	NASA's SpaceX GOES-U Booster Landing	2024-06-25

J.36 Chapter 35: Rocket Propulsion

TABLE J.36 Images in Chapter 35.

Image	NASA ID	Title	Date
Chapter opening	SSC20210116_GR8438	SSC20210116_GR8438	2021-01-16
fh27	KSC-20260830-PH-SPX02_0005	NASA's Nancy Grace Roman Space Telescope Liftoff, Launch Comple	2026-08-30
f1	6862832	Saturn Apollo Program	1968-01-01
ssme	2010-3891	KSC-2010-3891	2010-06-25
tubes	KSC-99pp0981	KSC-99pp0981	1999-07-28
amturbo	MSFC-1600485	3-D PRINTED TURBOPUMP TEST SET UP	2016-03-29
bole	C0180.00_01_41_23.Still002_Edit	NASA, Northrop Grumman Test Fire SLS BOLE Booster	2025-06-26
hybrid	KSC-2012-1013	KSC-2012-1013	2010-09-21
merlintest	KSC-20180102-PH_SPX01_0004	Commerical Crew Program - SpaceX	2018-01-02
rl10	KSC-20160812-PH_DNG01_0078	Commerical Crew Program (CCP) Astronauts visit Aerojet Rocketdyn	2016-08-12

J.37 Chapter 36: Launch Vehicles of the World

TABLE J.37 Images in Chapter 36.

Image	NASA ID	Title	Date
Chapter opening	KSC-20220816-PH-FMX01_0067	Artemis I Launch Rollout	2022-08-16
saturnv	6901000	Saturn Apollo Program	1969-07-16
shuttle	201107080006HQ	STS-135 Atlantis Launch	2011-07-08
ariane5	NHQ202112250015	James Webb Space Telescope Launch	2021-12-25
h2a	201402280004HQ	Global Precipitation Measurement (GPM) Mission	2014-02-28
sls	SLS_MAF_20260401_ArtemisIILaunch_05	Artemis II launch	2026-04-01
vulcan	KSC-20240108-PH-ILW01_0013	CLPS Peregrine Mission One (PM-1) Astrobotic/ULA Liftoff	2024-01-08
newglenn	KSC-20251113-PH-BLU01_0007	NASA's ESCAPADE Launch on Blue Origin's NG-2	2025-11-13
electron	KSC-20230530-PH-RKL02_0004	TROPICS Rocket Launch	2023-05-26
soyuz	NHQ202409110047	Expedition 72 Launch	2024-09-11

J.38 Chapter 37: Spaceports and Ground Infrastructure

TABLE J.38 Images in Chapter 37.

Image	NASA ID	Title	Date
Chapter opening	KSC-05pd2403	KSC-05pd2403	2005-11-02
soyuzrail	NHQ202309120007	Expedition 70 Soyuz Rollout	2023-09-12
soyuzerect	NHQ202409080023	Expedition 72 Soyuz Rollout	2024-09-08
h2a	201402270008HQ	Global Precipitation Measurement (GPM) Mission	2014-02-27
deflector	KSC-20180516-PH_KLS01_0024	Flame Deflector Complete at Launch Complex 39B	2018-05-16
wetflow	KSC-20190913-PH_KLS01_0030	Nominal Wet Flow Test at Pad 39B	2019-09-13
scape	KSC-2010-1055	KSC-2010-1055	2010-01-07
crawler	KSC-20220318-PH-KLS01_0025	Artemis I Arrives at Pad 39B	2022-03-18
lightning	KSC-20220610-PH-NAS01_0006	Lightning Strikes at Pad 39B	2022-06-10
pegasus	PIA15633	NuSTAR Hitches a Ride on the Stargazer	2012-06-04

J.39 Chapter 38: Space Telescope Engineering

TABLE J.39 Images in Chapter 38.

Image	NASA ID	Title	Date
Chapter opening	GSFC_20171208_Archive_e000440	By the Dozen: NASA's James Webb Space Telescope Mirrors	2017-12-08
hstmirror	7995584	History of Hubble Space Telescope (HST)	1979-03-01
actuator	GSFC_20171208_Archive_e001917	The Webb Telescope's Actuators: Curving Mirrors in Space	2017-12-08
hrma	9704166	History of Chandra X-Ray Observatory	1997-01-01
starshade	PIA20904	Half-Scale Starshade	2016-08-09
fpa	GSFC_20171208_Archive_e001055	Webb Instruments Perfected to Microscopic Levels	2017-12-08
sunshield	GSFC_20171208_Archive_e001023	NASA's Webb Sunshield Stacks Up to Test	2017-12-08
cryocooler	PIA20688	MIRI Cryocooler Packing	2016-06-13
euclid	PIA25783	Euclid Spacecraft in Cleanroom	2023-07-01
chambera	GSFC_20171208_Archive_e000032	NASA's Webb Telescope "chilling out" in Houston for the summer	2017-12-08
wfpc	9400368	History of Hubble Space Telescope (HST)	1993-12-01

J.40 Chapter 39: Great Observatories and Deep-Space Telescopes

TABLE J.40 Images in Chapter 39.

Image	NASA ID	Title	Date
Chapter opening	PIA18165	Hubble Against Earth Horizon 1997	1997-10-10
sts31	s31-03-009	STS-31 Hubble Space Telescope (HST) (SAs & HGAs deployed) is grappled	1990-04-25
gro	s37-99-056	STS-37 Gamma Ray Observatory (GRO) grappled by RMS	1991-04-07
chandrabay	sts093-705-020	View of the Chandra Observatory before its deployment from Columbia PL	1999-07-23
pillars	GSFC_20171208_Archive_e000842	Hubble Goes High Def to Revisit the Iconic 'Pillars of Creation'	2017-12-08
crab	GSFC_20171208_Archive_e000053	Observatories Combine to Crack Open the Crab Nebula	2017-12-08
spitzer	PIA23644	Spitzer Space Telescope Ready for Launch	2020-01-22
jwstmirror	GSFC_20171208_Archive_e000422	NASA's James Webb Space Telescope Primary Mirror Fully Assembled	2017-12-08
deepfield	webb_first_deep_field	Webb_first_deep_field_SMACS_0723	2022-07-12
kepler	PIA11188	NASA Kepler Spacecraft in a Clean Room	2008-09-23
romanfairing	KSC-20260821-PH-SER01_0002	Roman Space Telescope Encapsulation	2026-08-21

J.41 Chapter 40: In-Space Servicing, Assembly and Manufacturing

TABLE J.41 Images in Chapter 40.

Image	NASA ID	Title	Date
Chapter opening	iss071e361950	The Canadarm2 robotic arm above the Mozambique Channel	2024-07-23
solarmax	41c-34-1380	View of the damaged Solar Maximum Mission Satellite from the 41-C Chal	1984-04-08
intelsat	s49-91-026	STS-49 crew captures INTELSAT VI above OV-105's payload bay (PLB) duri	1992-05-13
hstgrapple	s125e006955	Hubble Space Telescope (HST) during rendezvous with STS-125 Shuttle At	2009-05-13
rrm	s135e007549	Fossum transfers Robotics Refueling Mission during EVA 1	2011-07-12
restorel	KSC-20200219-PH-CSH01_0001	Restore-L Testing	2020-02-19
s1truss	sts112-709-033	S1 truss installed on ISS	2002-10-12
printer	iss042e031282	3D Printer Coupon removal and stowage	2014-12-09

Image	NASA ID	Title	Date
metal	iss071e522127	NASA astronaut Jeanette Epps configures the Metal 3D printer	2024-08-21

J.42 Chapter 41: Space Resources and Asteroid Mining

TABLE J.42 Images in Chapter 41.

Image	NASA ID	Title	Date
Chapter opening	Bennu_Sample2	Vial with Sample of Asteroid Bennu	2024-07-22
ilmenite	S69-47900	PHOTO MICROGRAPH - LUNAR SAMPE 10022	1969-08-28
orange	as17-137-20989	View of the orange soil which Apollo 17 crewmen found at Station 4 dur	1972-12-12
marsice2	PIA26044	Ice-Exposing Impact Crater Surrounded by Polygon Terrain	2023-10-26
eros	PIA02923	Mosaic of Eros Northern Hemisphere	2000-06-10
psyche	KSC-20230626-PH-FMX01_0040	Psyche Prelaunch Spacecraft Processing	2023-06-26
moxie	PIA24176	Engineers Lower MOXIE into Perseverance	2020-11-24
ipex	KSC-20240830-PH-FMX01_0027	IPEx Testbed Filming	2024-08-30
rassor	KSC-20250603-PH-FMX01_0086	RASSOR Excavation Testing Swamp Works	2025-06-03
trailblazer	PIA25837	Lunar Trailblazer Spacecraft Nears Completion	2023-08-16
msolo	KSC-20200925-PH-GEB01_0004	MSolo Instrument Work	2020-09-25

J.43 Chapter 42: Advanced Propulsion

TABLE J.43 Images in Chapter 42.

Image	NASA ID	Title	Date
Chapter opening	GRC-2017-C-01056	Hall Thruster	2017-03-06
aeps	grc-2025-c-07151	Testing Gateway's AEPS Thruster at NASA Glenn (grc-2025-c-07151)	2025-09-02
ds1ion	PIA04238	Deep Space 1 Ion Engine	2002-12-21
kiwi	GRC-1964-C-69680	Technicians Manufacture a Nozzle for the Kiwi B-1-B Engine	1964-05-23
fsptdu	GRC-2015-C-06315	Fission Surface Power Technology Demonstration Unit	2016-11-09
neascout	MSFC-1601350	NEAR EARTH ASTERIOD SCOUT SOLAR SAIL	2015-01-08
acs3	CamC_Seq109_2024-08-28_17-17-13Z_V2_S595171	Advanced Composite Solar Sail System (ACS3) Onboard View of Sail/Boom	2024-08-28

Image	NASA ID	Title	Date
x43nose	EC04-0324-146	The small size of the X-43A scramjet is evident in this nose-on view w	2004-11-15
rdre	09-27-2023-HF-12 _062	3D-printed Rotating Detonation Rocket Engine Test	2023-09-27

J.44 Chapter 43: AI Foundations: Machine Learning for Space Engineers

TABLE J.44 Images in Chapter 43.

Image	NASA ID	Title	Date
Chapter opening	ACD24-0072-012	NASA Advanced Supercomputing Modular Facility Hyperwall Visualiz	2024-05-23
inco	S88-38979	JSC MCC Bldg 30 Instrumentation and Communications Officer (INCO) RTDS	1988-06-02
hubble	s31-04-027	Hubble Space Telescope (HST) above OV-103's PLB during STS-31 deployme	1990-04-25
smap	KSC-314D-0492_0074	SMAP During Weighing	2014-11-07
aegis	PIA20762	Autonomous Selection of a Rover Laser Target on Mars	2016-07-21
rockies	GSFC_20171208_Archive_e001576	Rocky Mountain Snow	2017-12-08
eo1	PIA18796	NASA Spacecraft Eyes Iceland Volcanic Eruption	2014-09-03
lovell	S69-35099	ASTRONAUT LOVELL, JAMES A., JR. - APOLLO VIII (GUIDANCE & NAVIGATION [	1969-05-25
fires	GSFC_20171208_Archive_e000668	Series of Wildfires in Northern California Continue Blazing	2017-12-08

J.45 Chapter 44: Deep Learning, Foundation Models and Generative AI

TABLE J.45 Images in Chapter 44.

Image	NASA ID	Title	Date
Chapter opening	ACD15-0062-021	ACD15-0062-021	2015-04-10
marscraters	PIA24040	AI Spots a Cluster of Mars Craters: HiRISE's view	2020-10-01
ai4mars	PIA23897	Martian Terrain as Viewed in AI4Mars	2020-06-12
milton	iss072e029135	Hurricane Milton is pictured in the Gulf of Mexico	2024-10-08
landsat9b	KSC-20210816-PH-RNB01_0090	Landsat 9 Encapsulation	2021-08-16
sdo	KSC-2009-6419	KSC-2009-6419	2009-11-16
roverdrivers	PIA24724	3D Glasses Used for Rover Driving	2021-07-01

J.46 Chapter 45: Data, Computing Hardware and Edge AI in Orbit

TABLE J.46 Images in Chapter 45.

Image	NASA ID	Title	Date
Chapter opening	iss065e009485	Spaceborne Computer-2	2021-04-29
discover	GSFC_20171208_Archive_e002034	Discover Supercomputer 1	2017-12-08
himasssee	iss032e016954	Spacecraft Single Event Enviroments at High Shielding Mass (HiMassSEE)	2012-08-11
ingenuity	PIA24547	Mastcam-Z Gives Ingenuity a Close-up	2021-04-06
cloudyearth	cloudy-earth_16810027983_o	Cloudy Earth	2015-05-08
laserdownlink	PIA26148	DSN's Experimental Hybrid Antenna Tracks DSOC's Laser Downlink	2024-02-08
issflyaround	s132e013215	Flyaround view of ISS after undocking	2010-05-23
cubesats	iss038e044994	NanoRacks CubeSat Deployment	2014-02-11

J.47 Chapter 46: AI for Earth Observation

TABLE J.47 Images in Chapter 46.

Image	NASA ID	Title	Date
Chapter opening	GSFC_20171208_Archive_e001699	Landsat View: Pearl River Delta, China	2017-12-08
kazakh	GSFC_20171208_Archive_e001371	Fall Harvest in Kazakhstan	2017-12-08
arcdefor	PIA26111	Composite of Brazil's 'Arc of Deforestation' Shows Imagery NISAR Will	2023-10-27
suez	PIA24553	Suez Canal Crisis	2021-04-06
idai	PIA23142	Satellites Image Mozambique Flooding after Cyclone Idai	2019-04-01
mariadpm	PIA21964	NASA Damage Map Aids FEMA's Hurricane Maria Rescue Operation in Puerto	2017-09-26
grasshopper	iss075e0001471	Thick smoke emanates from the Grasshopper wildfire burning near Mount	2026-07-31
tanager	PIA26416	Tanager-1 First Methane and Carbon Dioxide Plume Detections	2024-10-10

J.48 Chapter 47: AI for Weather, Climate and Space Weather

TABLE J.48 Images in Chapter 47.

Image	NASA ID	Title	Date
Chapter opening	GSFC_20171208_Archive_e000204	Hurricane Matthew from Space	2017-12-08
pleiades	ARC-2008-ACD08-0272-005	ARC-2008-ACD08-0272-005	2008-12-02

Image	NASA ID	Title	Date
melissa	iss073e1044319	Category 5 Hurricane Melissa is pictured in the Caribbean Sea	2025-10-28
sunspot	GSFC_20171208_Archive_e000923	NASA's SDO Observes Largest Sunspot of the Solar Cycle	2017-12-08
activeregion	PIA22582	Rambunctious Active Region	2018-07-05
aurora	iss074e0150531	A wispy, multi-colored aurora created by a geomagnetic storm	2026-01-19
imap	KSC-20250922-PH-SPX01_0008	NASA/SpaceX IMAP Vertical at LC 39A (Sunrise)	2025-09-22

J.49 Chapter 48: AI for Autonomy, Navigation and Robotics

TABLE J.49 Images in Chapter 48.

Image	NASA ID	Title	Date
Chapter opening	PIA24723	Computer Simulation of Perseverance's First Autonav Drive	2021-07-01
atvjv	iss016e034177	ATV during Demonstration Day 2 Rendezvous Test	2008-03-31
dm2	iss063e021460	Historic SpaceX DM-2 Approach and Docking to the ISS	2020-05-31
aegis	PIA25289	Perseverance's SuperCam Uses AEGIS For the First Time	2022-05-31
astrobee	iss071e046284	A free-flying Astrobee robot pictured aboard the International Space S	2024-05-01
robonaut	iss034e013990	Robonaut 2 in the U.S. Laboratory	2013-01-02
cadre	PIA25665	CADRE Rover Testing in JPL's Mars Yard	2023-08-02

J.50 Chapter 49: AI for Mission Operations, Communications and Networks

TABLE J.50 Images in Chapter 49.

Image	NASA ID	Title	Date
Chapter opening	jsc2026e022251	Artemis II Flight Control Team During Splashdown	2026-04-10
hubbleops	GSFCRR2021071501	NASA Begins Switch to Backup Spacecraft Hardware	2021-07-15
batteryeva	iss061e005514	iss061e005514	2019-10-11
optimism	PIA24527	OPTIMISM Ready for Testing	2021-11-17
goldstone	PIA26716	Goldstone's 70-Meter Antenna Under the Stars	2026-01-27
tdrs	KSC-2014-1075	Shots of TDRS-L Spacecraft Prior to Media Day	2014-01-03
scantb	GRC-2011-C-03408	GRC-2011-C-03408	2007-09-07

J.51 Chapter 50: AI for Space Situational Awareness, Defence and Security

TABLE J.51 Images in Chapter 50.

Image	NASA ID	Title	Date
Chapter opening	PIA24561	Goldstone Radar Observations of Asteroid 2001 FO32	2021-04-05
2018la	PIA22468	Asteroid 2018 LA	2018-06-03
shelter	iss030e173969	Expedition 30 crewmembers in the SM	2012-03-24
atv3	iss033e009232	ATV reentry	2012-10-03
kuwait	s37-152-092	Kuwait Oil Fires, Kuwait	1991-04-11
sarship	PIA01799	Space Radar Image of North Atlantic Ocean	1999-04-15
fdo	jsc2022e090753	Flight Day 16: Mission Control during Distant Retrograde Departure	2022-12-01

J.52 Chapter 51: AI for Space Science, Planetary Exploration and Human Spaceflight

TABLE J.52 Images in Chapter 51.

Image	NASA ID	Title	Date
Chapter opening	iss069e010815	Astronaut Sultan Alneyadi observes a free-flying Astrobee robotic assi	2023-05-16
tess	KSC-20180212-PH-JBS03_0030	TESS Spacecraft Lift to Work Stand	2018-02-12
jezero	PIA23380	Jezero Crater Minerals	2019-11-12
pixl	PIA26365	Perseverance's PIXL Instrument Examines 'Thunderbolt Peak'	2024-07-16
biomonitor	iss058e003984	Bio-Monitor	2019-01-16
cimon	iss073e0384102	JAXA astronaut Takuya Onishi sets up the CIMON AI-powered robotic assi	2025-07-18
callisto	art001e000192 callisto	Inside Orion With Callisto	2022-11-18
hololens	iss046e043637	HoloLens	2016-02-20

J.53 Chapter 52: AI in Spacecraft Design, Manufacturing, Launch and the Space Business

TABLE J.53 Images in Chapter 52.

Image	NASA ID	Title	Date
Chapter opening	MSFC-201900153	NASA Deputy Administrator, Jim Morhard, Tours MSFC Additive Manu	2019-03-25
st5	KSC-06pd0434	KSC-06pd0434	2006-02-14
excite	GSFC2024excite02	GSFC2024excite02	2024-08-29

Image	NASA ID	Title	Date
shaker	PIA26713	NEO Surveyor's Spacecraft Bus on the 'Shaker Table'	2026-01-14
landing	NHQ20261001_admin_0012	Falcon 9 Landing Following NASA's SpaceX Crew-13 Launch	2026-10-01
rs25	SSC-20240229-s00309	NASA Caps Off February with RS-25 Engine Certification Test	2024-02-29
lcc	KSC-20251008-PH-GEB01_0136	Artemis II Terminal Count Simulation	2025-10-18

J.54 Chapter 53: Trustworthy AI for Space: Assurance, Security, Governance and Outlook

TABLE J.54 Images in Chapter 53.

Image	NASA ID	Title	Date
Chapter opening	ivv	NASA Renames Facility in Honor of 'Hidden Figure' Katherine Johnson	2019-02-21
morph3	KSC-2014-4818	KSC-2014-4818	2014-12-15
artemisfcr	jsc2026e021924	Artemis II Flight Control Team during Lunar Flyby	2026-04-06
lax	iss074e0622240	Los Angeles International Airport on Southern California's Pacific coa	2026-05-29
cities	iss073e0842595	City lights from Tokyo to Osaka and Hiroshima	2025-09-15
ltv	jsc2026e417568	Crewed Lunar Terrain Vehicle Testing	2026-09-15
jett	jsc2024e035443	JETT 5 - jsc2024e035443	2024-05-16